Model Structure (26.4_Spat_3_Reg):

This is a Spatial model (do_spatial >= 2).
There are 2 sexes, 2 fishing fleets, 3 survey fleets, and 3 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post−1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre−1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A new weight curve is estimated with a single block (do_rec_WAA == 0).
For fishery composition data aggregation catch−weighting, the newly estimated sex−disaggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 1).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations were NOT included in the LLS RPN index (drop_WAI_LLS == 1).

The longline survey used off−year extrapolations for the BS (even years) or AI (odd years) based on the year over year change in the GOA RPNs.
Starting in 2025, carryover values for the BS+AI (odd years) or GOA (even years) were used (drop_LLS_extrap == 0 & do_LLS_carry == 0).
The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

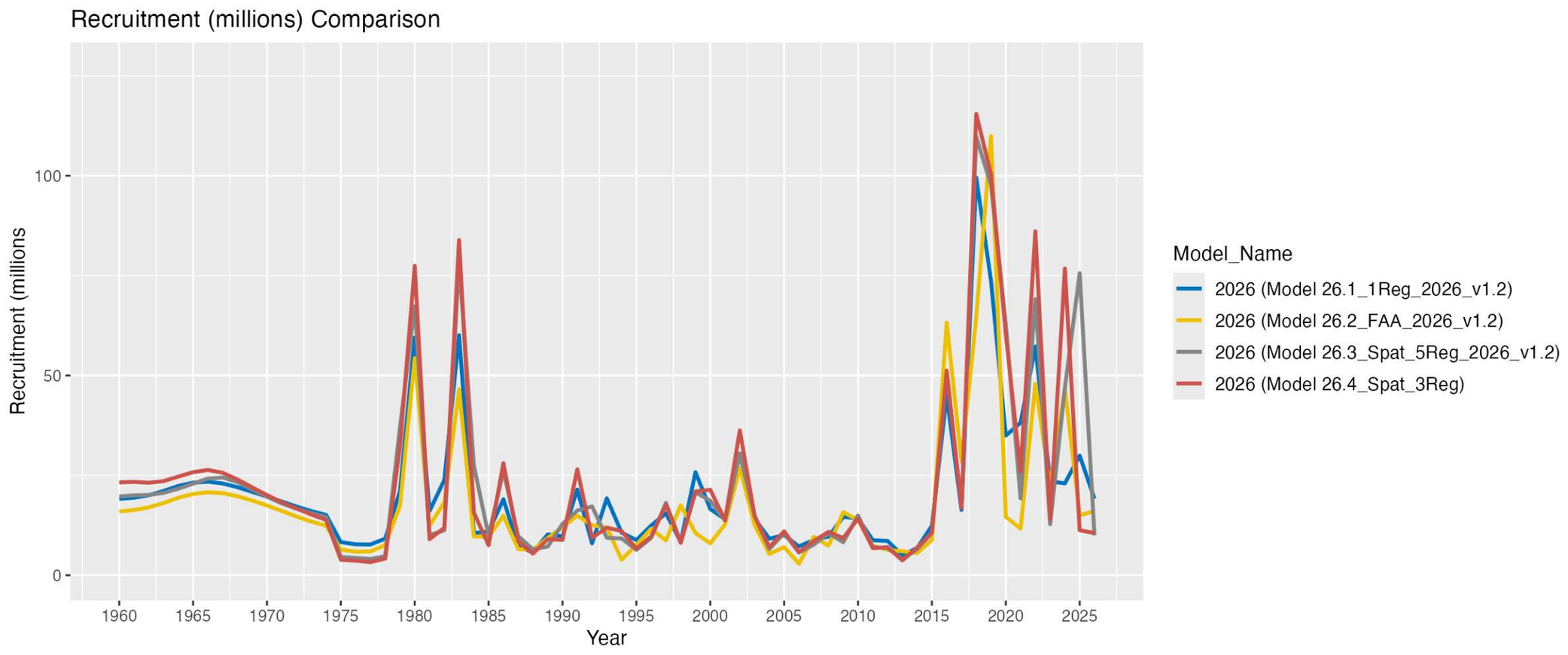
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years) (do_spatial > 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is scaled to a maximum value across years/sexes/regions to address interannual variability (do_spatial >1 & do_scale_ISS == 1).

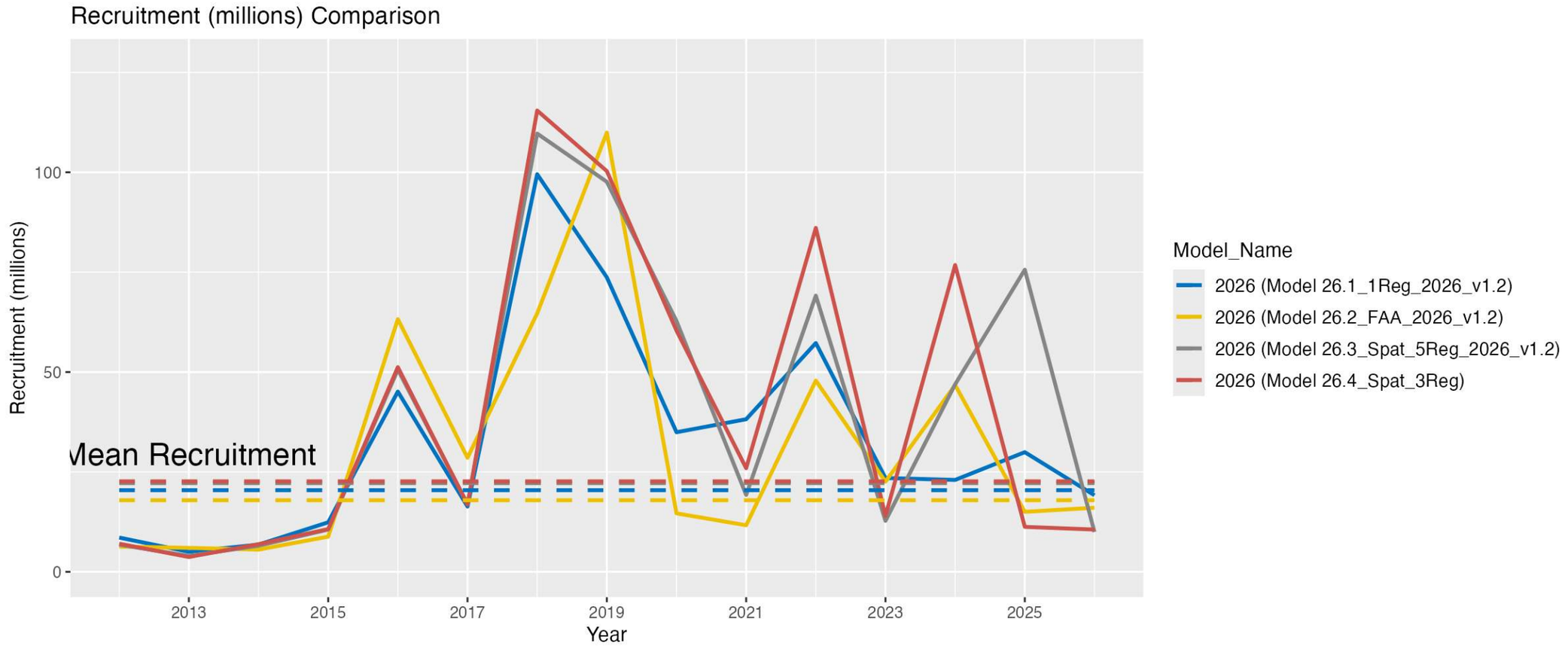
Length Composition Data Use:

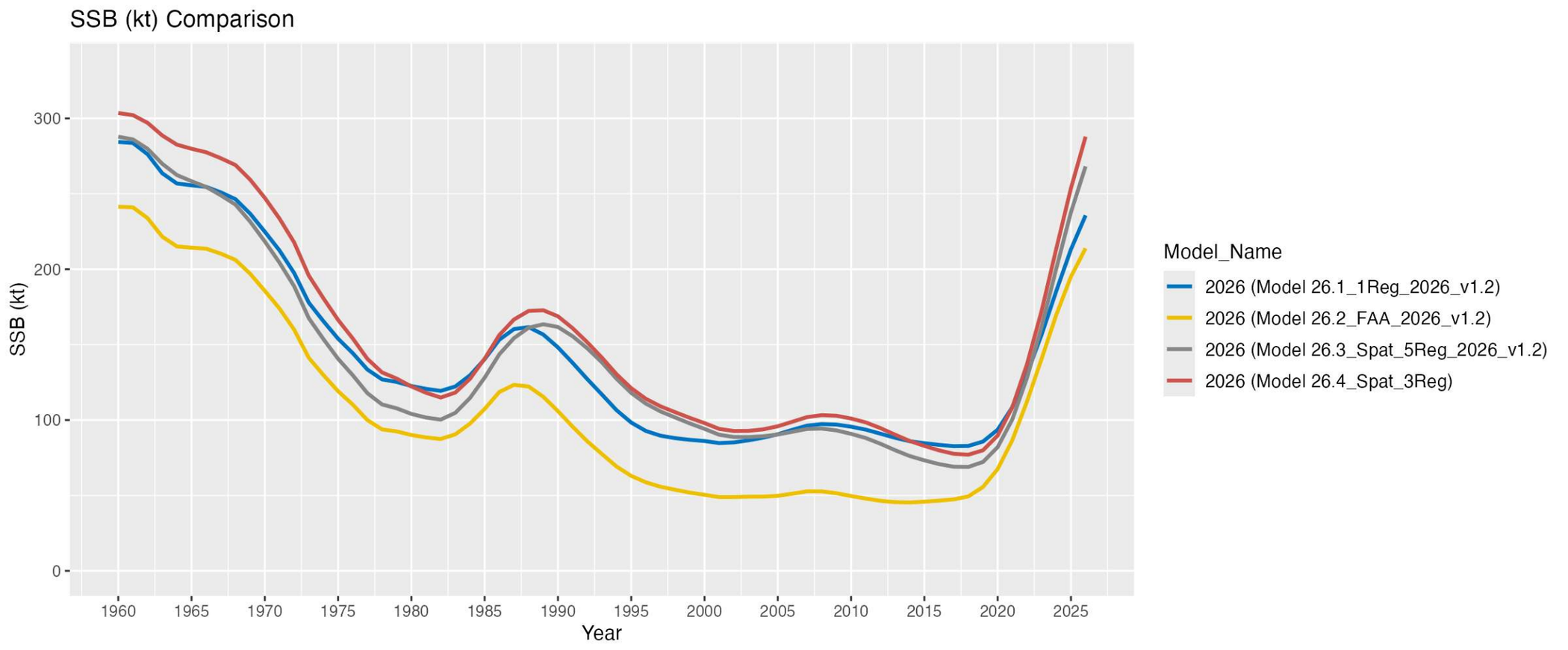
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data (LLS_length_comp_source == 1).
No longline survey length data are fit in the model (this does not apply to JPN LLS lengths) −− only age data are used (fit_LLS_len == 0).
No fixed gear fishery length data are fit in the model (this does not apply to trawl fishery lengths) −− only age data are used (fit_LLF_len == 0).

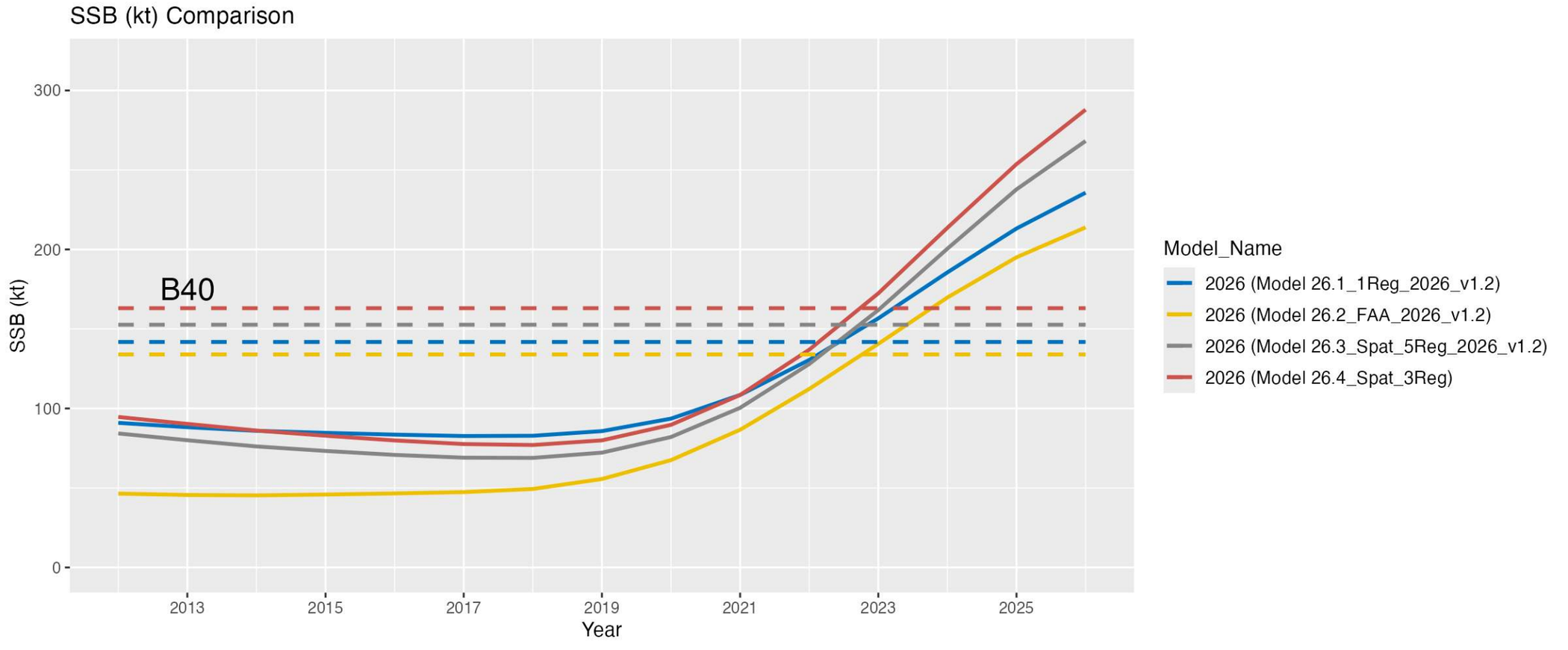
General Data Fitting Notes:

Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index was NOT fit in this model (use_CPUE == 0).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where all sources of removals were available for terminal year (add_non_fish_catch == 1 & non_fsh_catch_updated == 1).


Recruitment (millions) Comparison
Recruitment (millions
100
50
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.3_Spat_5Reg_2026_v1.2)
2026 (Model 26.4_Spat_3Reg)


Recruitment (millions) Comparison
Recruitment (millions)
100
50
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.3_Spat_5Reg_2026_v1.2)
2026 (Model 26.4_Spat_3Reg)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.3_Spat_5Reg_2026_v1.2)
2026 (Model 26.4_Spat_3Reg)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
B40
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.3_Spat_5Reg_2026_v1.2)
2026 (Model 26.4_Spat_3Reg)

Population — Population 1 Model — 26.4_Spat_3Reg

Female | Male

BS | AI | GOA

Spawning Weight-at-Age (kg)

0.100
0.075
0.050
0.025
0.000

0 10 20 30

Age

Population — Population 1 Model — 26.4_Spat_3Reg

Female | Male | BS | AI | GOA

Maturity at Age

1.00
0.75
0.50
0.25
0.00

0 10 20 30

Age

Population — Population 1 Model — 26.4_Spat_3Reg

Female | Male

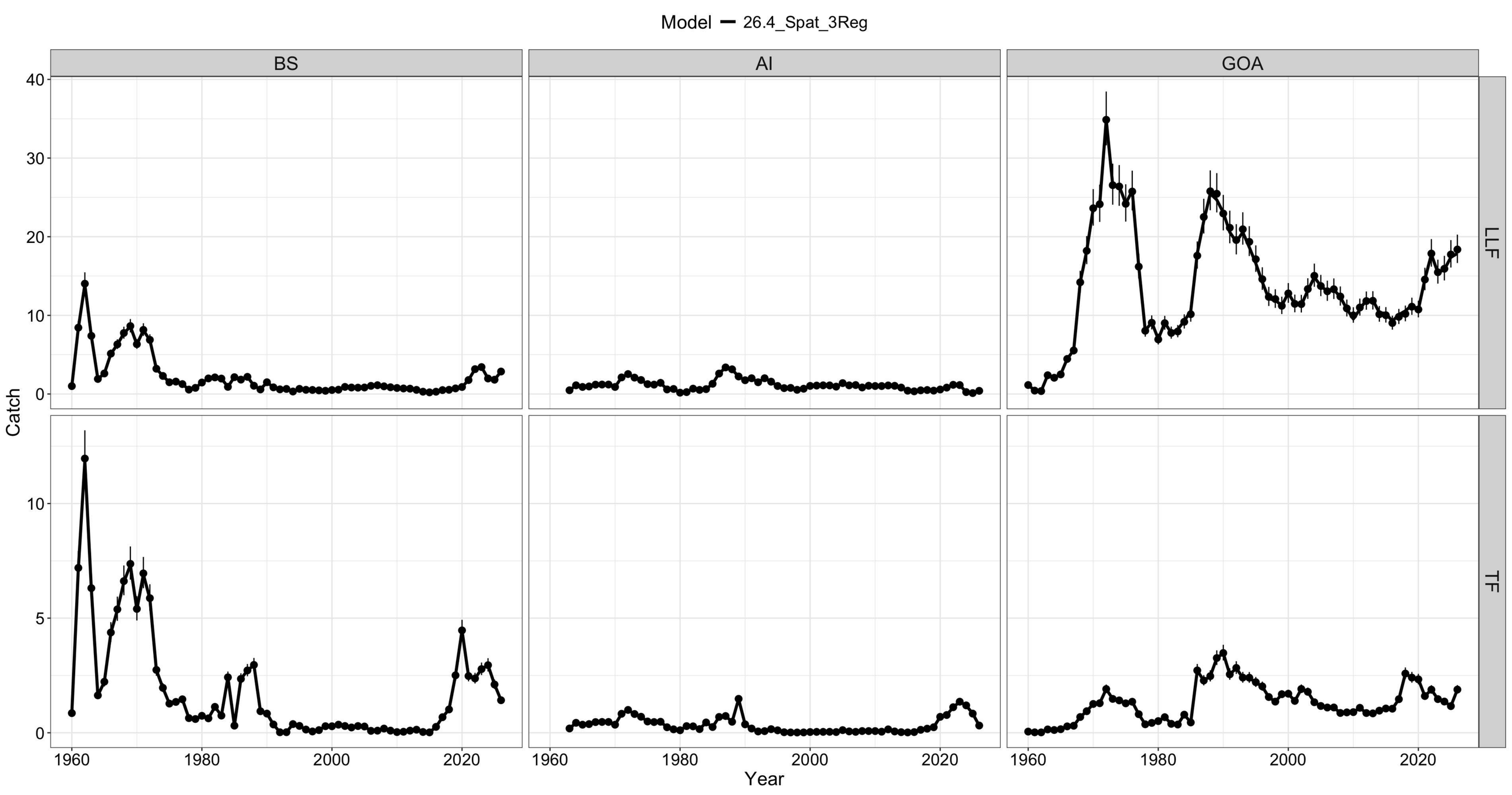
BS | AI | GOA

Spawning Weight at Age

0 2 4 6

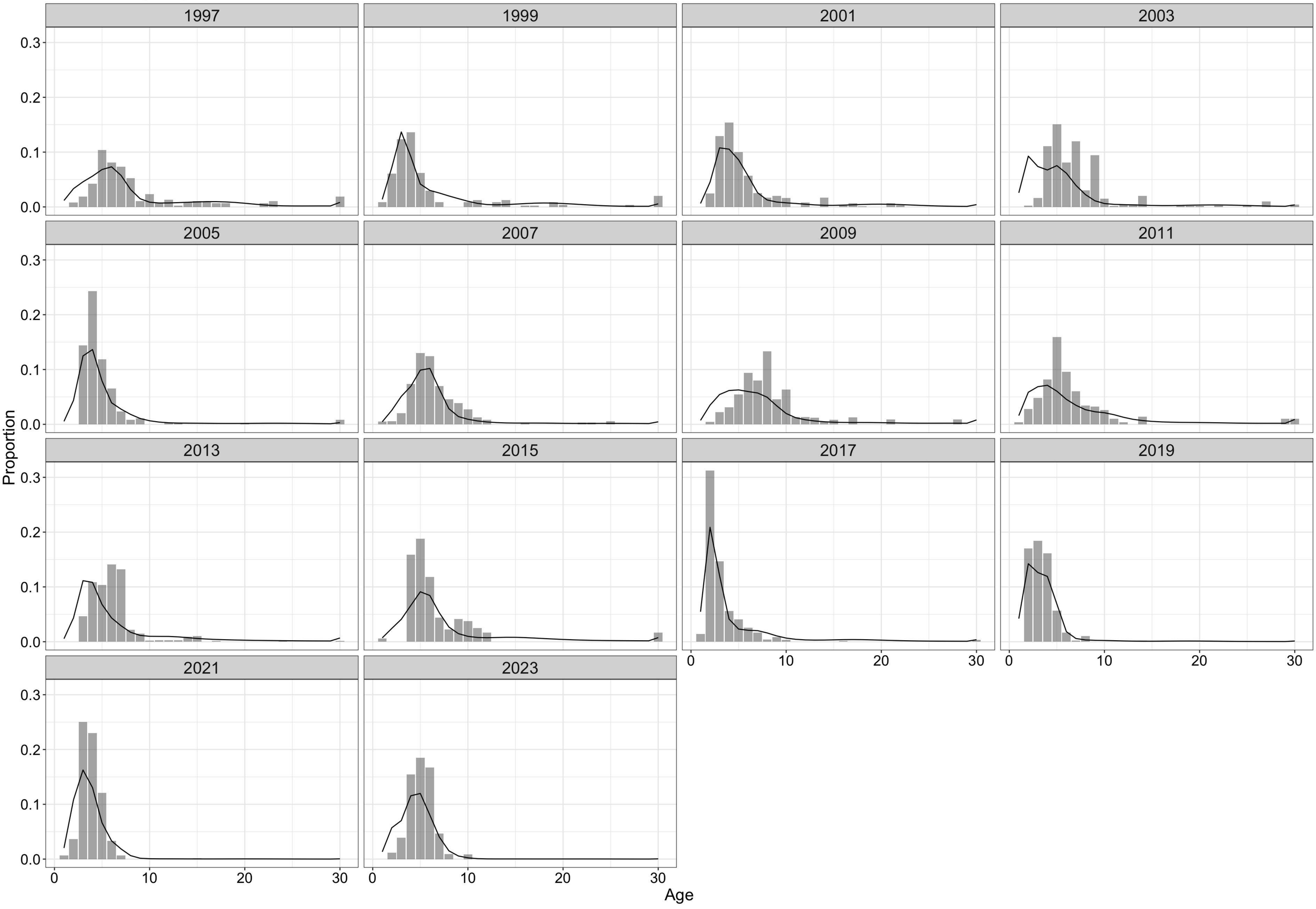
0 10 20 30

Age


Model 26.4_Spat_3Reg
BS
AI
GOA
LLF
TF
Catch
Year
0
10
20
30
40
5
1960
1980
2000
2020

BS
AI
GOA
26.4_Spat_3Reg
Tagging
JPN LLS Index
LLS Index
JPN LLS Ages
LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
1960
1980
2000
2020
Year

LL Survey Age Comps Female BS

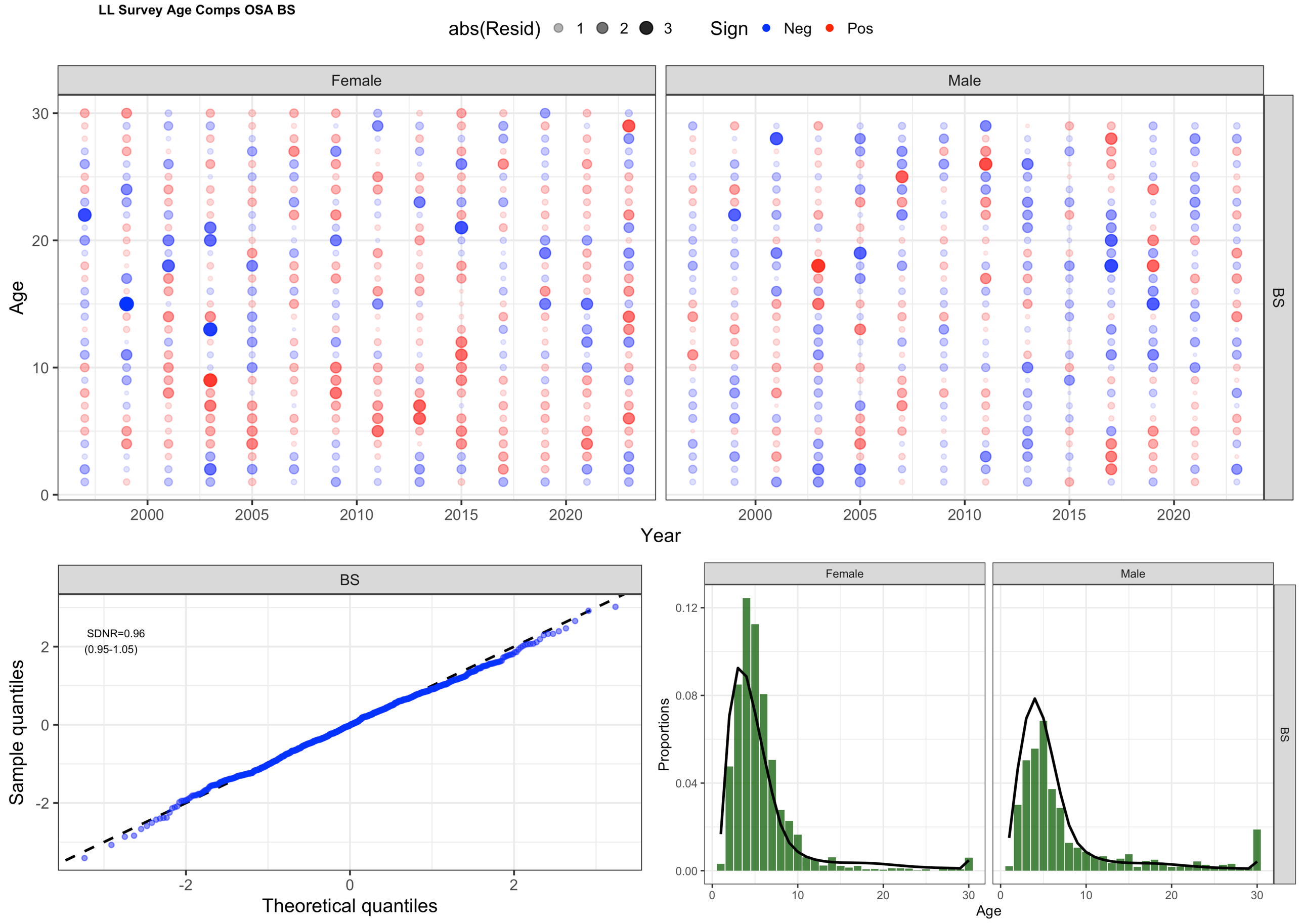


LL Survey Age Comps Male BS

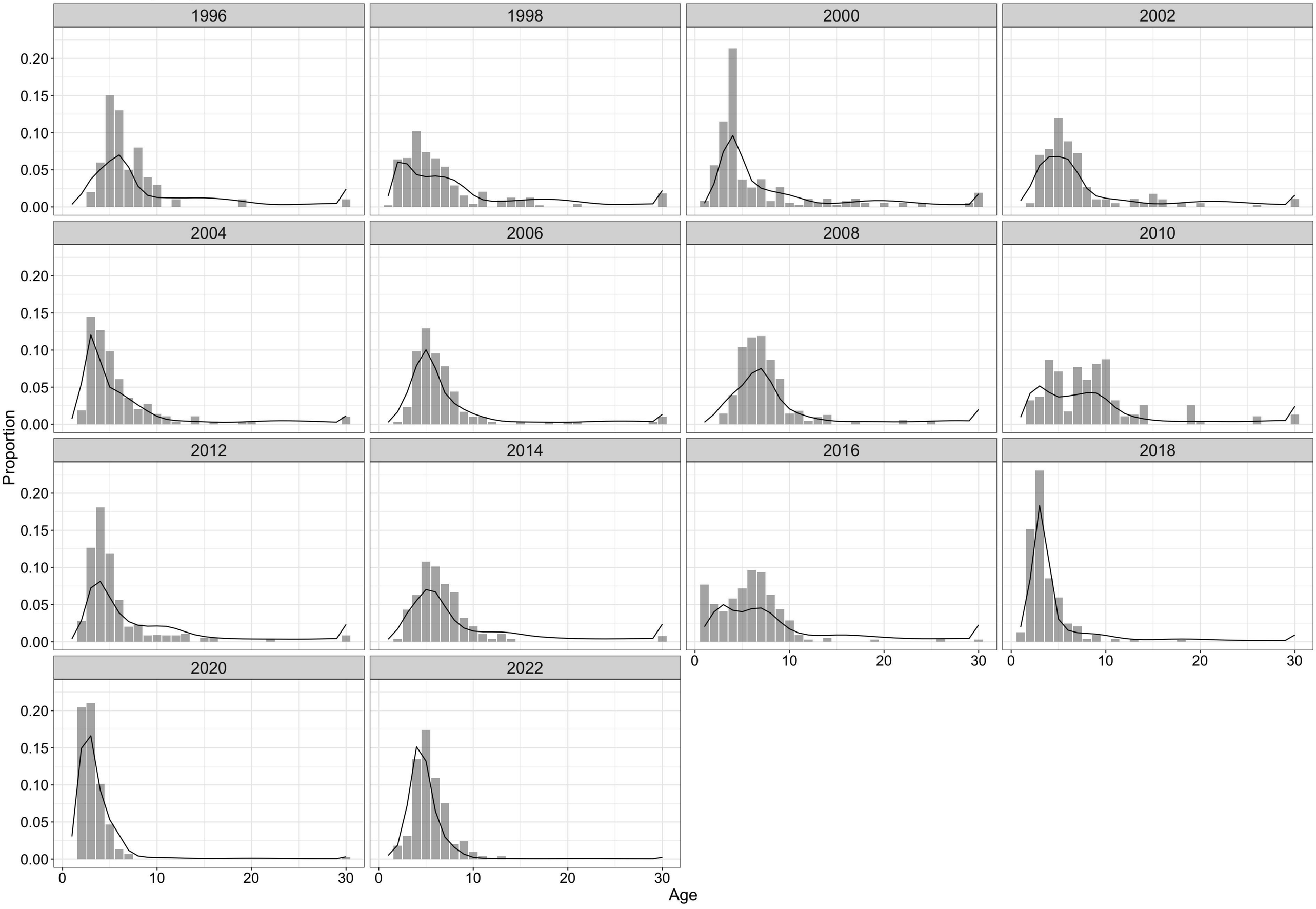
1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

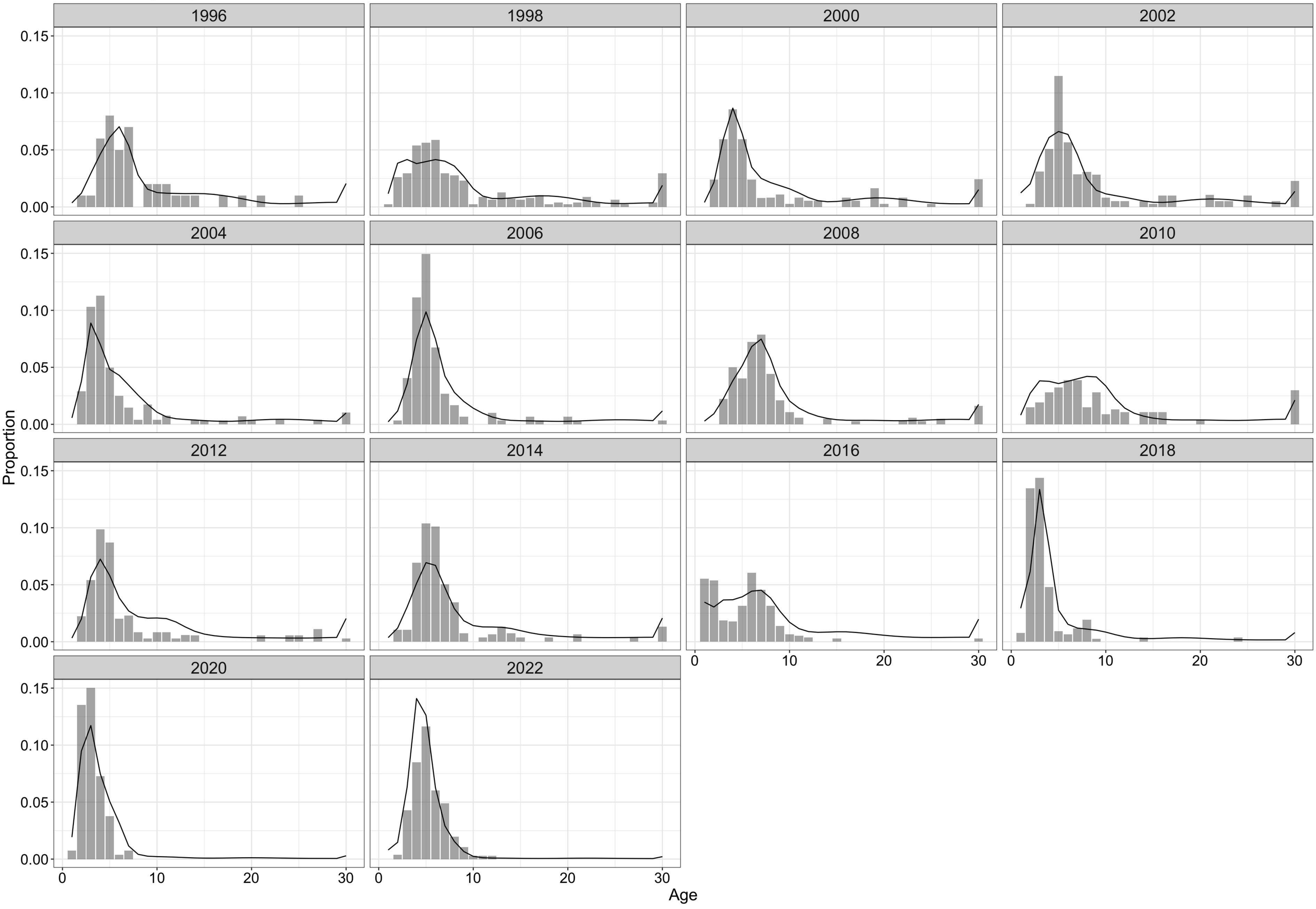
LL Survey Age Comps OSA BS



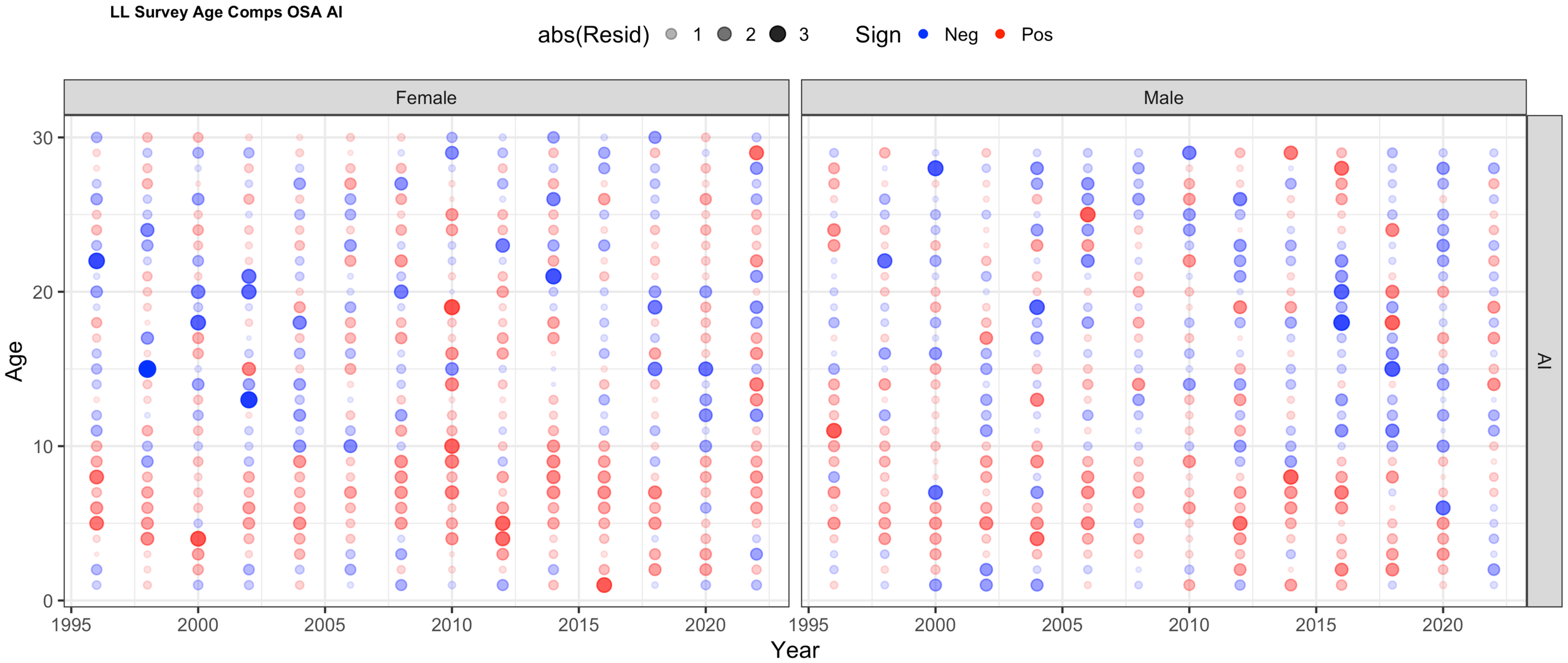
LL Survey Age Comps Female AI

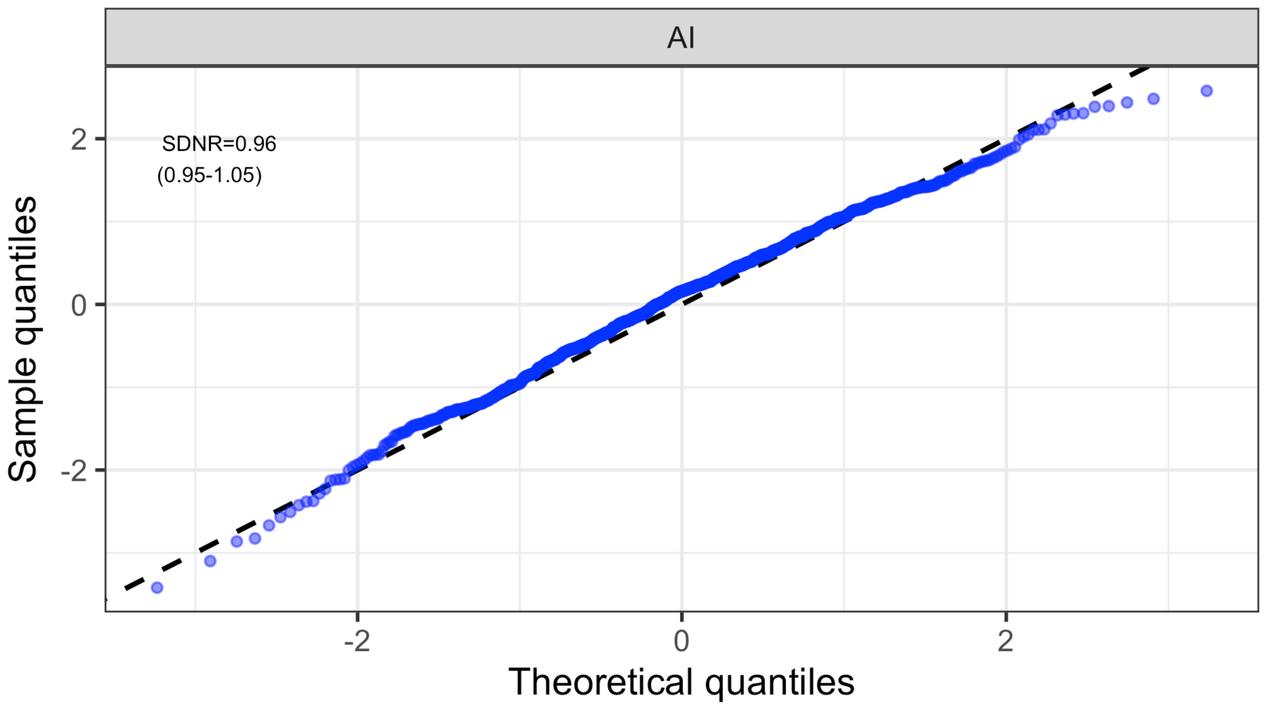


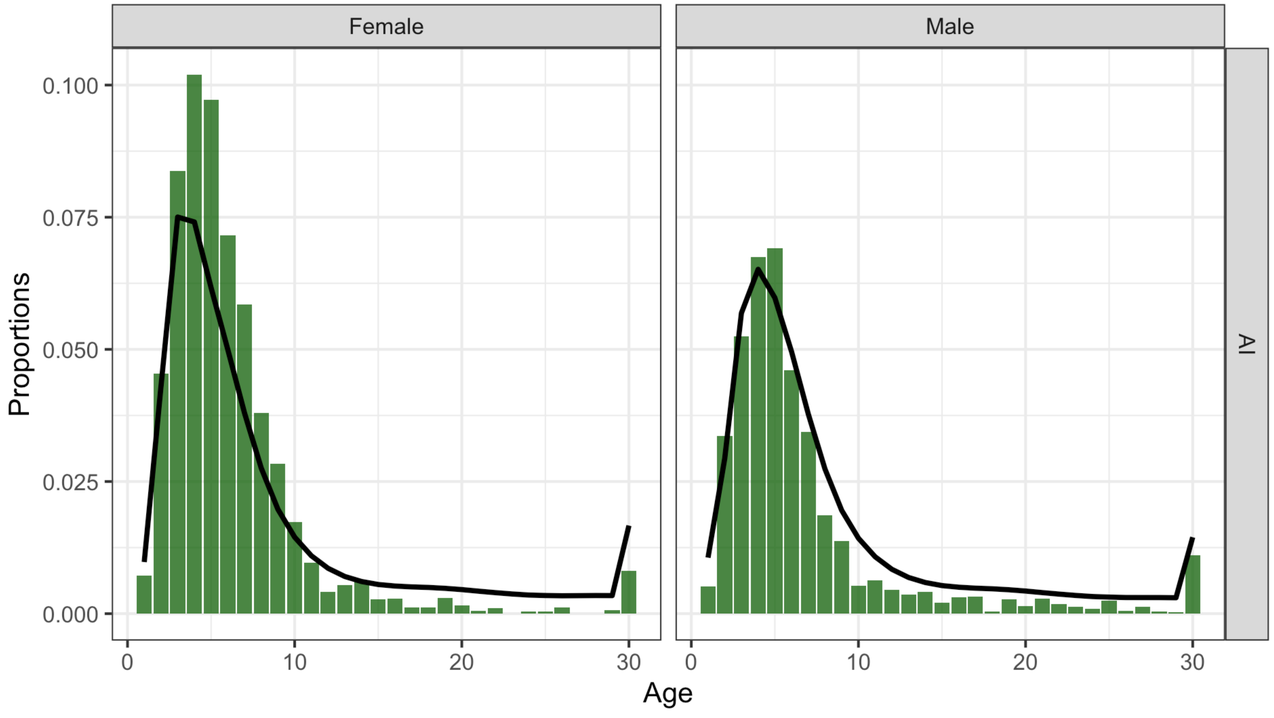
LL Survey Age Comps Male AI



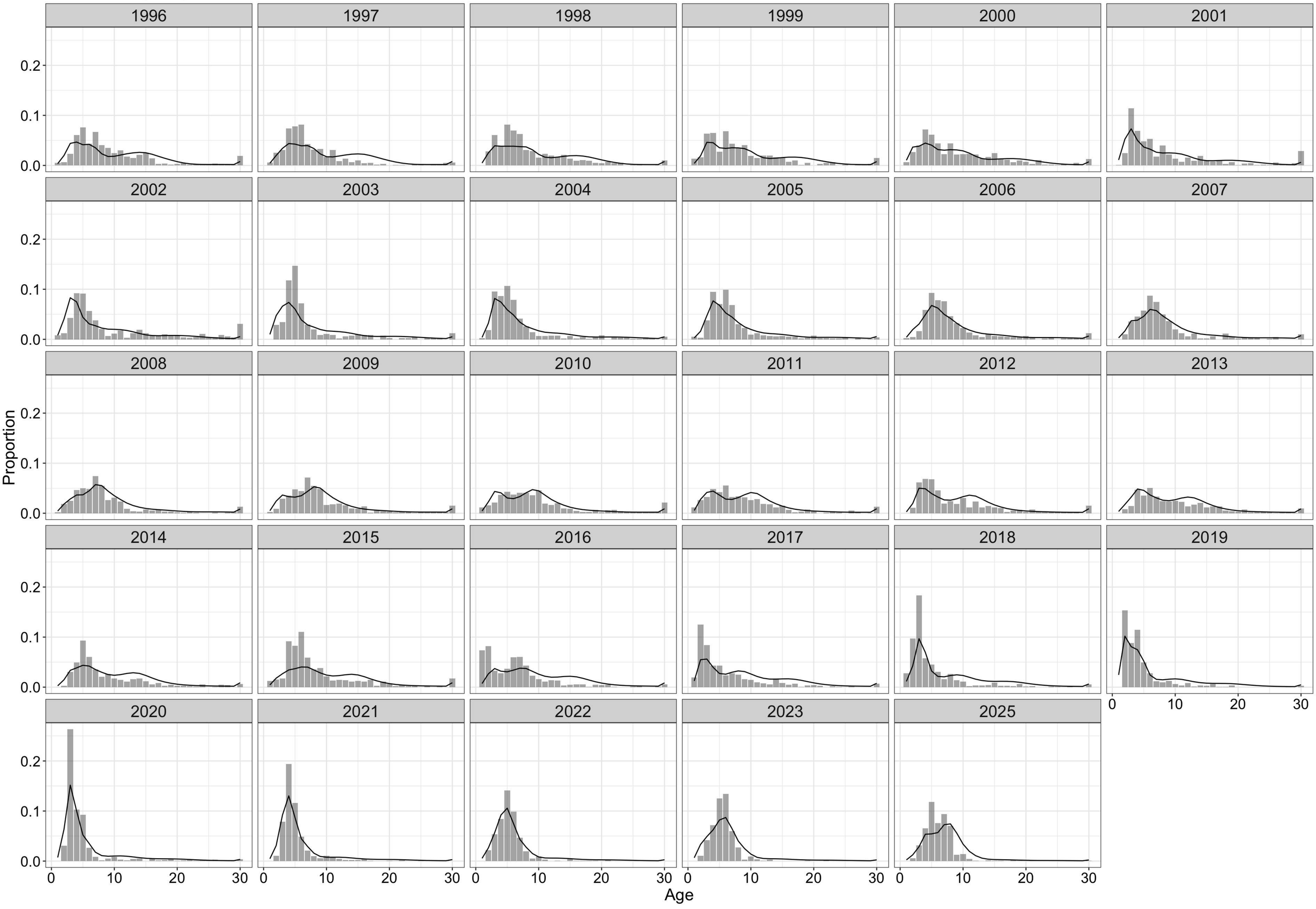
LL Survey Age Comps OSA AI







LL Survey Age Comps Female GOA



LL Survey Age Comps Male GOA

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

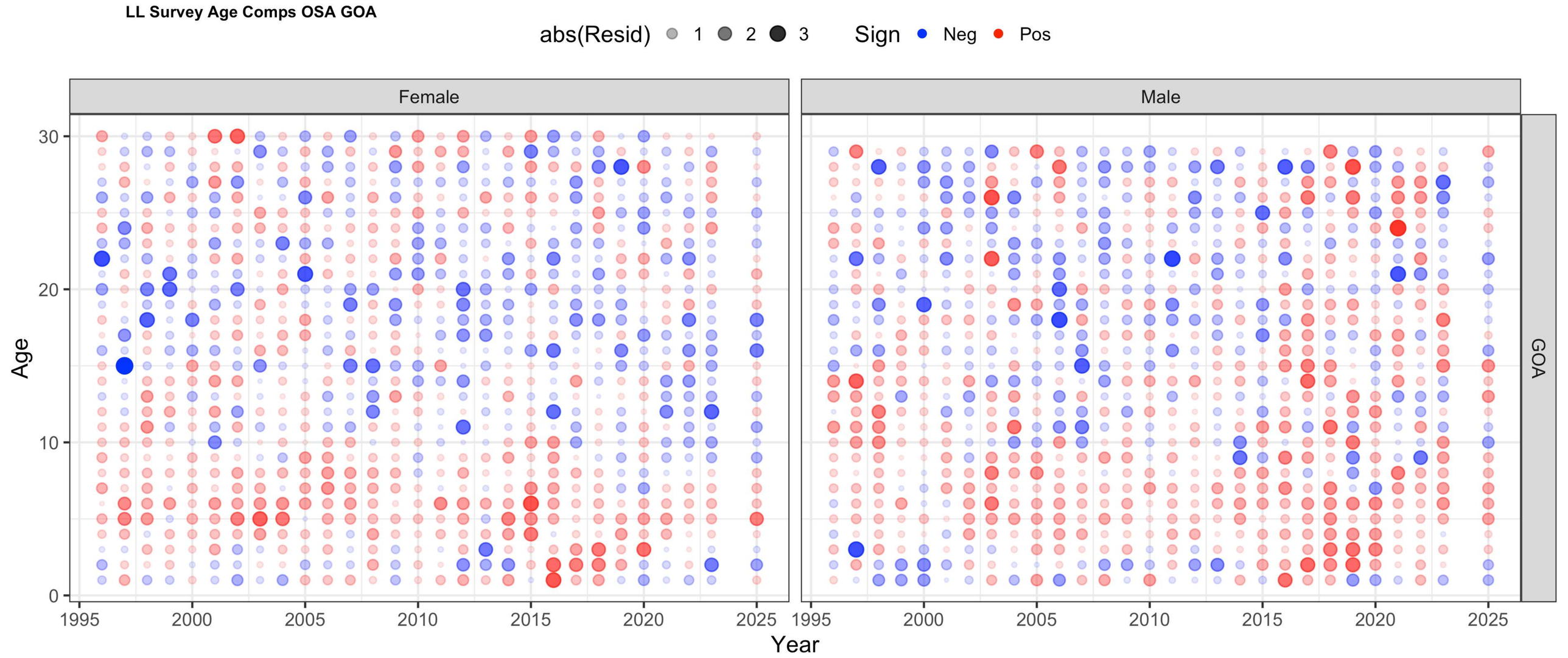
Proportion

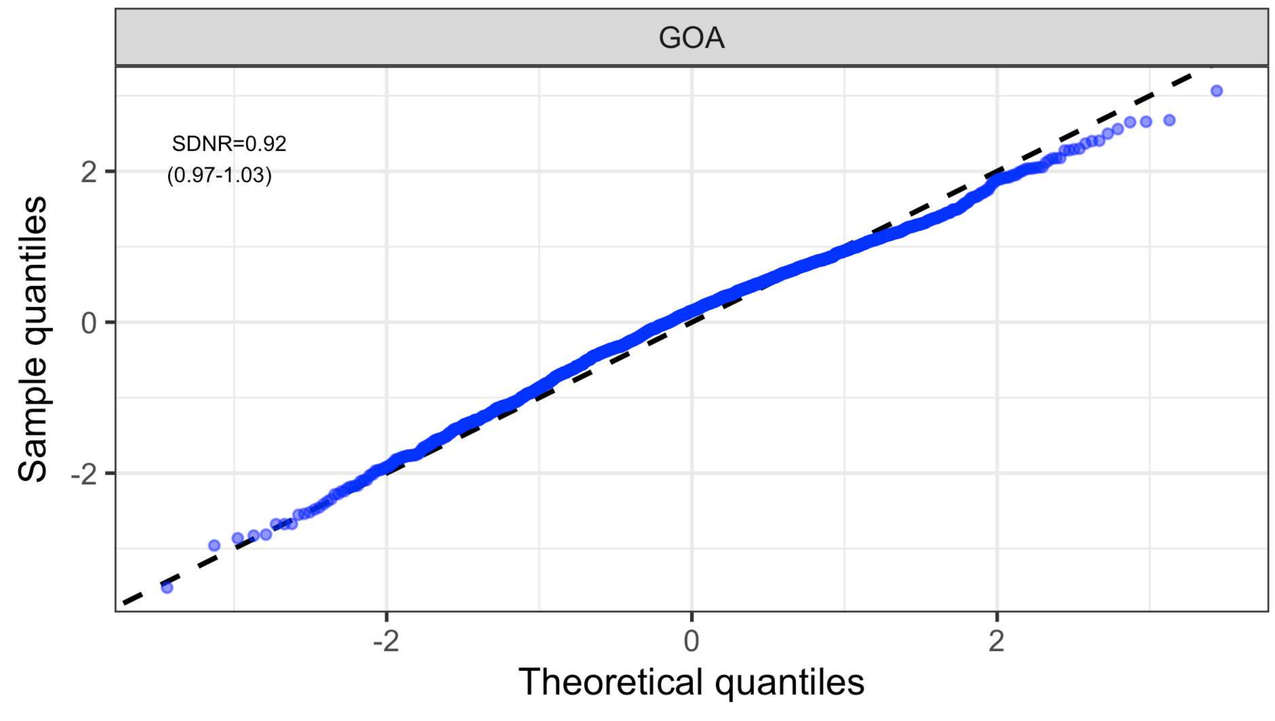
0.00 0.05 0.10 0.15

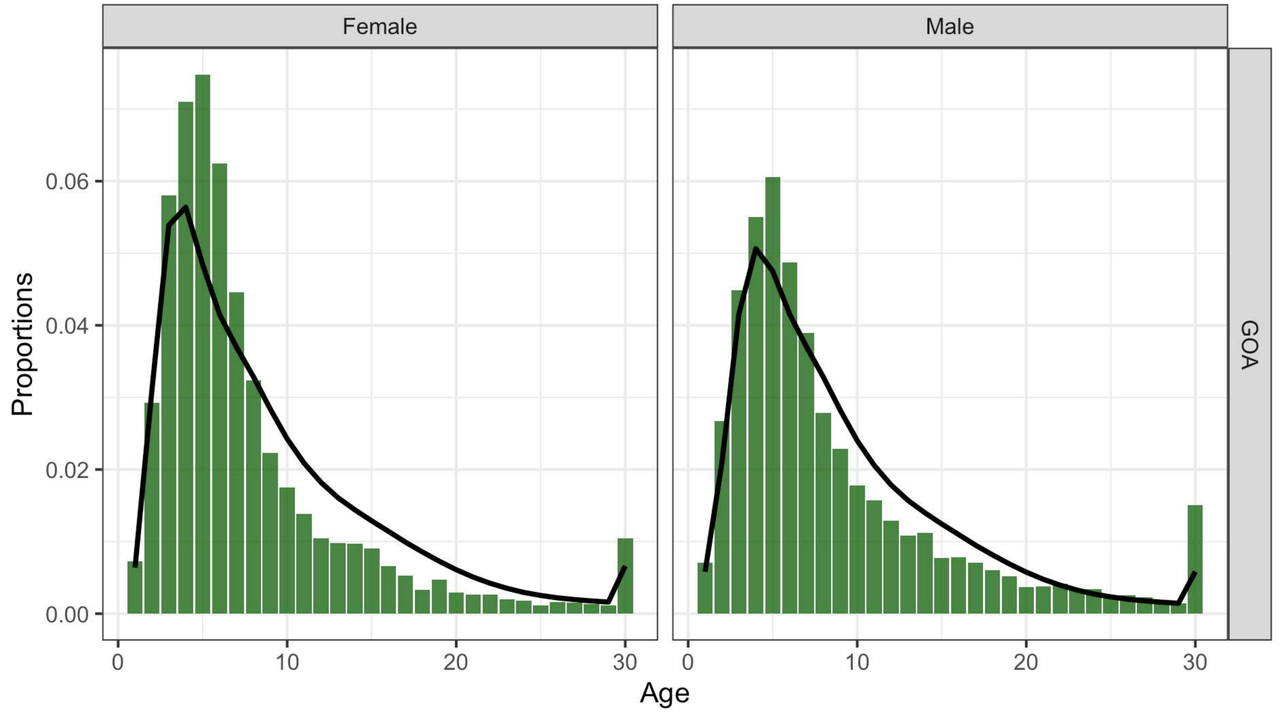
0 10 20 30

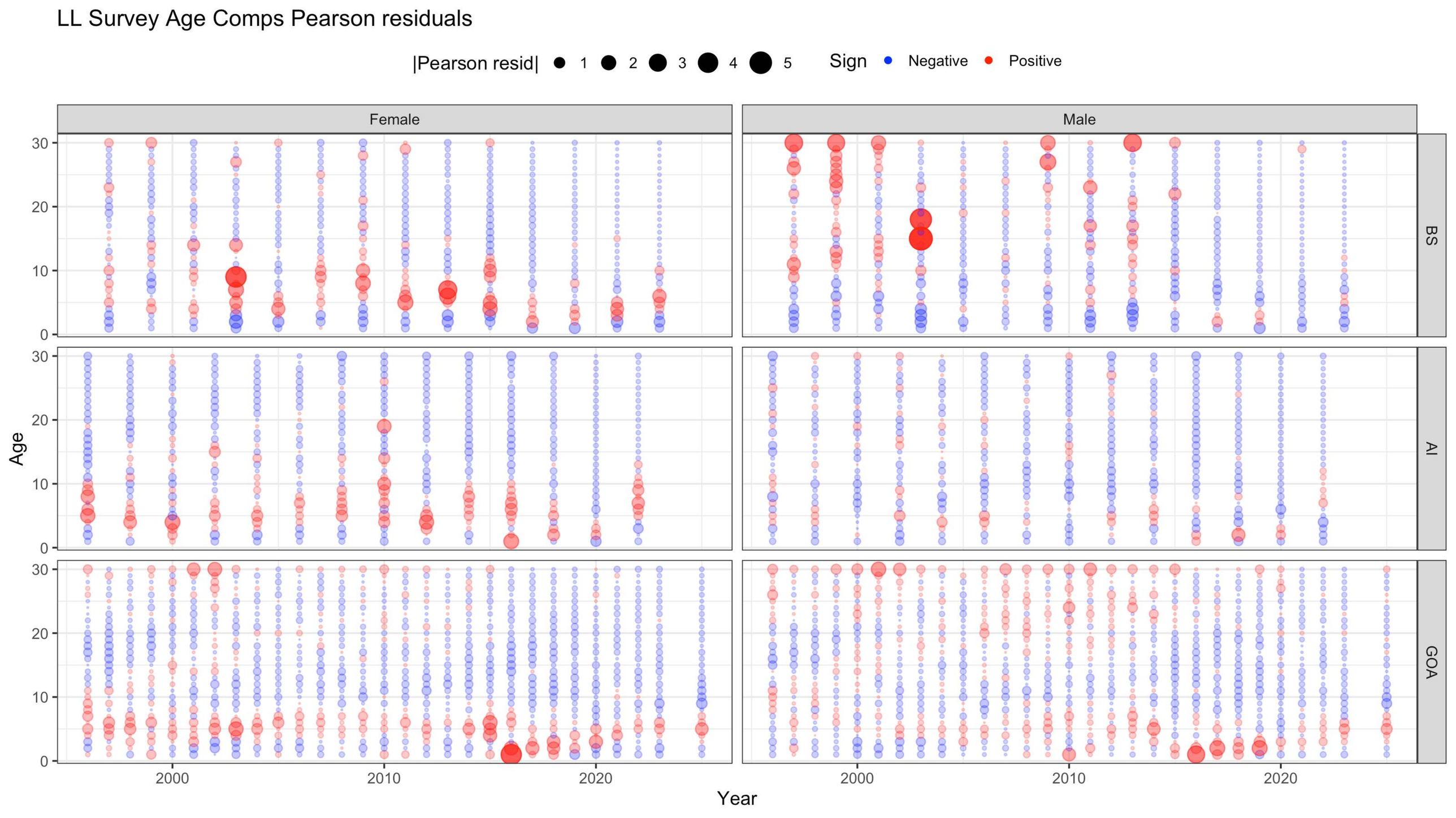
Age

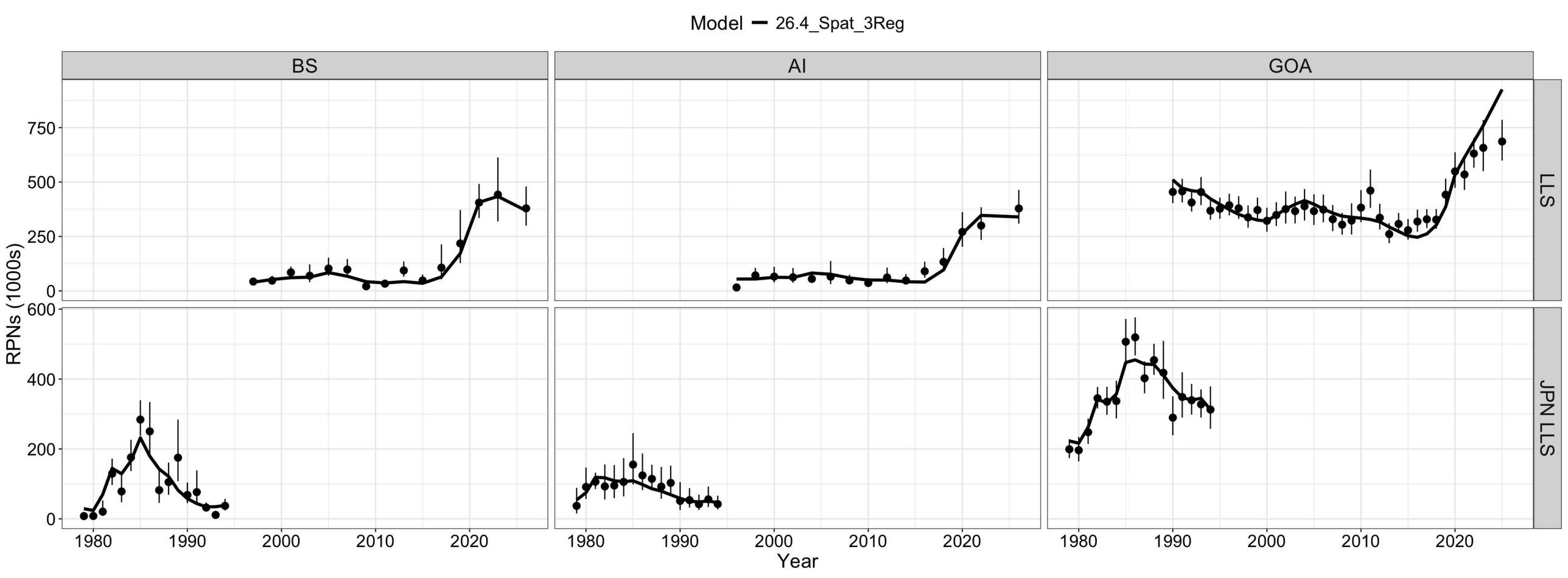
LL Survey Age Comps OSA GOA








LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
3
4
5
Sign
Negative
Positive
Female
Male
BS
AI
GOA
Age
Year
0
10
20
30
2000
2010
2020


Model — 26.4_Spat_3Reg
BS
AI
GOA
LLS
JPN LLS
RPNs (1000s)
0
250
500
750
0
200
400
600
1980
1990
2000
2010
2020
Year

Jitter SSB and Recruitment

Region 1 | Region 2 | Region 3

Recruitment (millions)

SSB (kt)

Val

ue

40
20
0

250
200
150
100
50
0

tion Converged: 1
L < original jnLL: 0

tion Converged: 1
L < original jnLL: 0

tion Converged: 1
L < original jnLL: 0

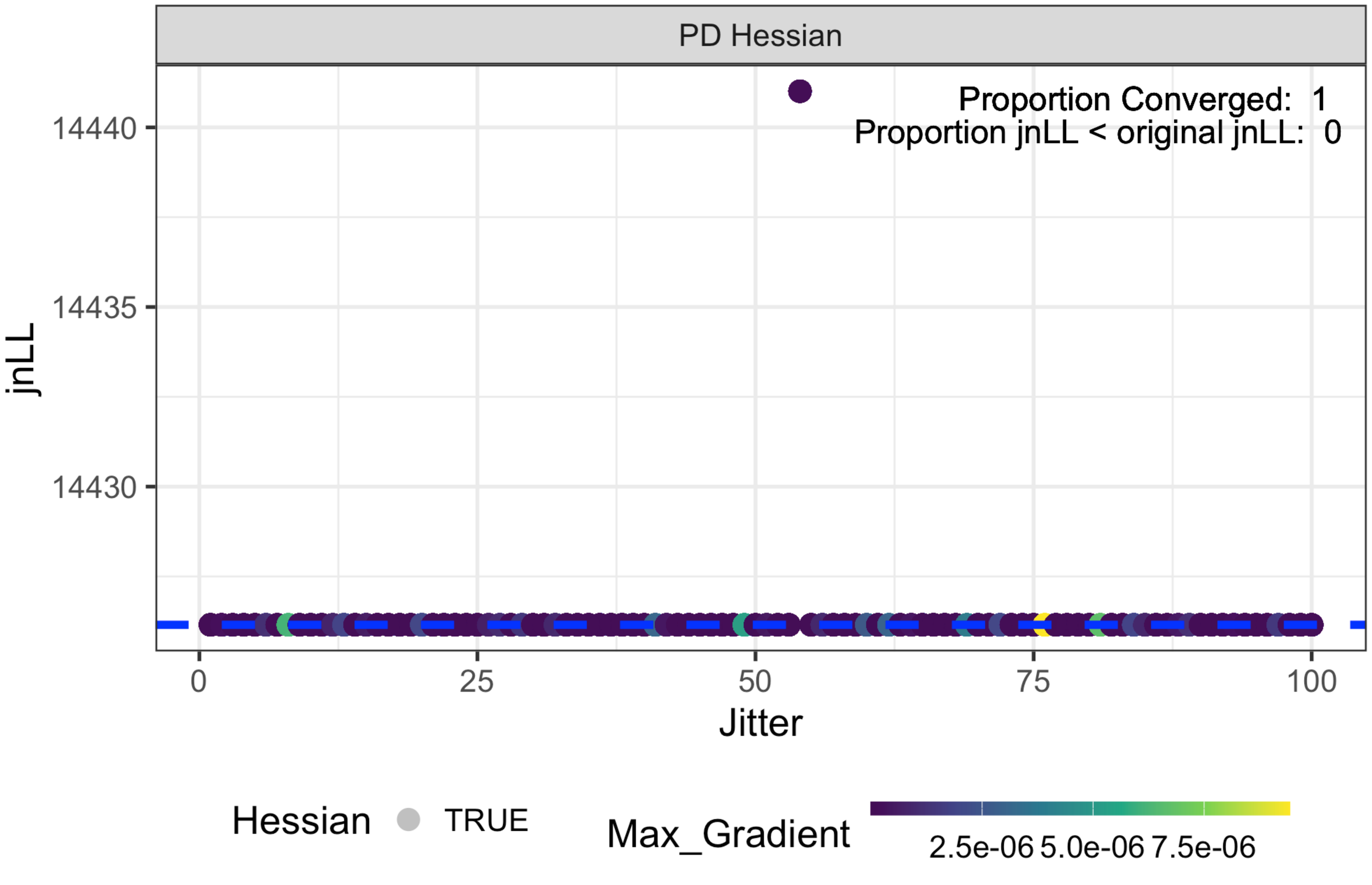
1960 1980 2000 2020

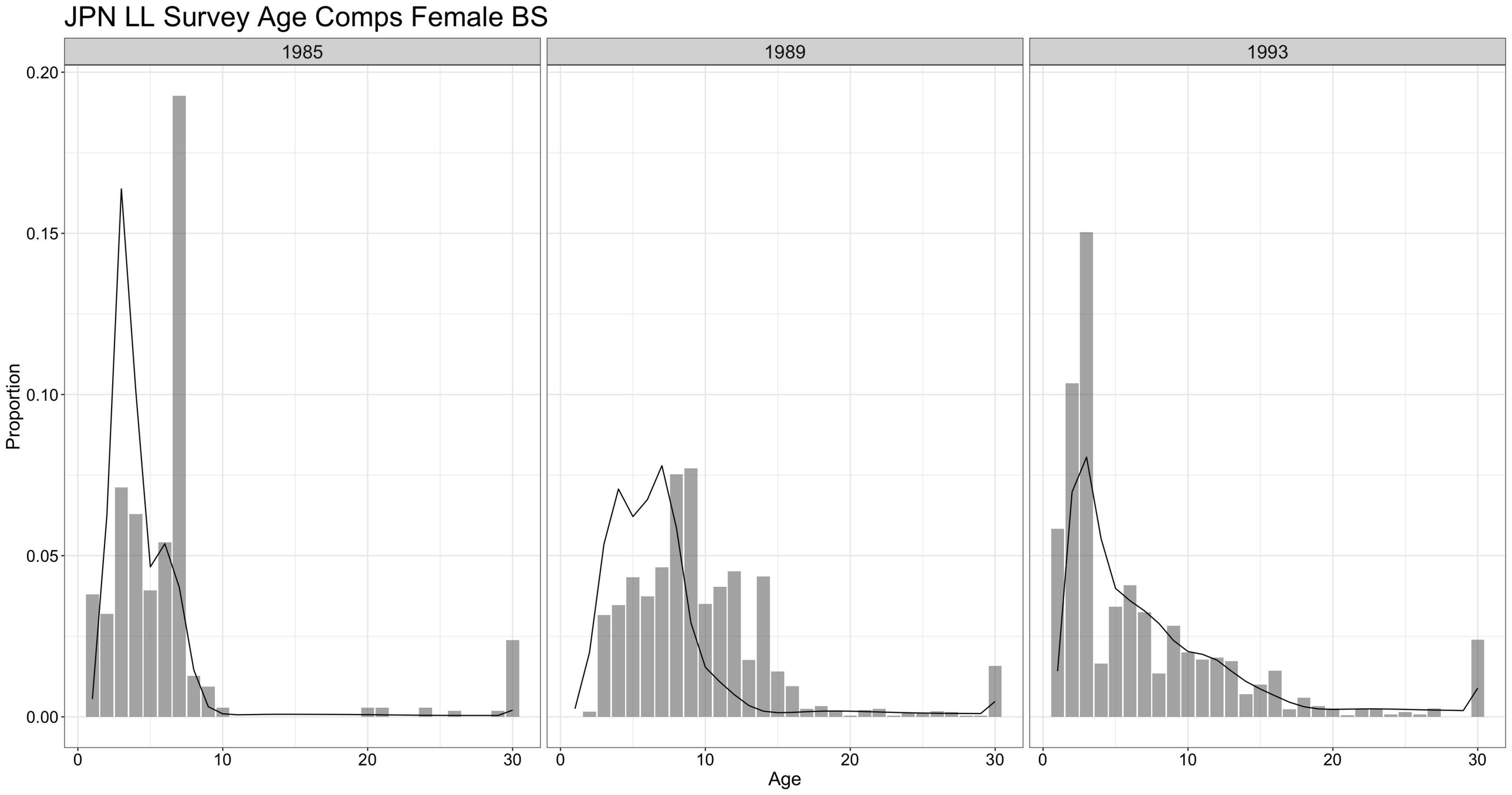
Year

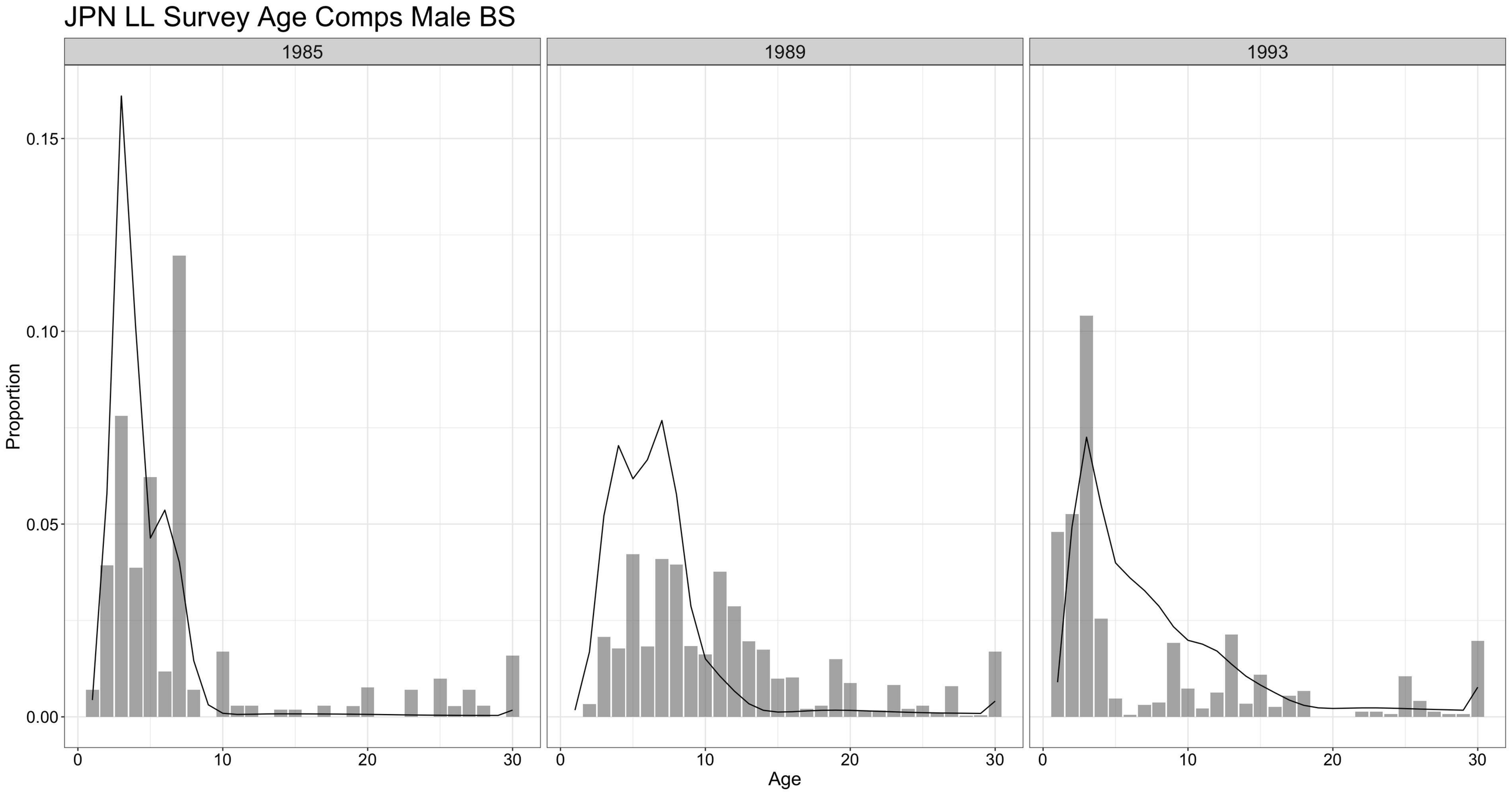
Hessian

TRUE

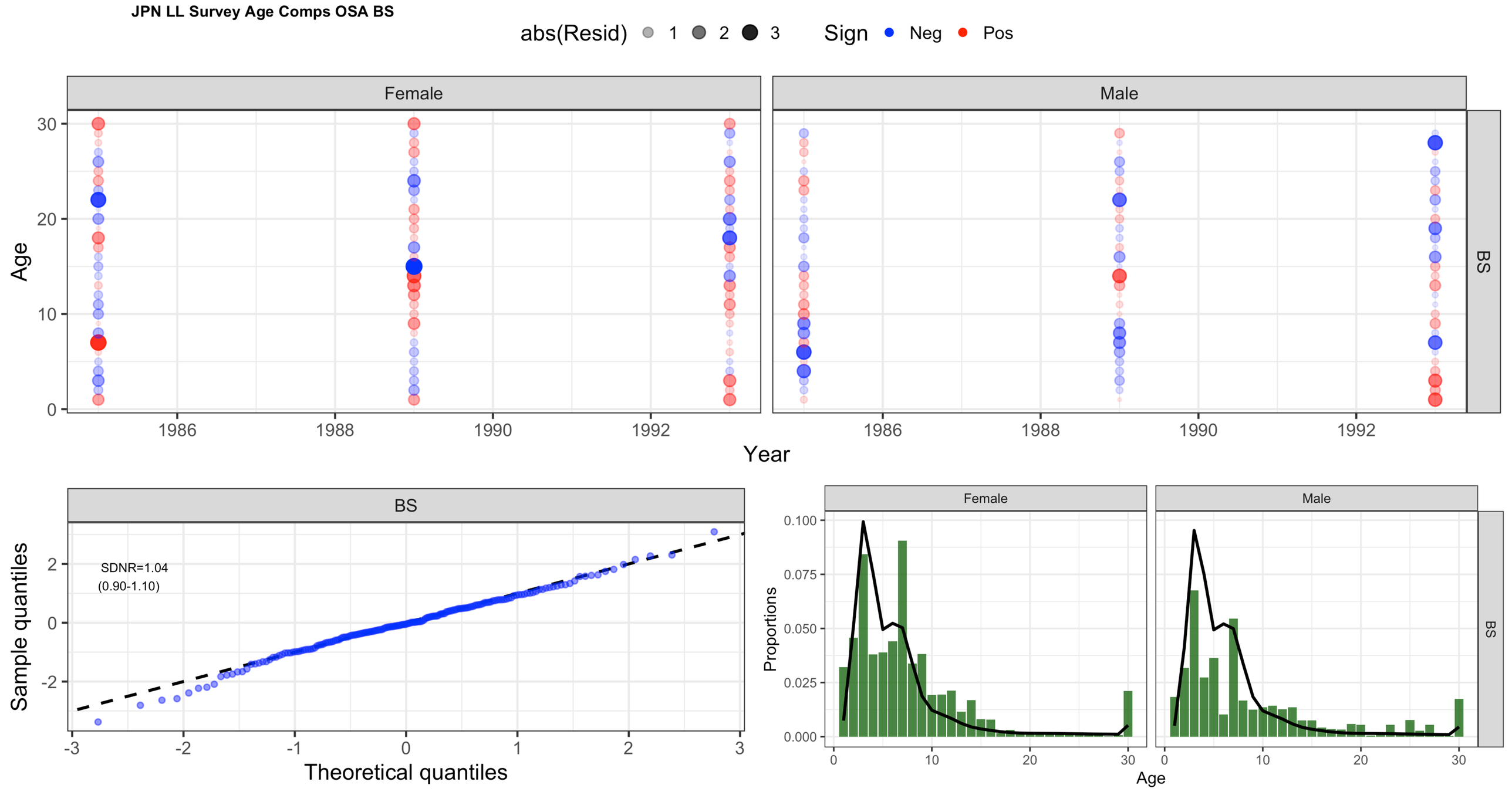
Jitter nLL and Convergence

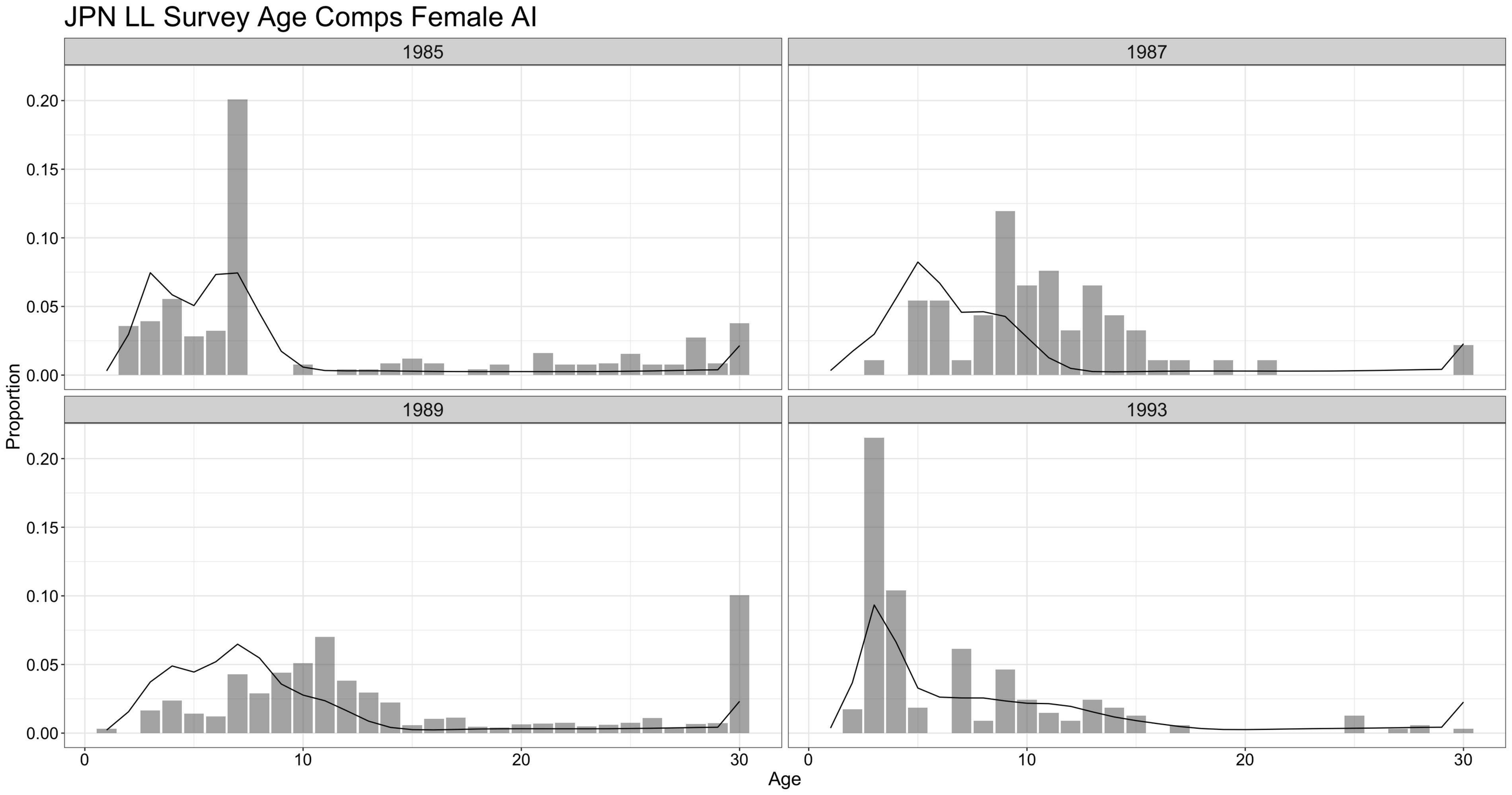


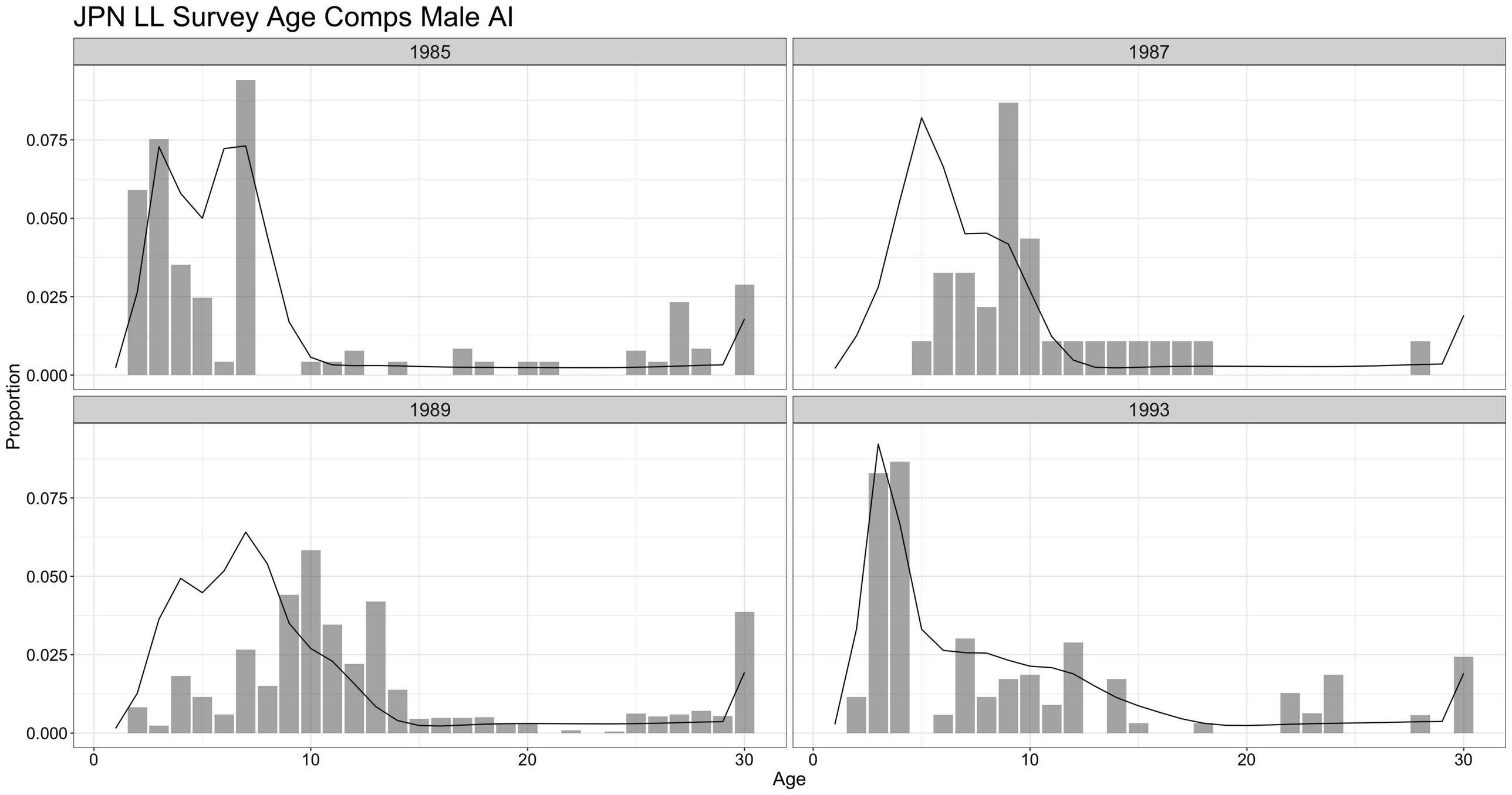

JPN LL Survey Age Comps Female BS
1985
1989
1993
Proportion
0.20
0.15
0.10
0.05
0.00
0
10
20
30
Age

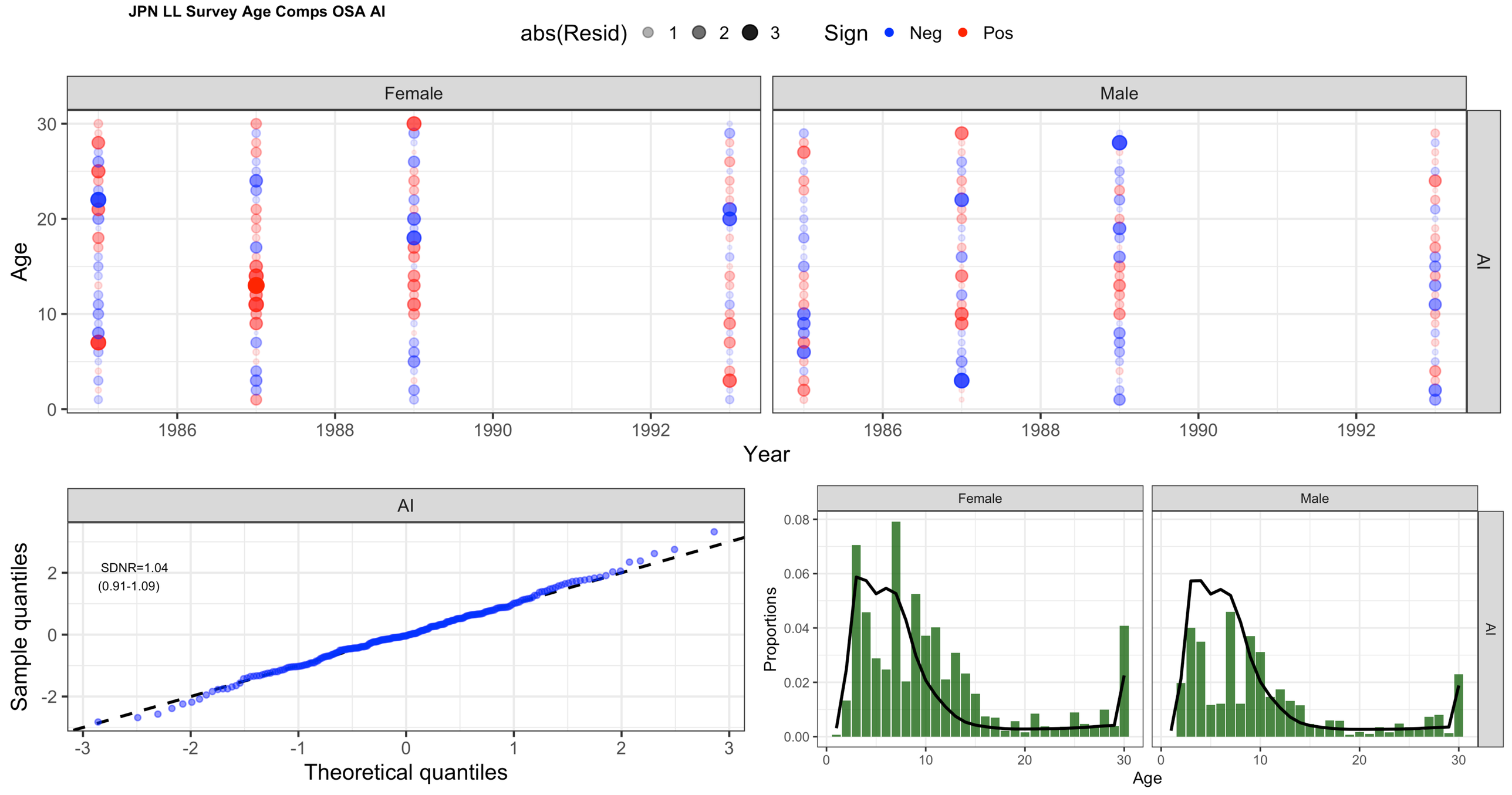

JPN LL Survey Age Comps Male BS
1985
1989
1993
0.15
0.10
0.05
0.00
Proportion
0
10
20
30
Age

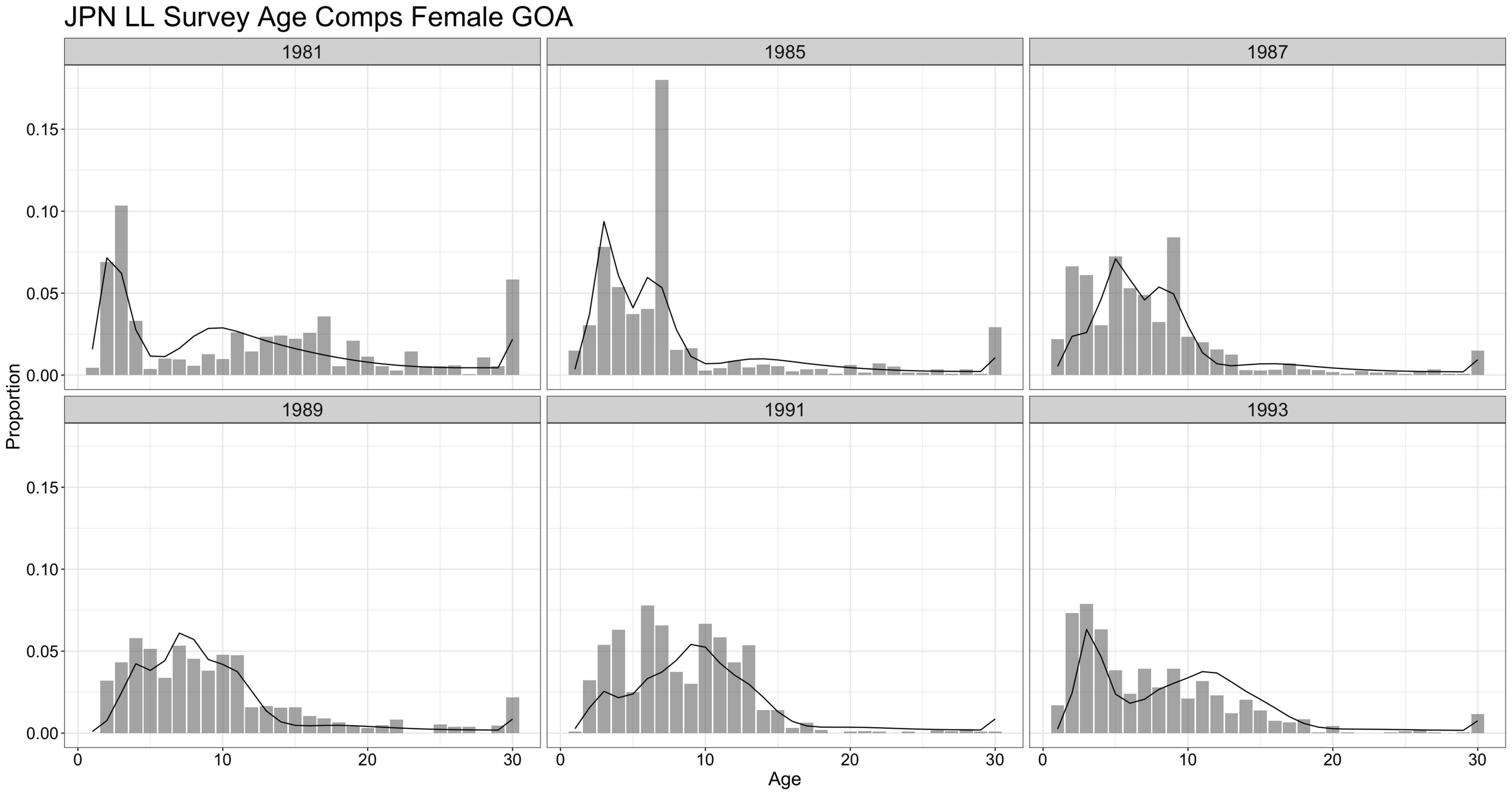
JPN LL Survey Age Comps OSA BS

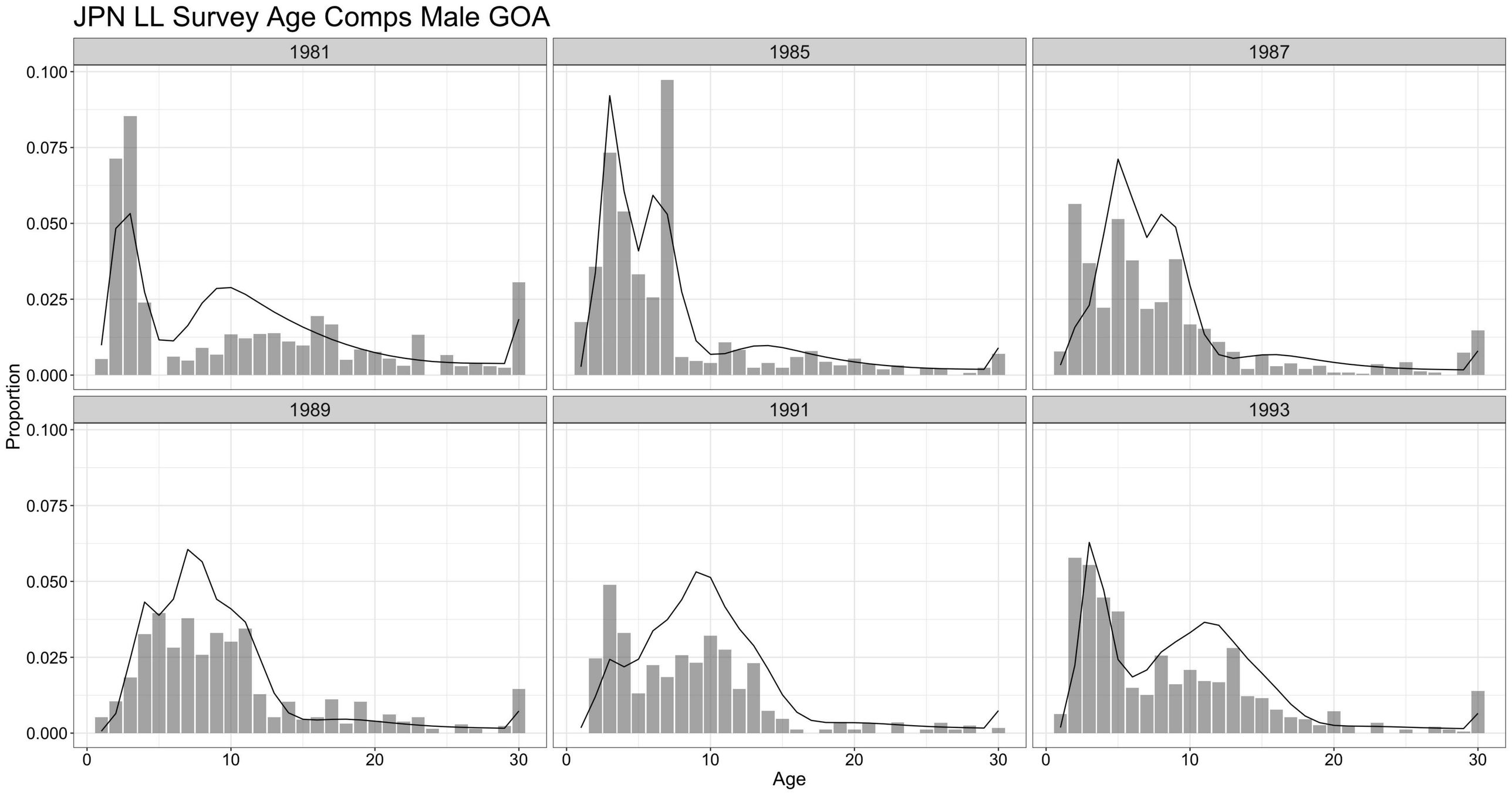



JPN LL Survey Age Comps Female AI
1985
1987
1989
1993
Proportion
0.00
0.05
0.10
0.15
0.20
0
10
20
30
Age

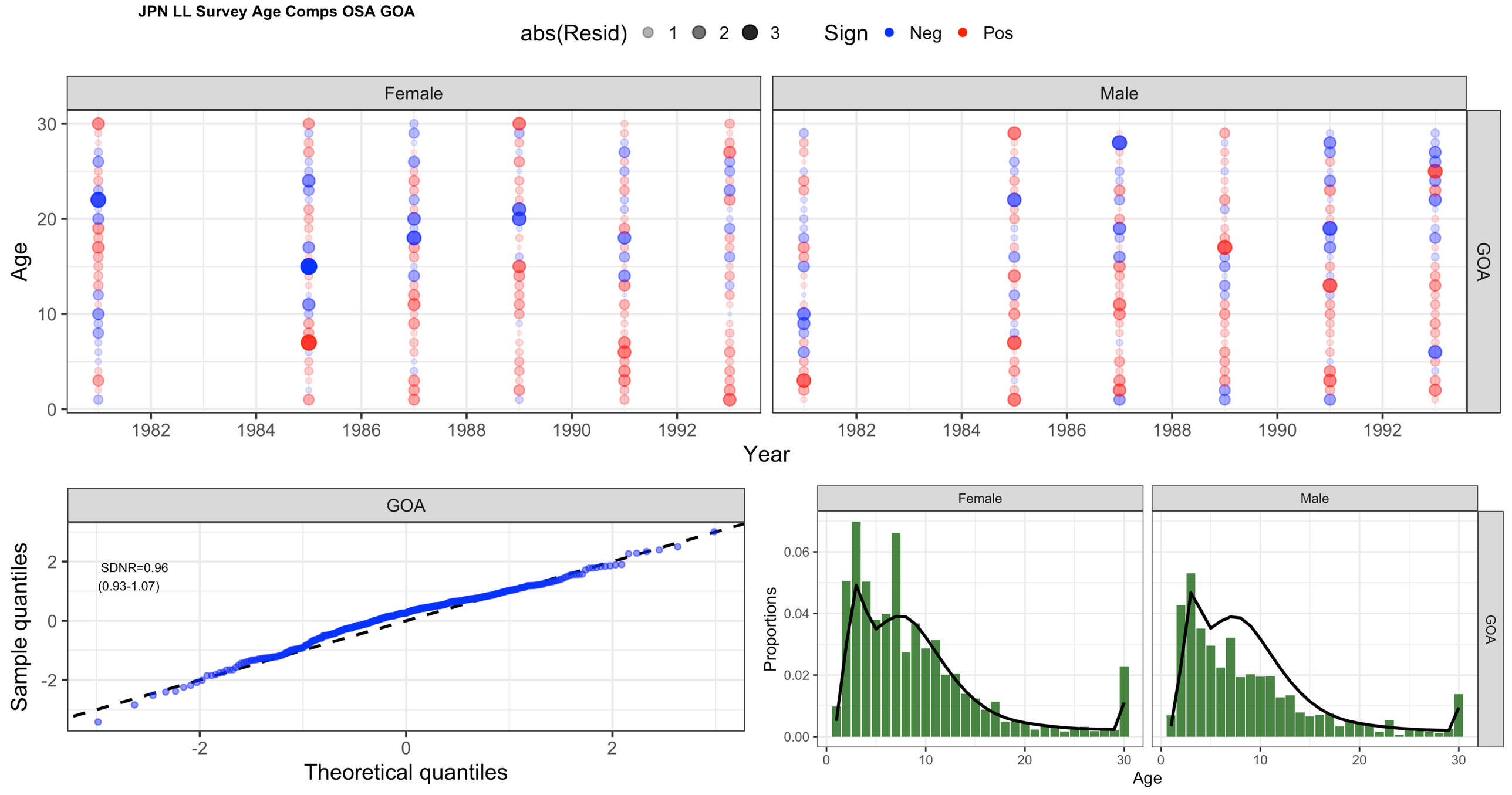

JPN LL Survey Age Comps Male AI
1985
1987
1989
1993
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

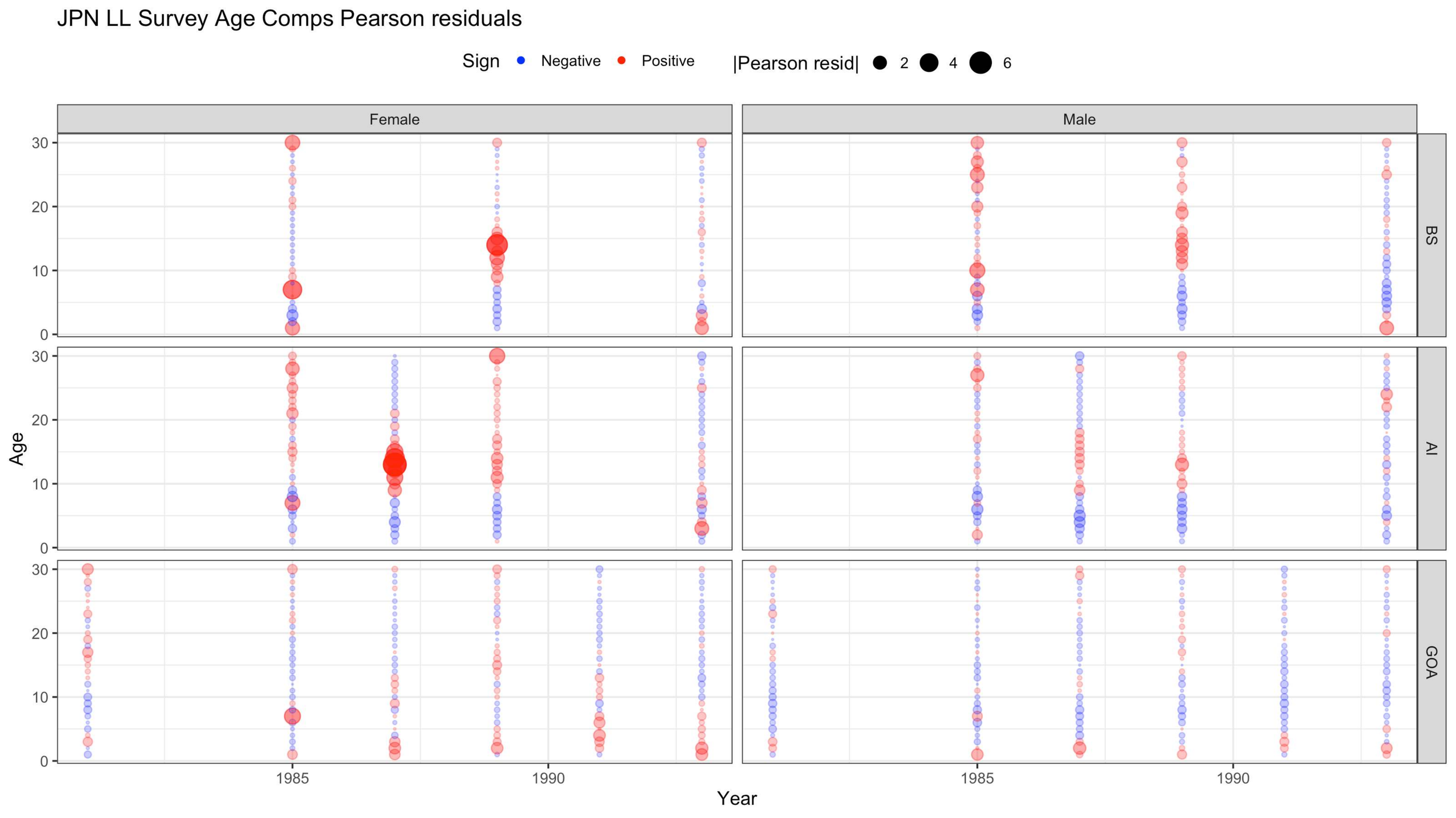

JPN LL Survey Age Comps OSA AI
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AI
Age
Year
1986
1988
1990
1992
0
10
20
30
AI
SDNR=1.04
(0.91-1.09)
Sample quantiles
Theoretical quantiles
-3
-2
-1
0
1
2
3
Female
Male
Proportions
0.00
0.02
0.04
0.06
0.08
Age


JPN LL Survey Age Comps Female GOA
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age

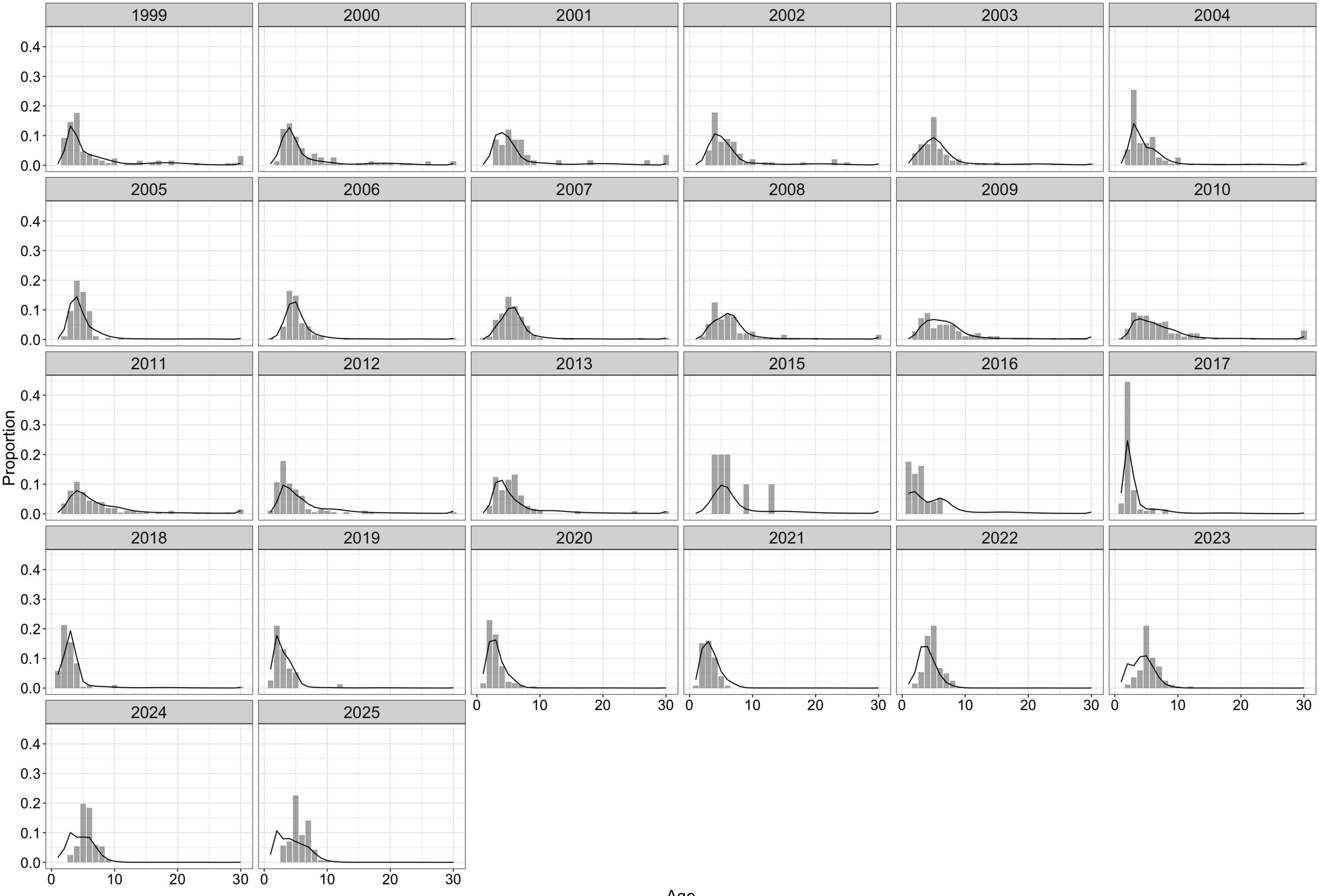

JPN LL Survey Age Comps Male GOA
1981
1985
1987
1989
1991
1993
Proportion
0.100
0.075
0.050
0.025
0.000
0
10
20
30
Age

JPN LL Survey Age Comps OSA GOA

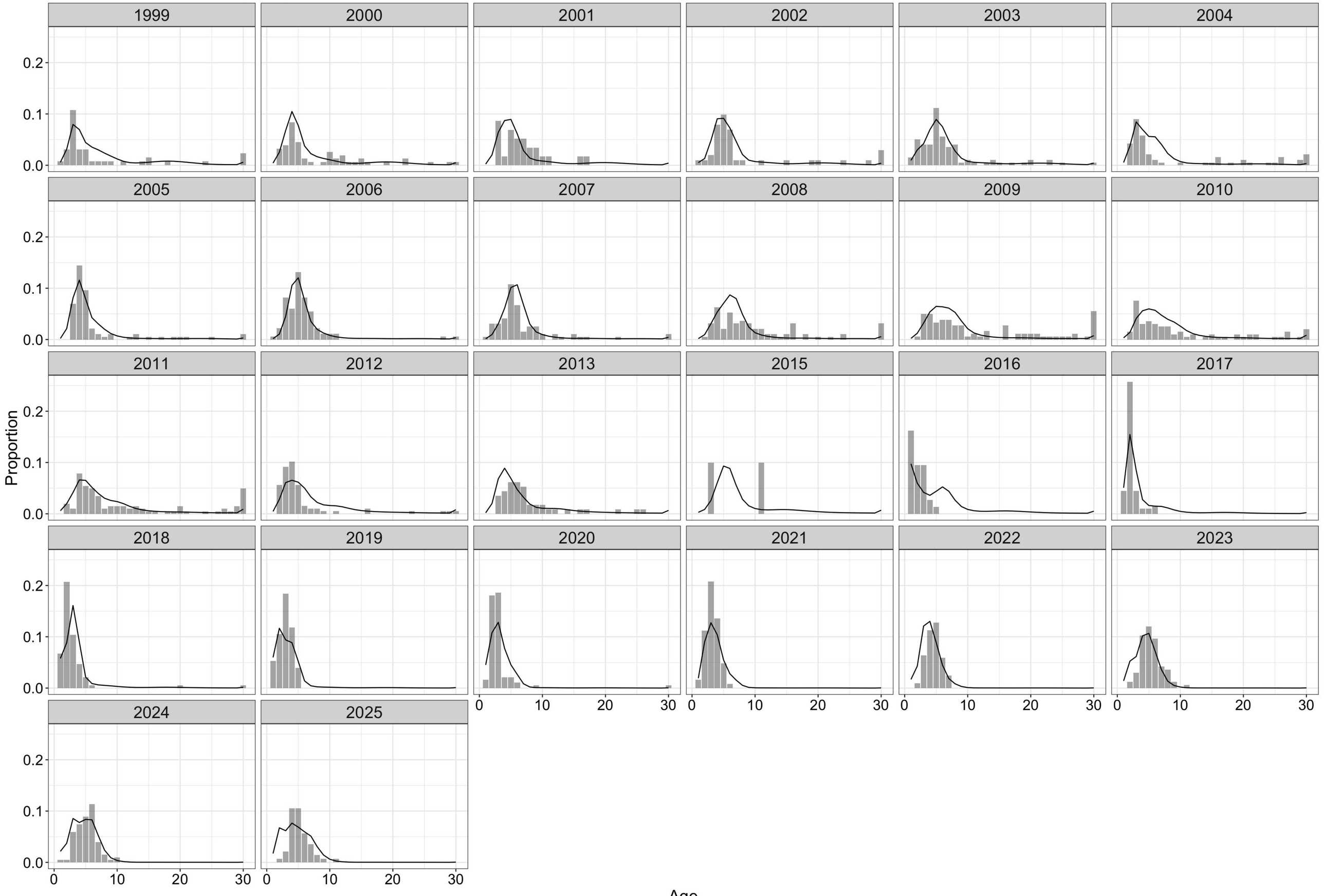



JPN LL Survey Age Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
2
4
6
Female
Male
BS
AI
GOA
Age
Year
0
10
20
30
1985
1990

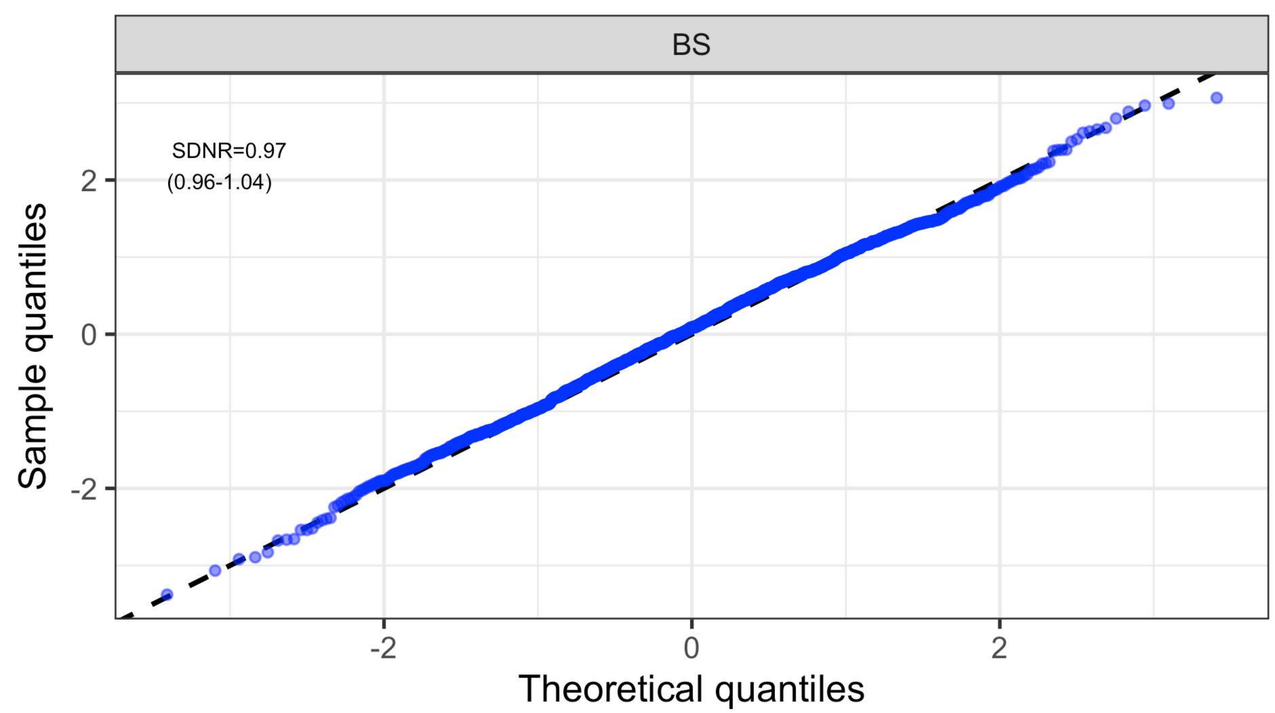
Fixed Gear Fishery Age Comps Female BS

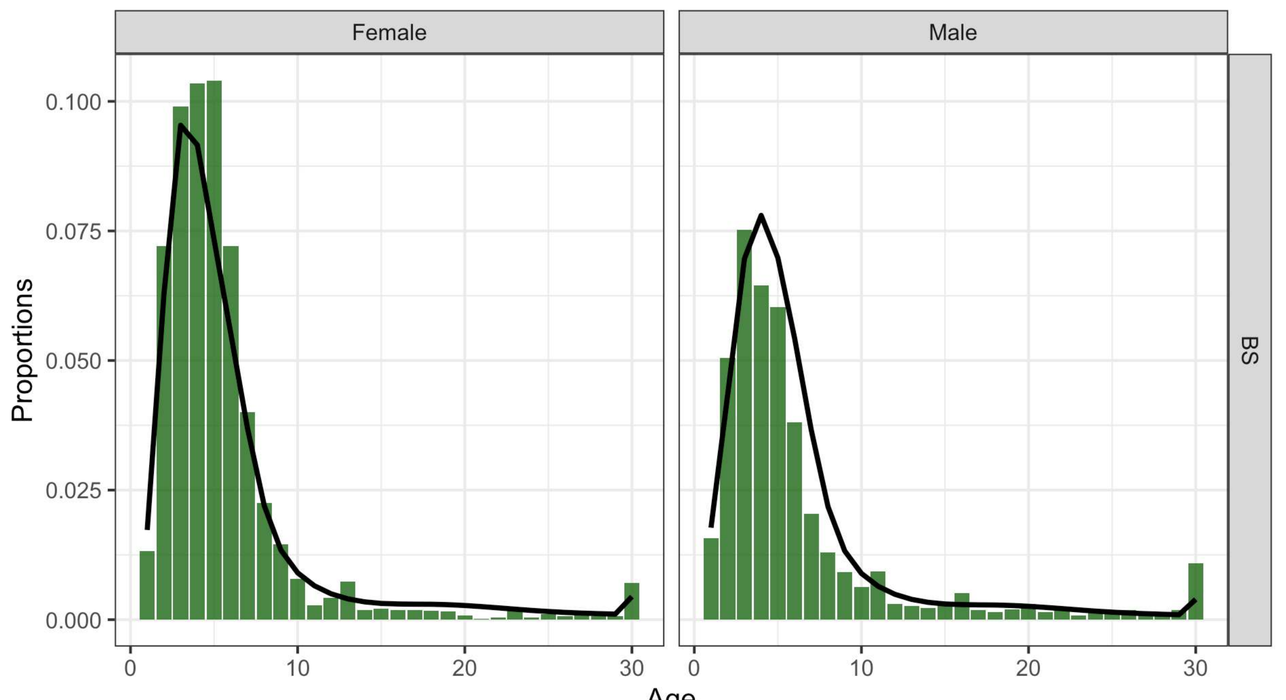


Fixed Gear Fishery Age Comps Male BS

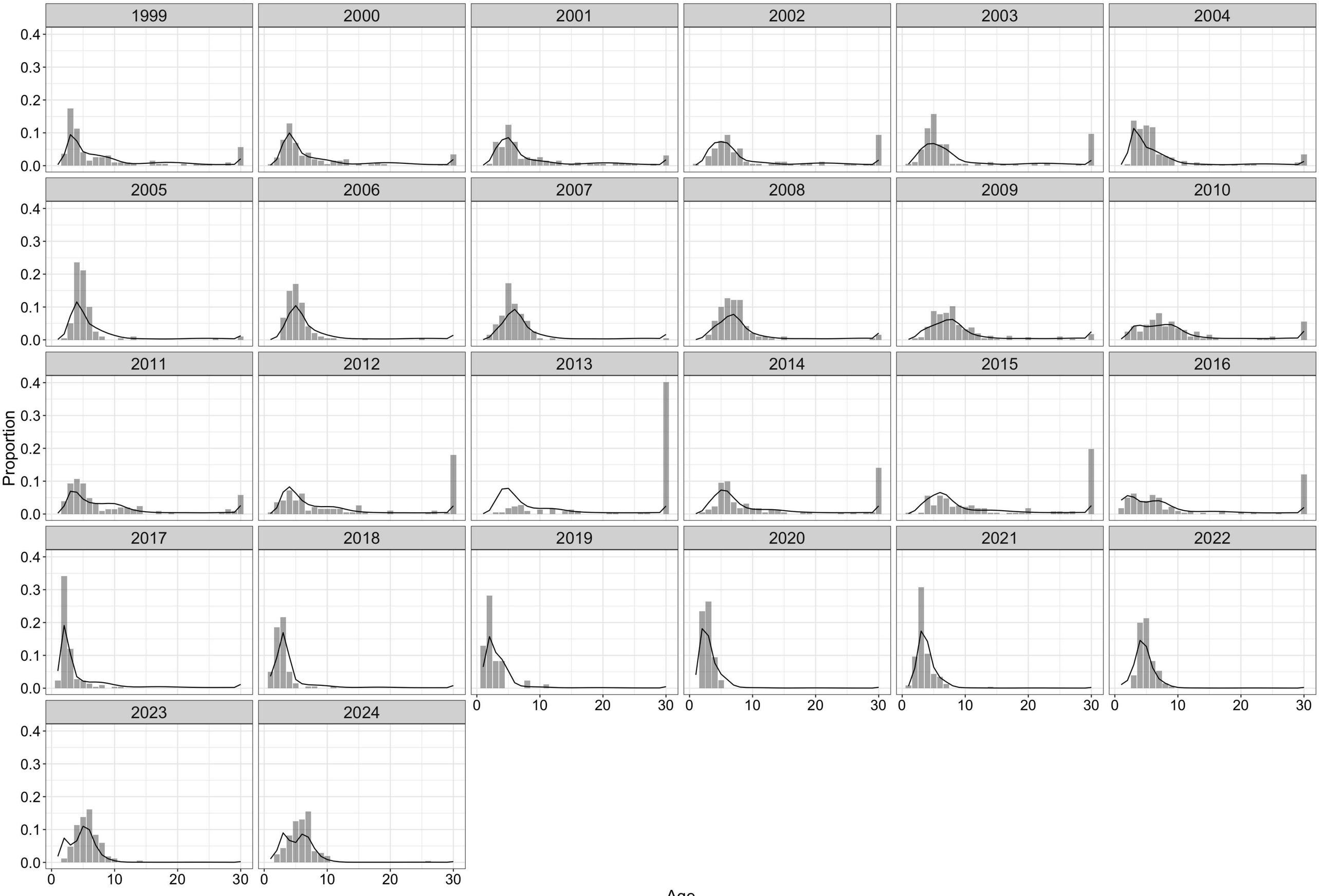


Fixed Gear Fishery Age Comps OSA BS

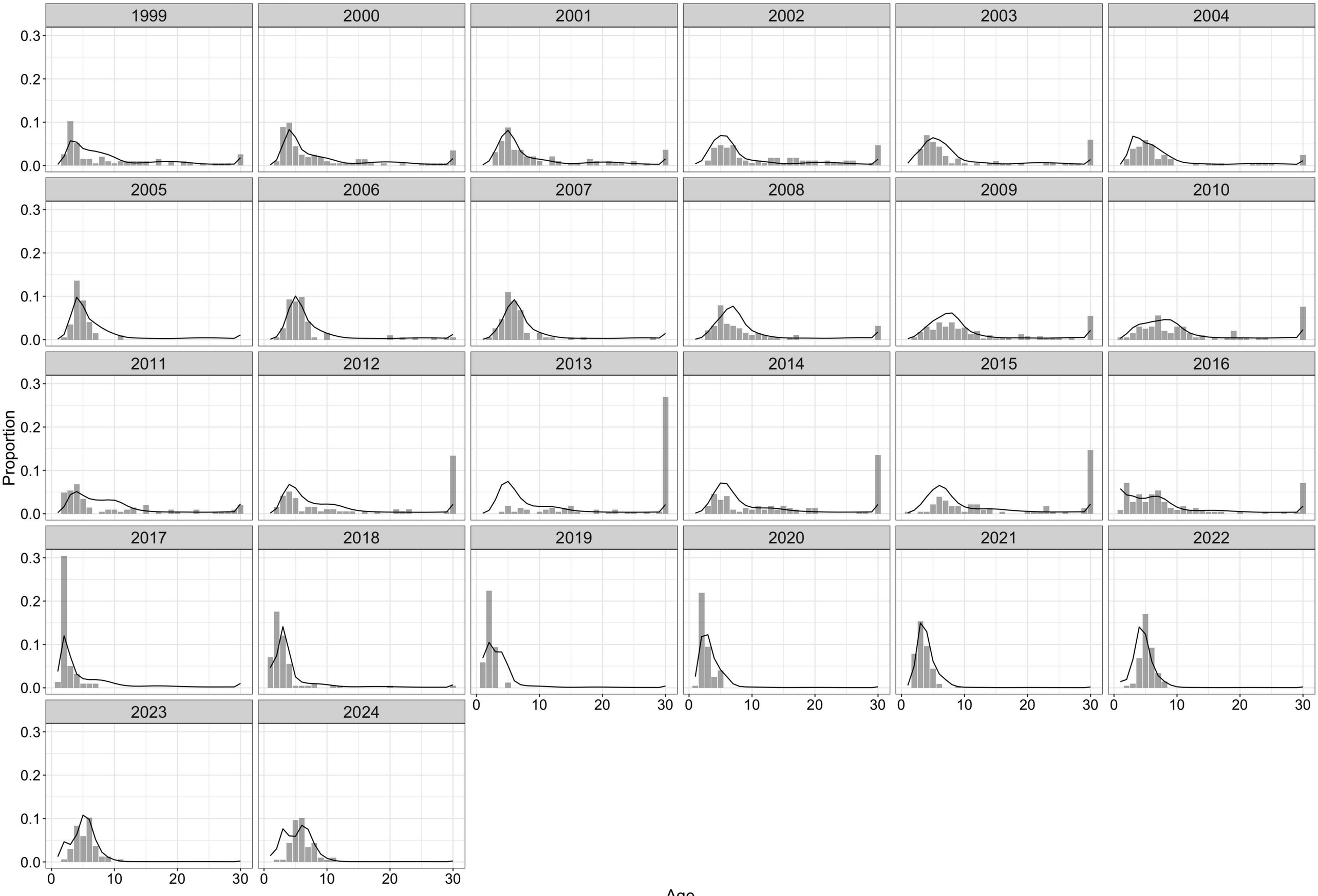




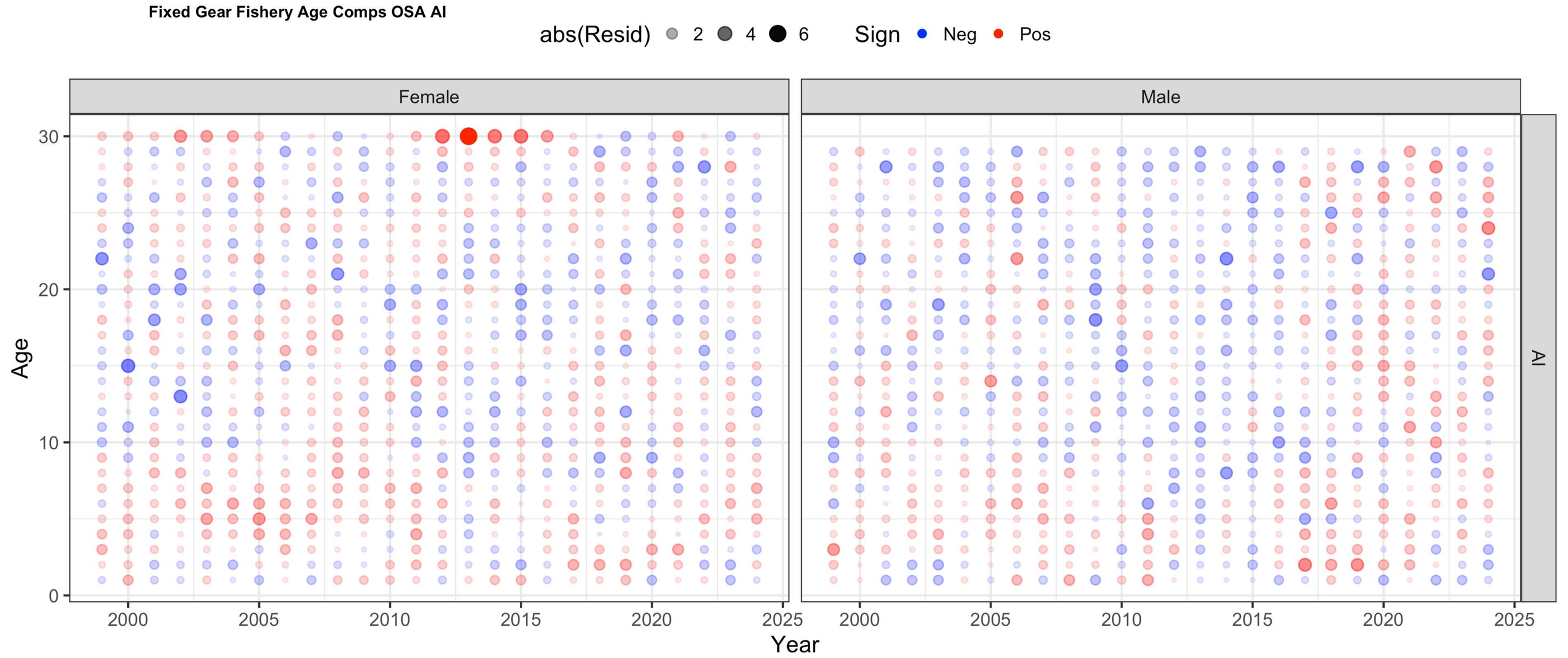
Fixed Gear Fishery Age Comps Female AI

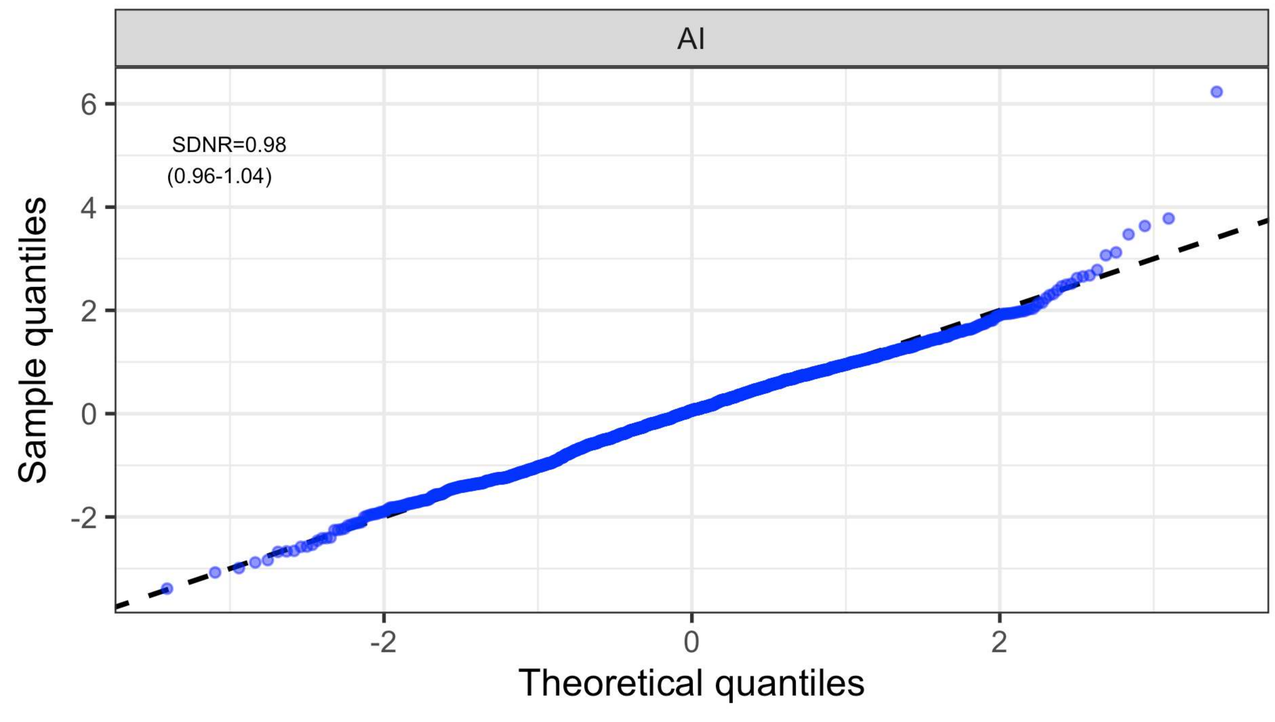


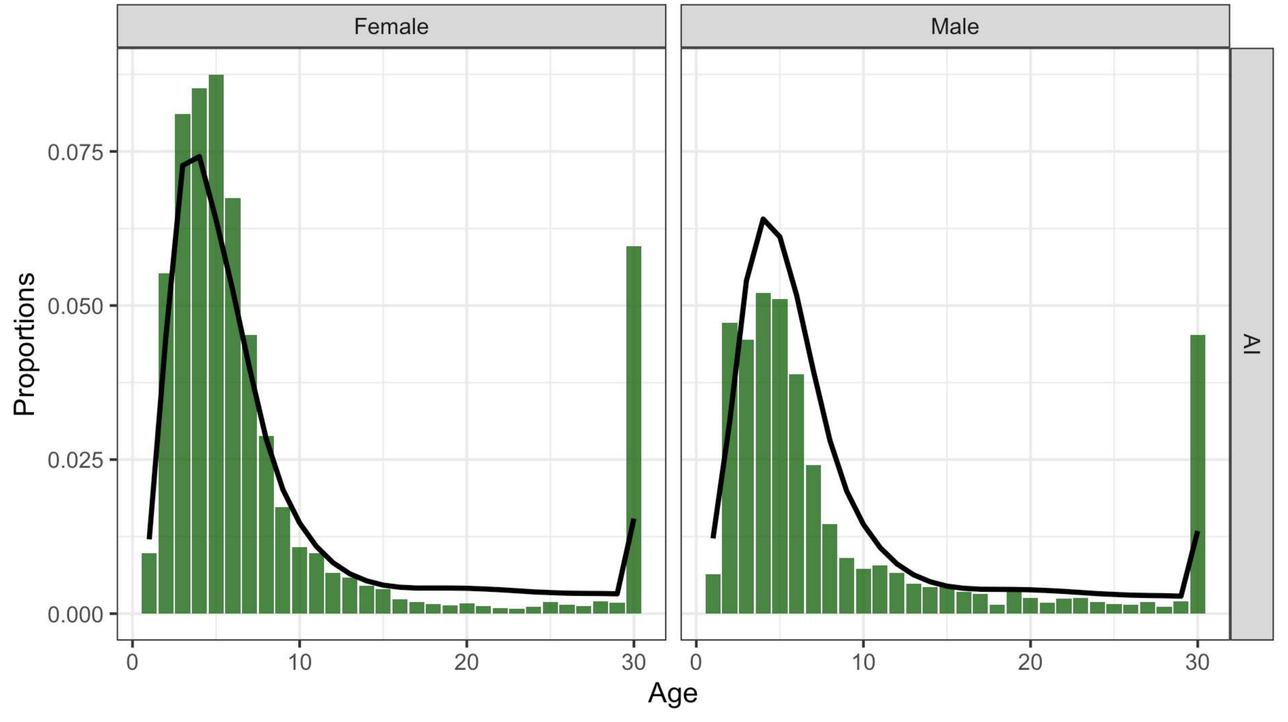
Fixed Gear Fishery Age Comps Male AI



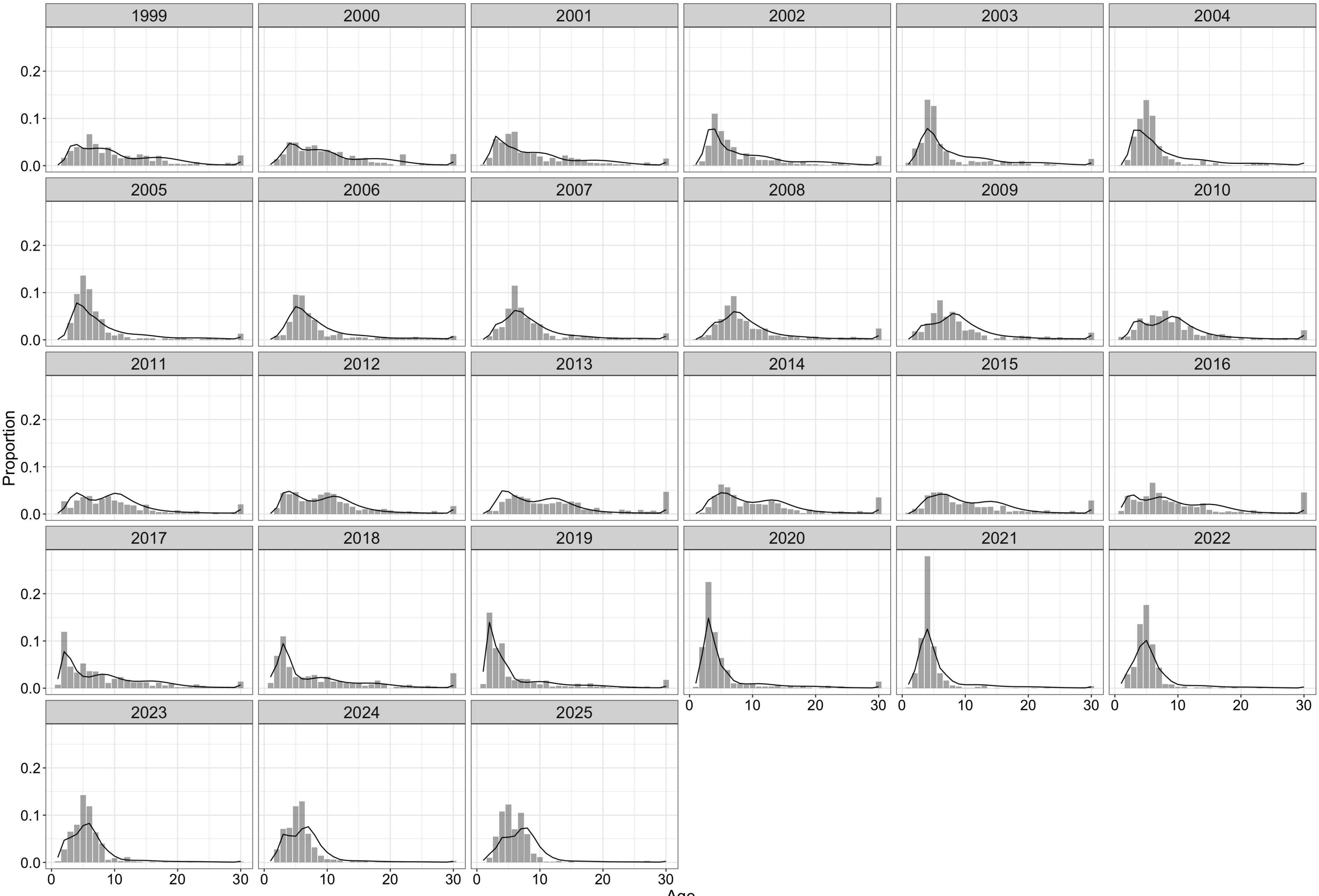
Fixed Gear Fishery Age Comps OSA AI



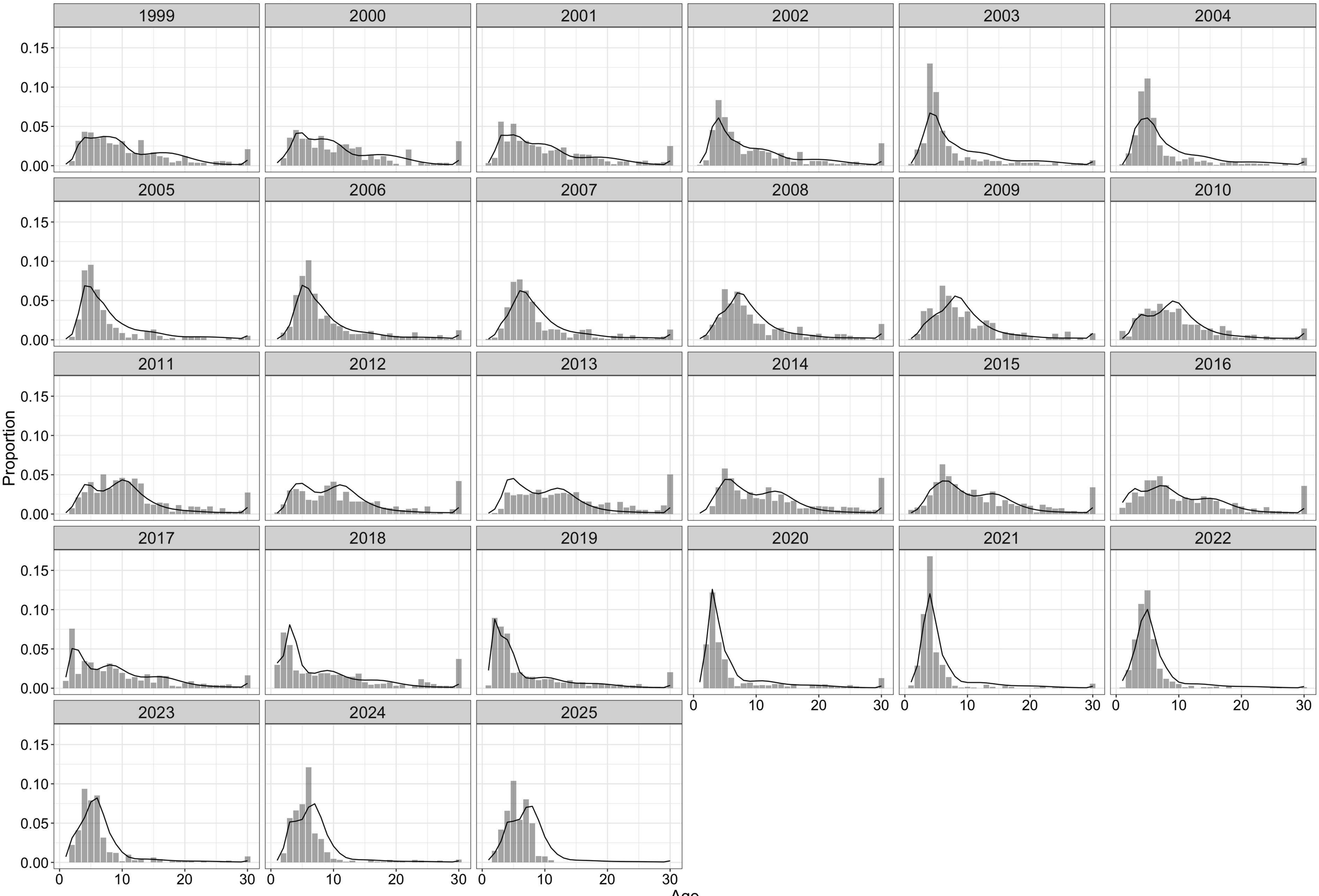




Fixed Gear Fishery Age Comps Female GOA



Fixed Gear Fishery Age Comps Male GOA



Fixed Gear Fishery Age Comps OSA GOA

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

GOA

Age

Year

GOA

SDNR=0.88
(0.97-1.03)

Sample quantiles

Theoretical quantiles

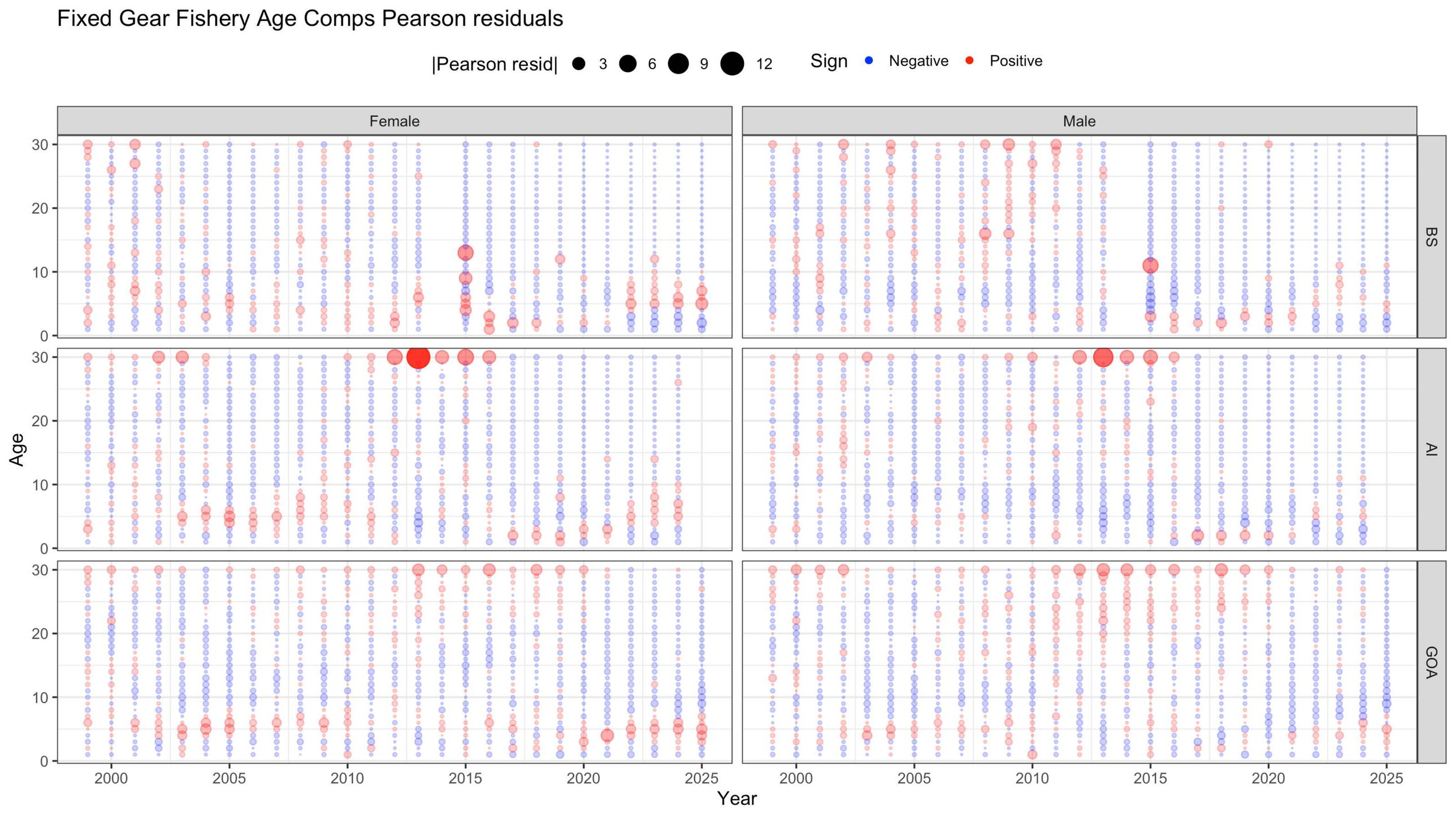
Female

Male

GOA

Proportions

Age


Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
3
6
9
12
Sign
Negative
Positive
Female
Male
BS
AI
GOA
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

Type: Age, Index, Length, Prior, Catch, jnLL, Penalty, Tagging

26.4_Spat_3Reg

Likelihood

15000
10000
5000
0

Catch nLL
Conventional Tagging nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Tag Reporting Prior

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year

Recruitment (R0) profile

Scaled nLL

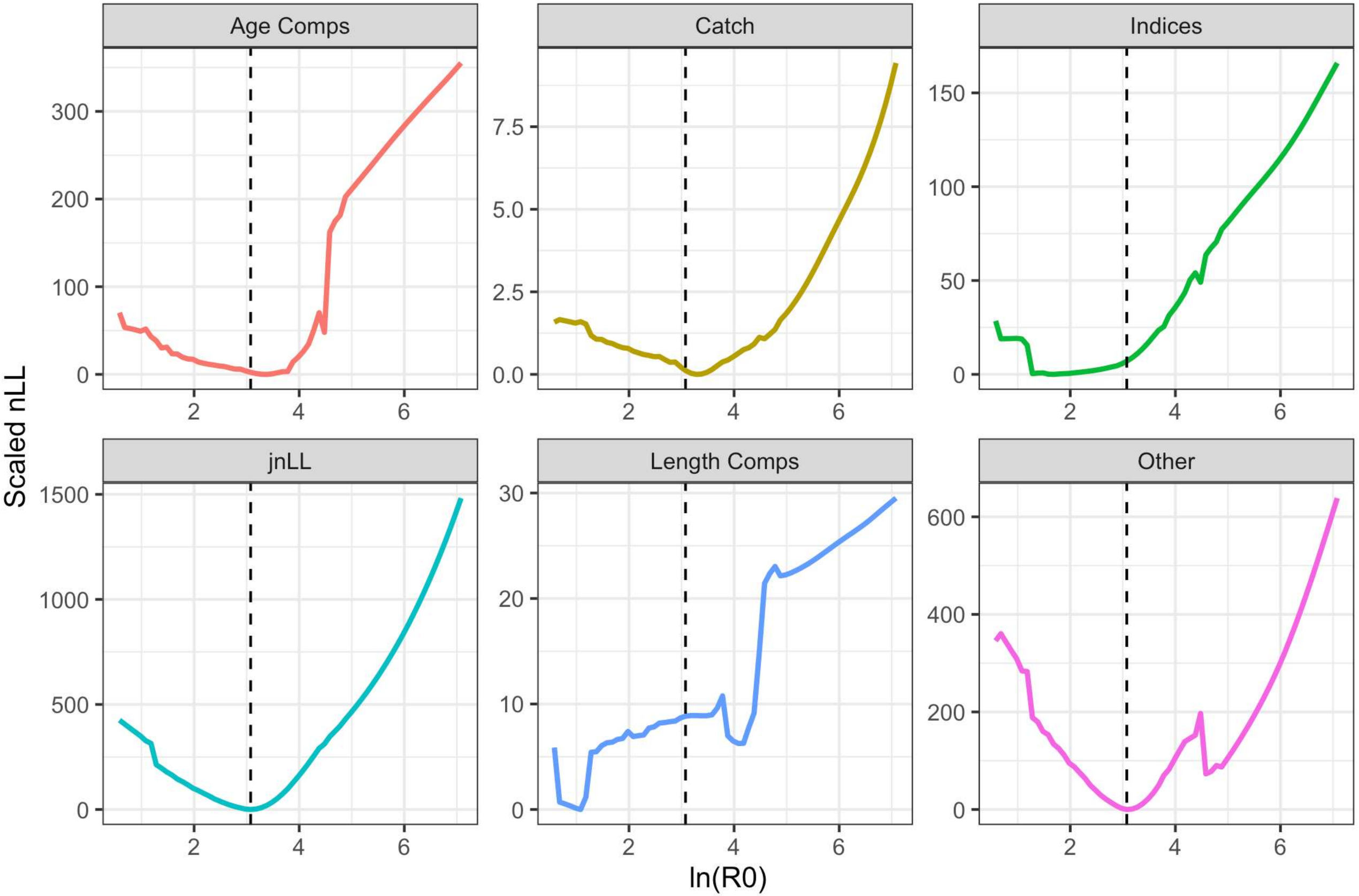
1500
1000
500
0

2
4
6

ln(R0)

Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

Recruitment (R0) profile



Retrospective Year
1 2 3 4 5 6 7

Recruitment
SSB
BS
AI
GOA

Mohns Rho: -0.4898
Mohns Rho: 0.3042
Mohns Rho: -0.5648
Mohns Rho: 0.1123
Mohns Rho: -0.1659
Mohns Rho: 0.1648

Relative Difference from Terminal Year

3
2
1
0
-1
20
15
10
5
0
1.0
0.5
0.0
-0.5

1960
1980
2000
2020

Year

Peel 1 2 3 4 5 6 7

BS Recruitment

BS SSB

AI Recruitment

AI SSB

GOA Recruitment

GOA SSB

Value

Year

BS
AI
GOA
100
50
0
Recruitment (millions)
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment
SSB
BS
AI
GOA
Mohns Rho: -0.239
Mohns Rho: 0.3111
Mohns Rho: -0.6916
Mohns Rho: 0.1364
Mohns Rho: 2.5331
Mohns Rho: 0.1775
Relative Difference from Terminal Year
Year

Peel 1 2 3 4 5 6 7

BS
Recruitment

BS
SSB

AI
Recruitment

AI
SSB

GOA
Recruitment

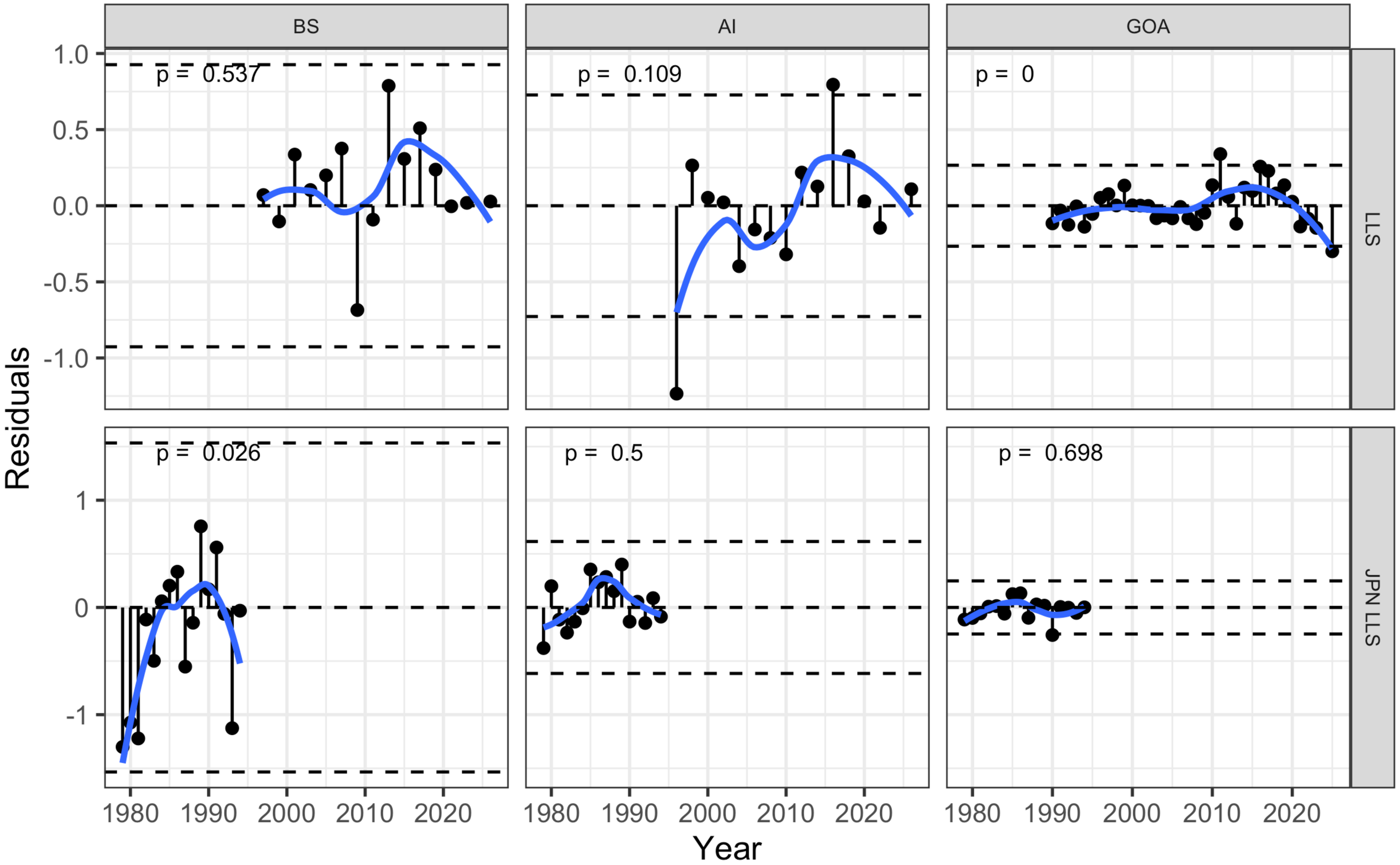
GOA
SSB

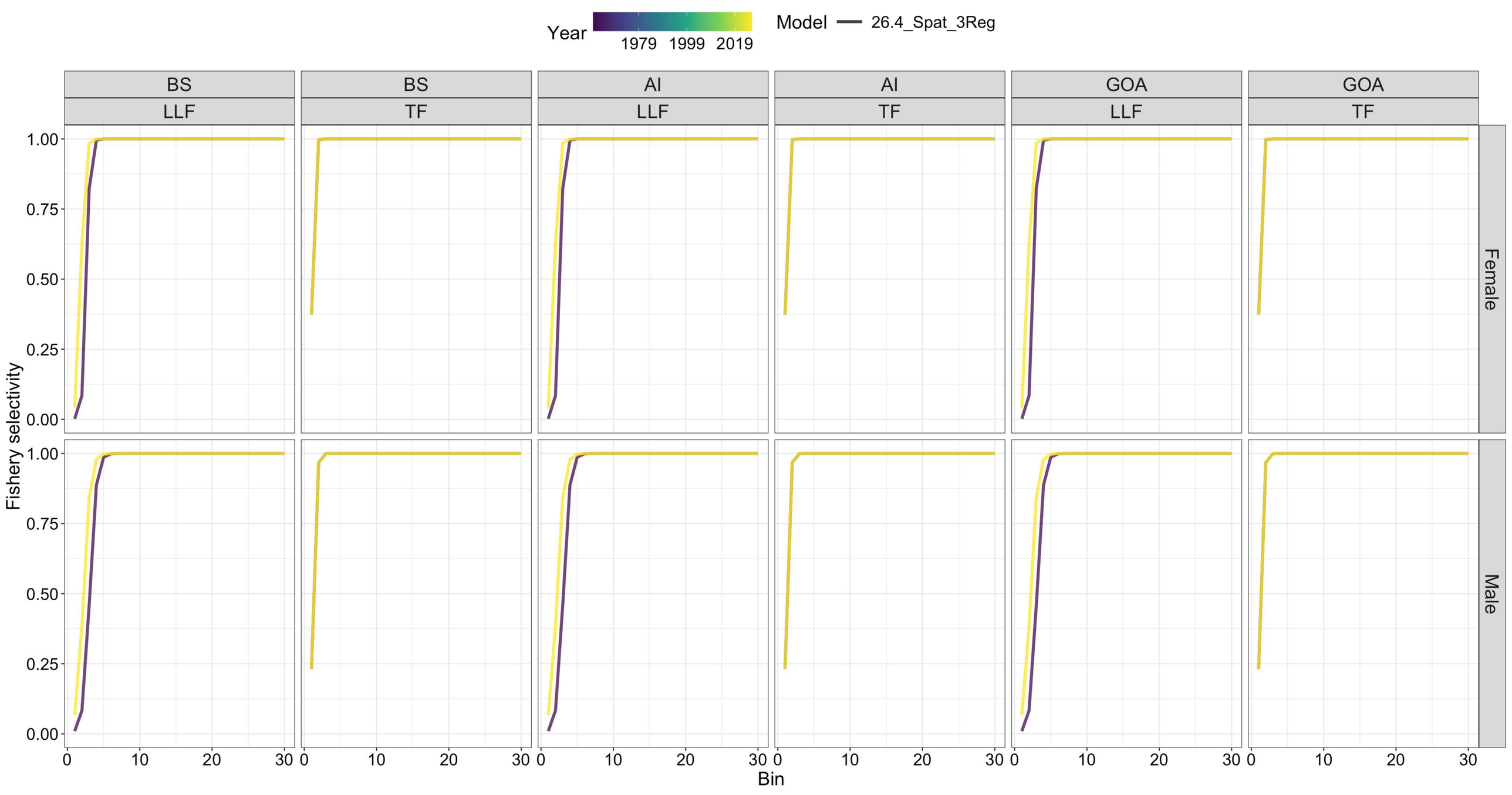
Value

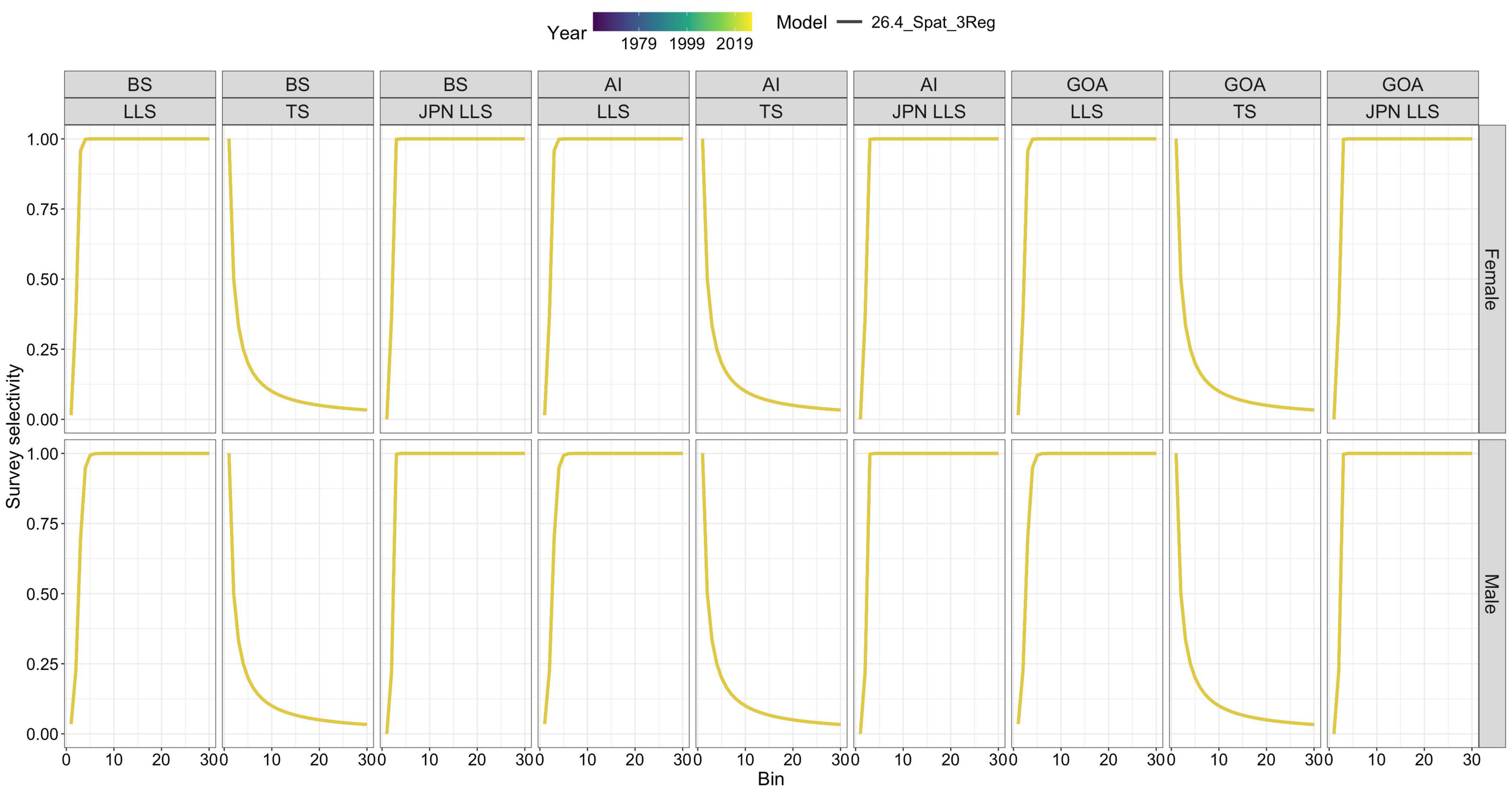
Year

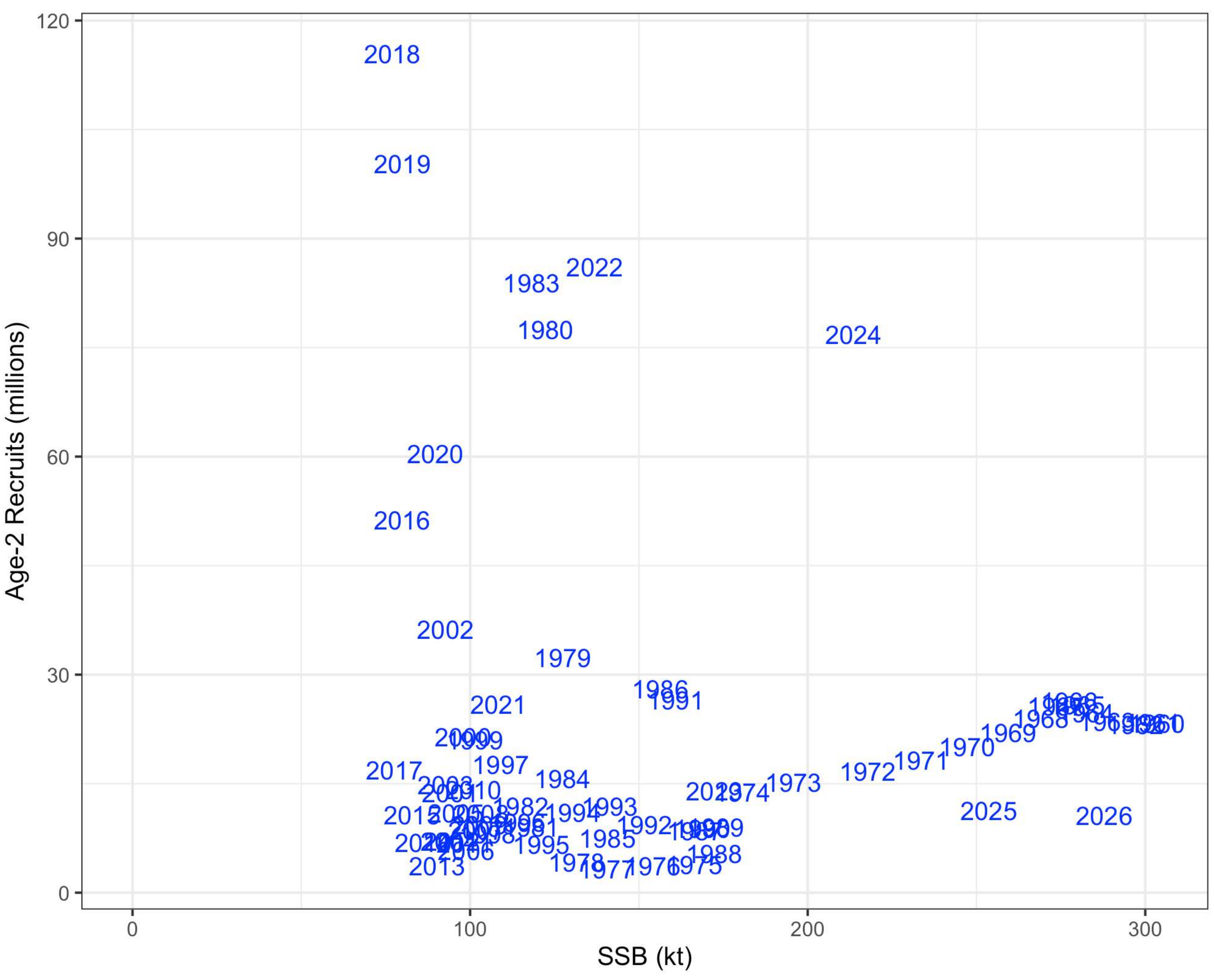
BS
AI
GOA
Recruitment (millions)
100
50
0
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test

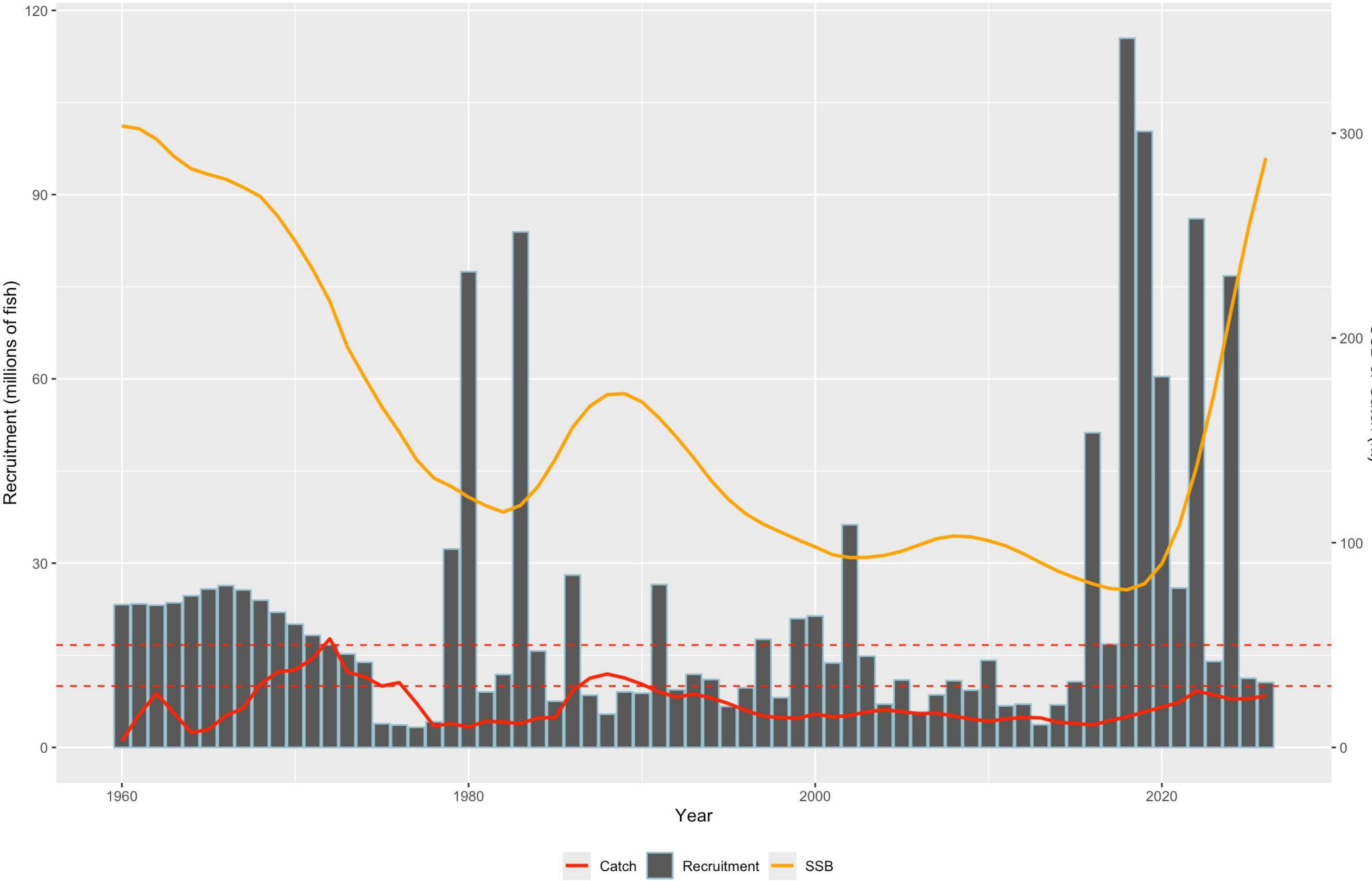


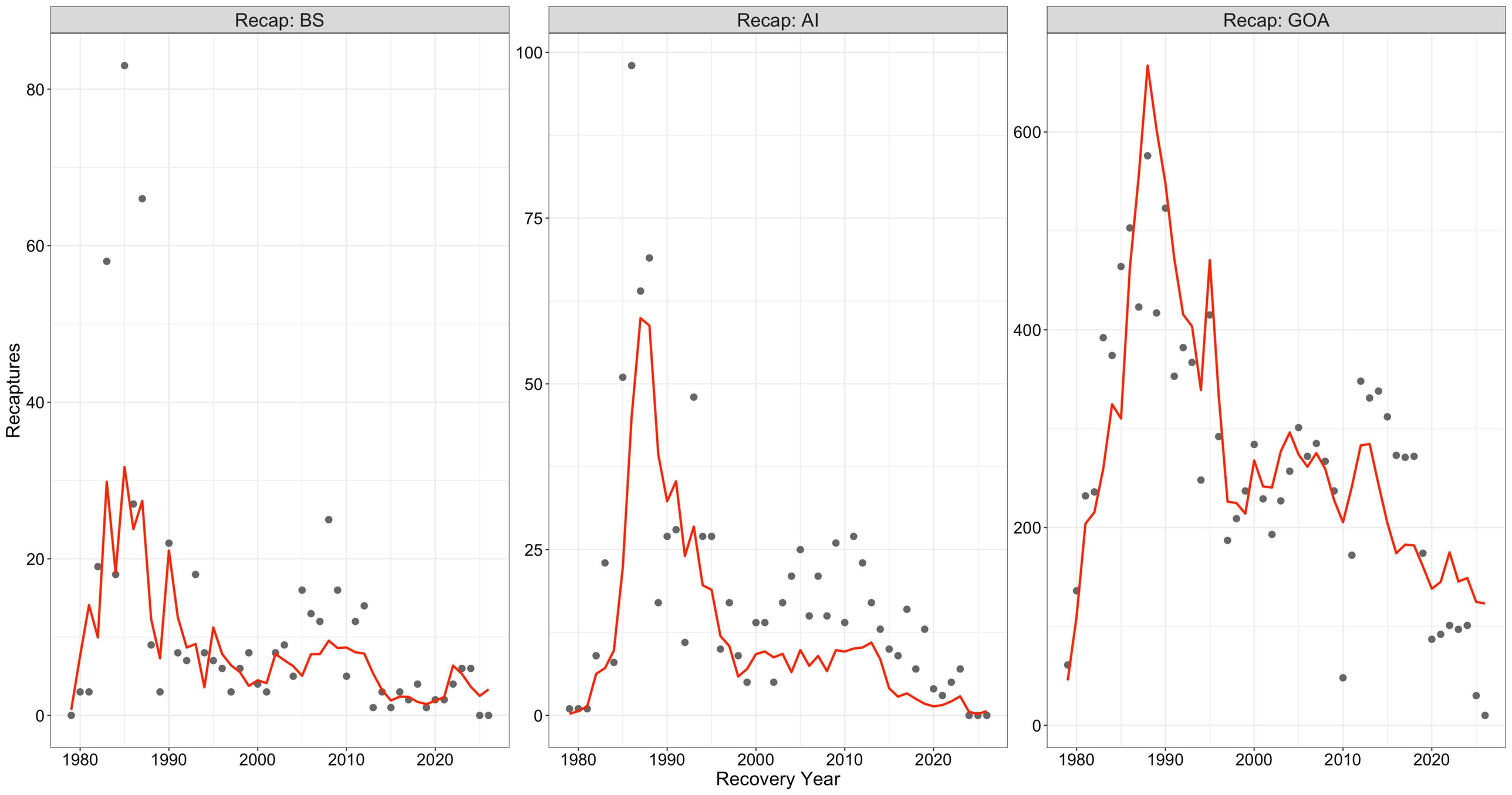

Year
1979
1999
2019
Model
26.4_Spat_3Reg
BS
LLF
BS
TF
AI
LLF
AI
TF
GOA
LLF
GOA
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Year
1979
1999
2019
Model
26.4_Spat_3Reg
BS
LLS
BS
TS
BS
JPN LLS
AI
LLS
AI
TS
AI
JPN LLS
GOA
LLS
GOA
TS
GOA
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
SSB (kt)
120
90
60
30
0
0
100
200
300
2018
2019
2016
2002
1979
1986
1991
2021
2000
1999
1997
2017
1984
1983
2022
1980
2024
1973
1972
1971
1970
1969
2025
2026
1995
1985
1988
2013

Population and Catch Trends




Recap: BS
Recap: AI
Recap: GOA
Recaptures
Recovery Year
0
20
40
60
80
0
25
50
75
100
0
200
400
600
1980
1990
2000
2010
2020

Model 26.4_Spat_3Reg

BS

AI

GOA

Dynamic SSB0

Recruitment

SSB

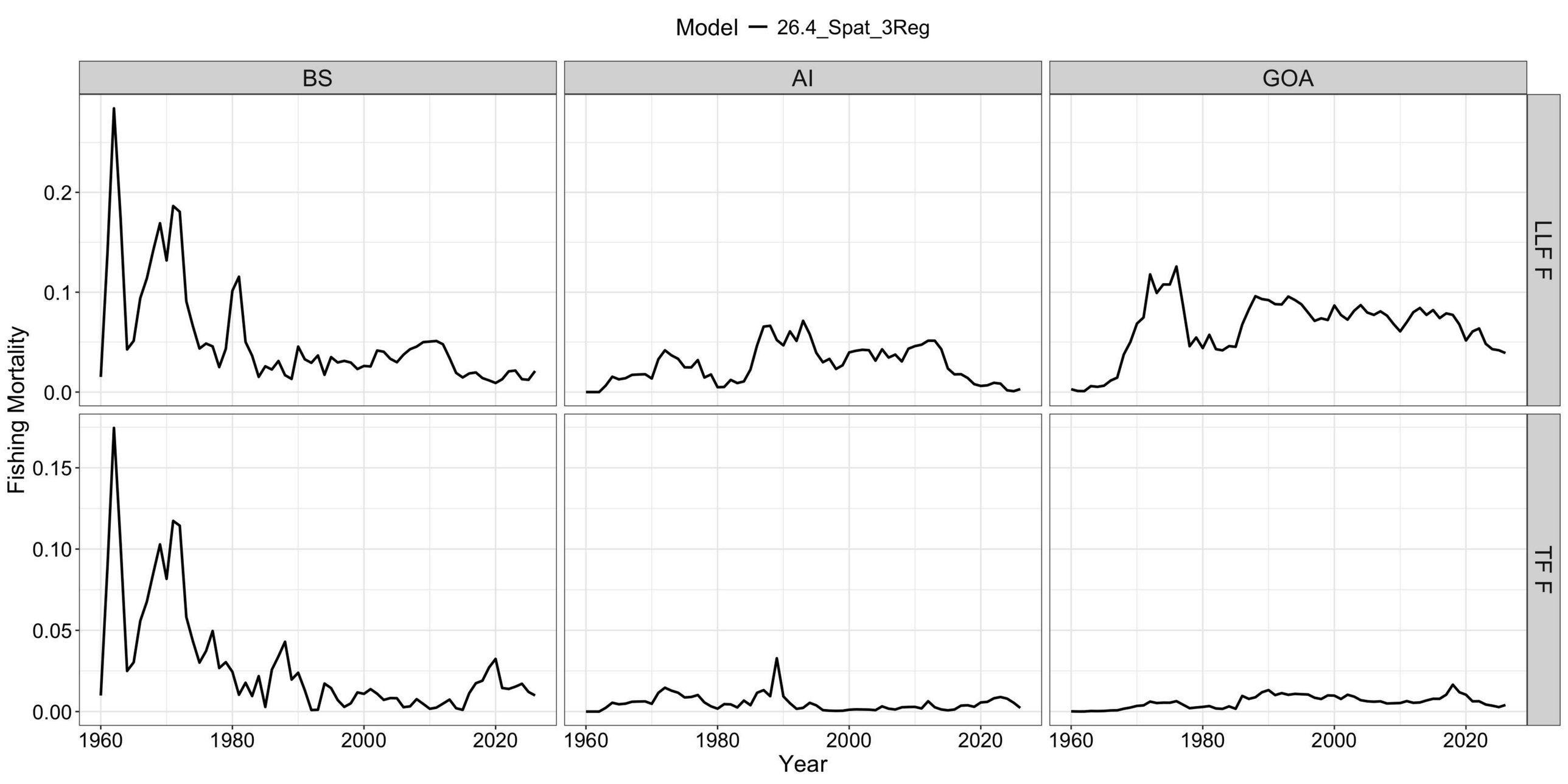
Total Biom

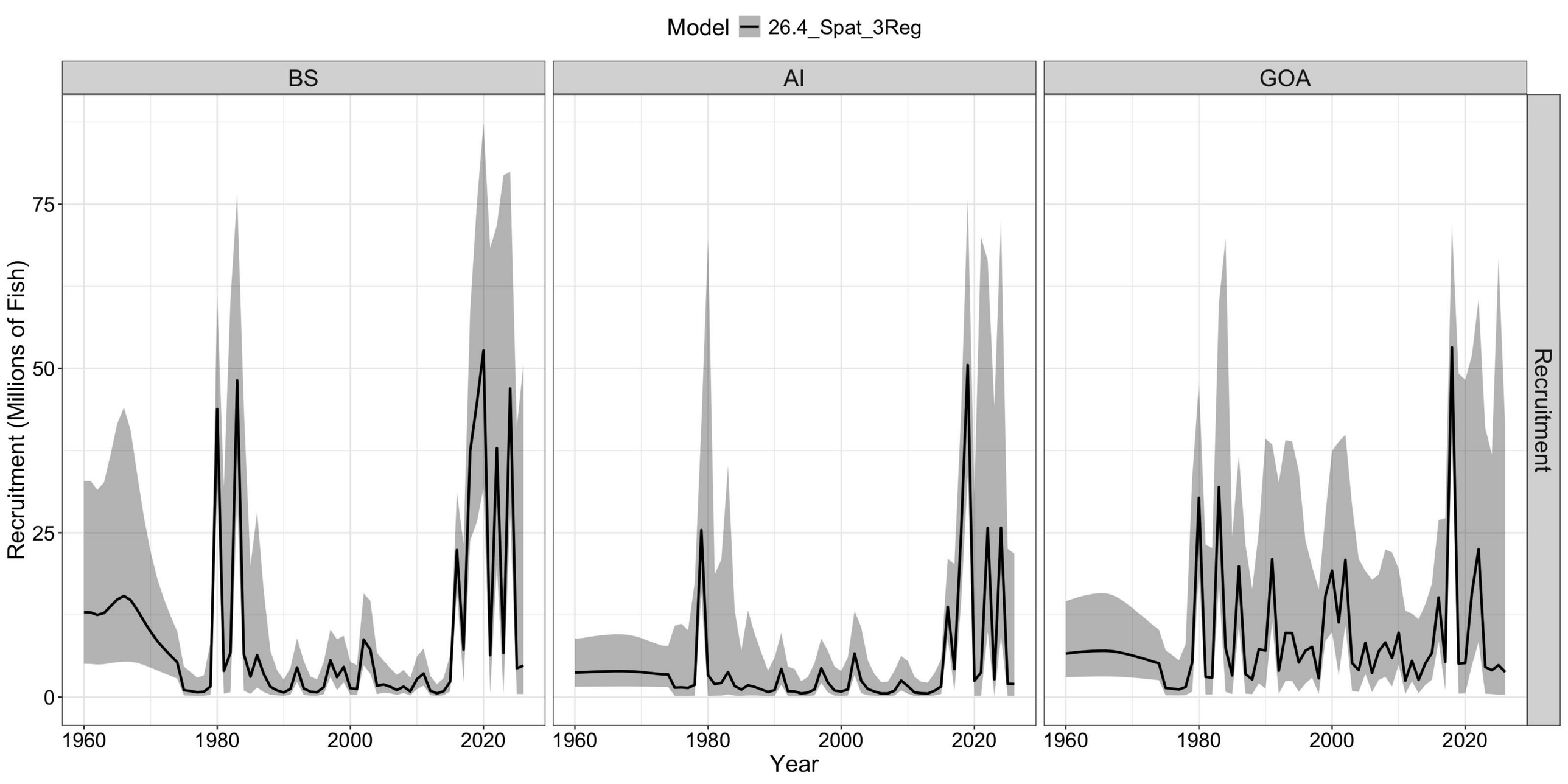
LLF F

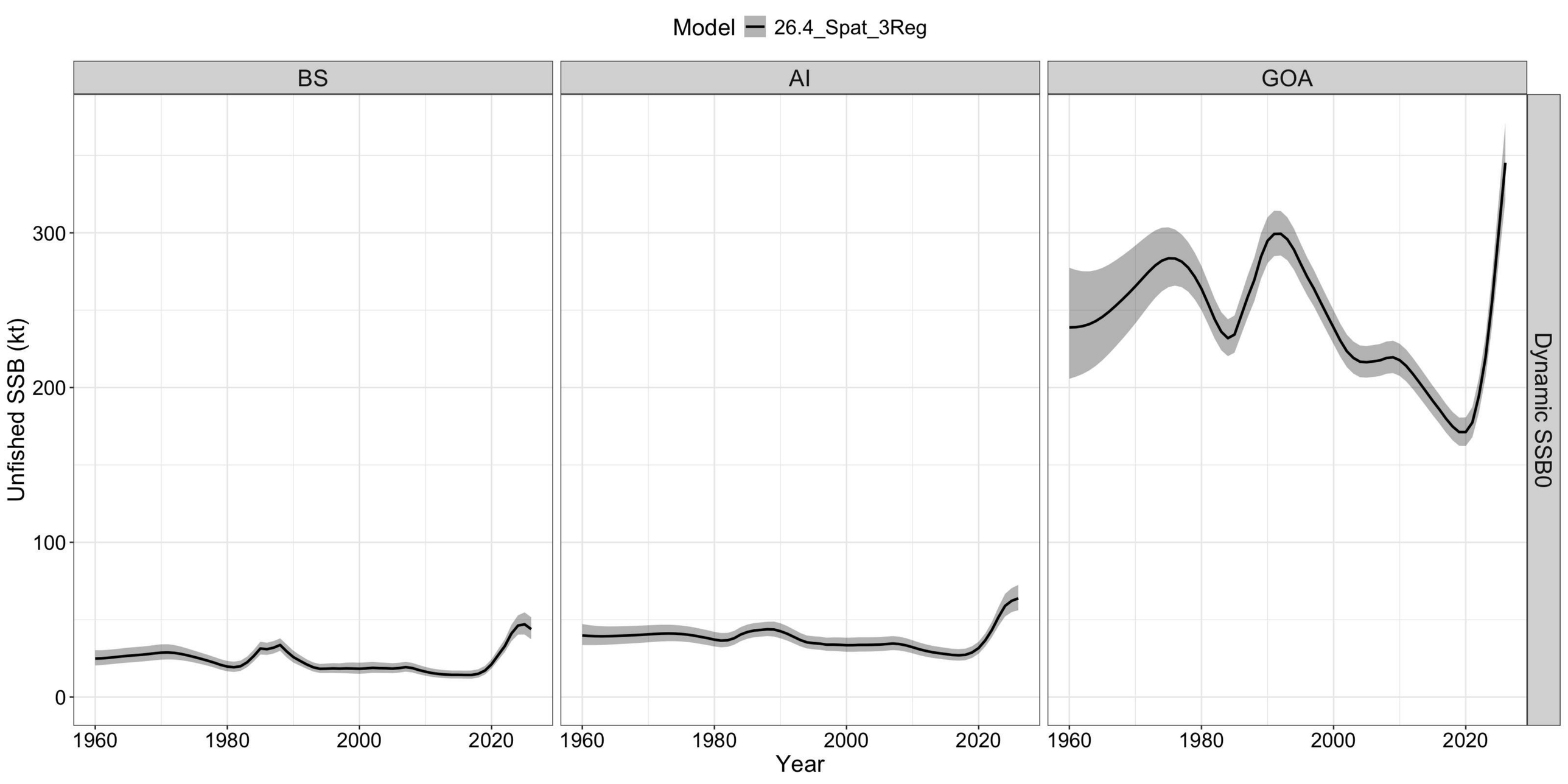
TF F

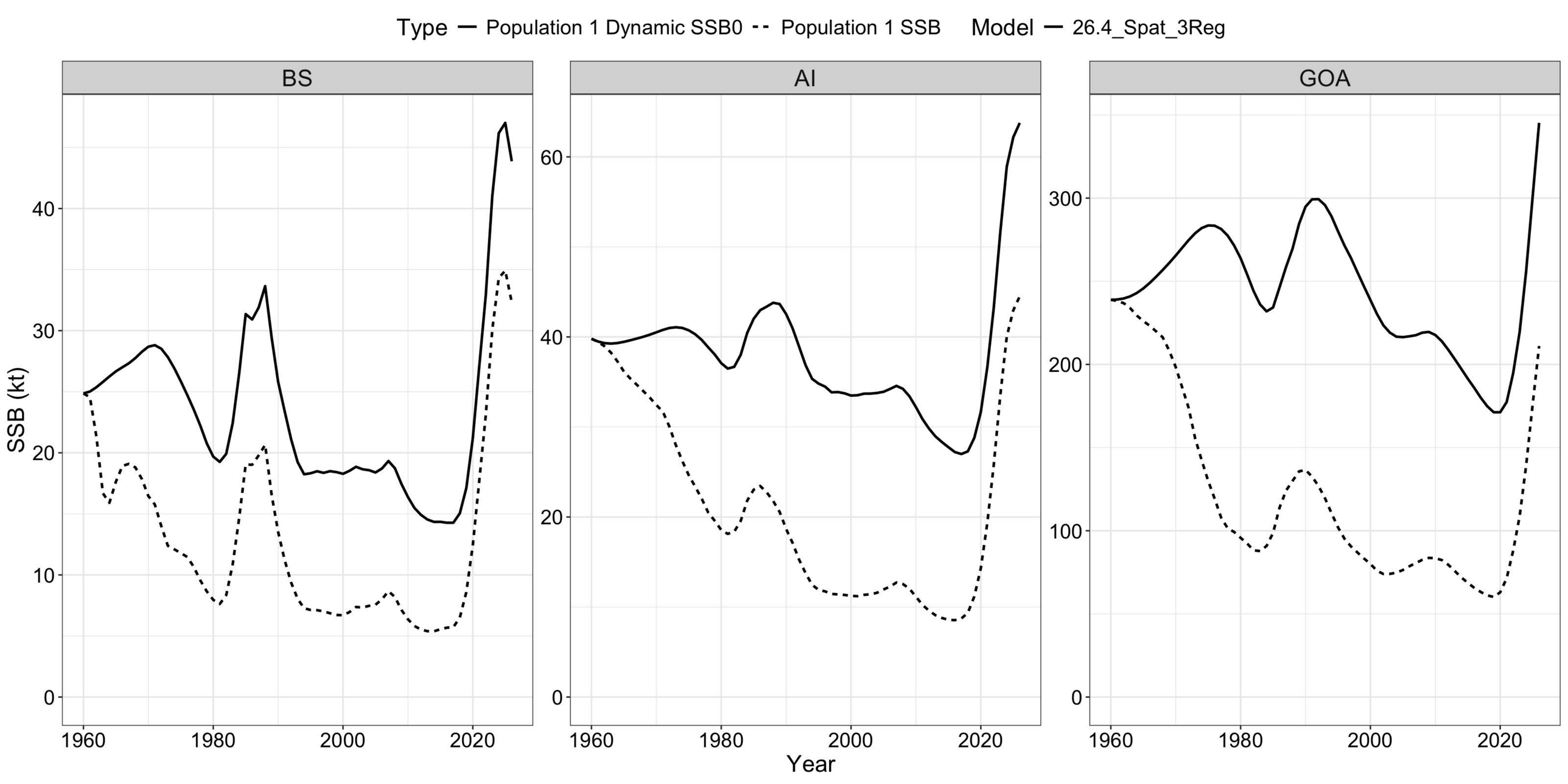
Value

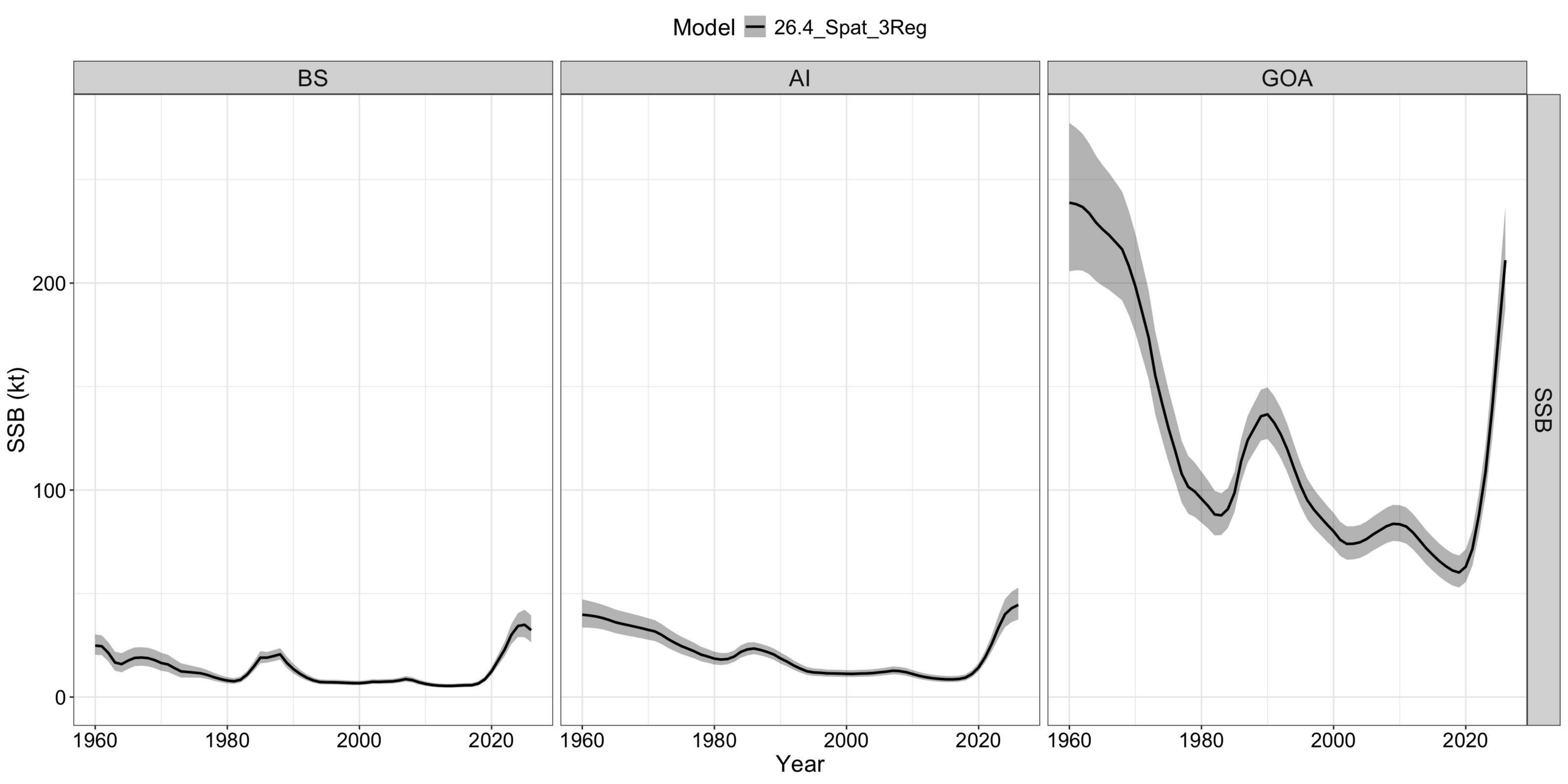
Year

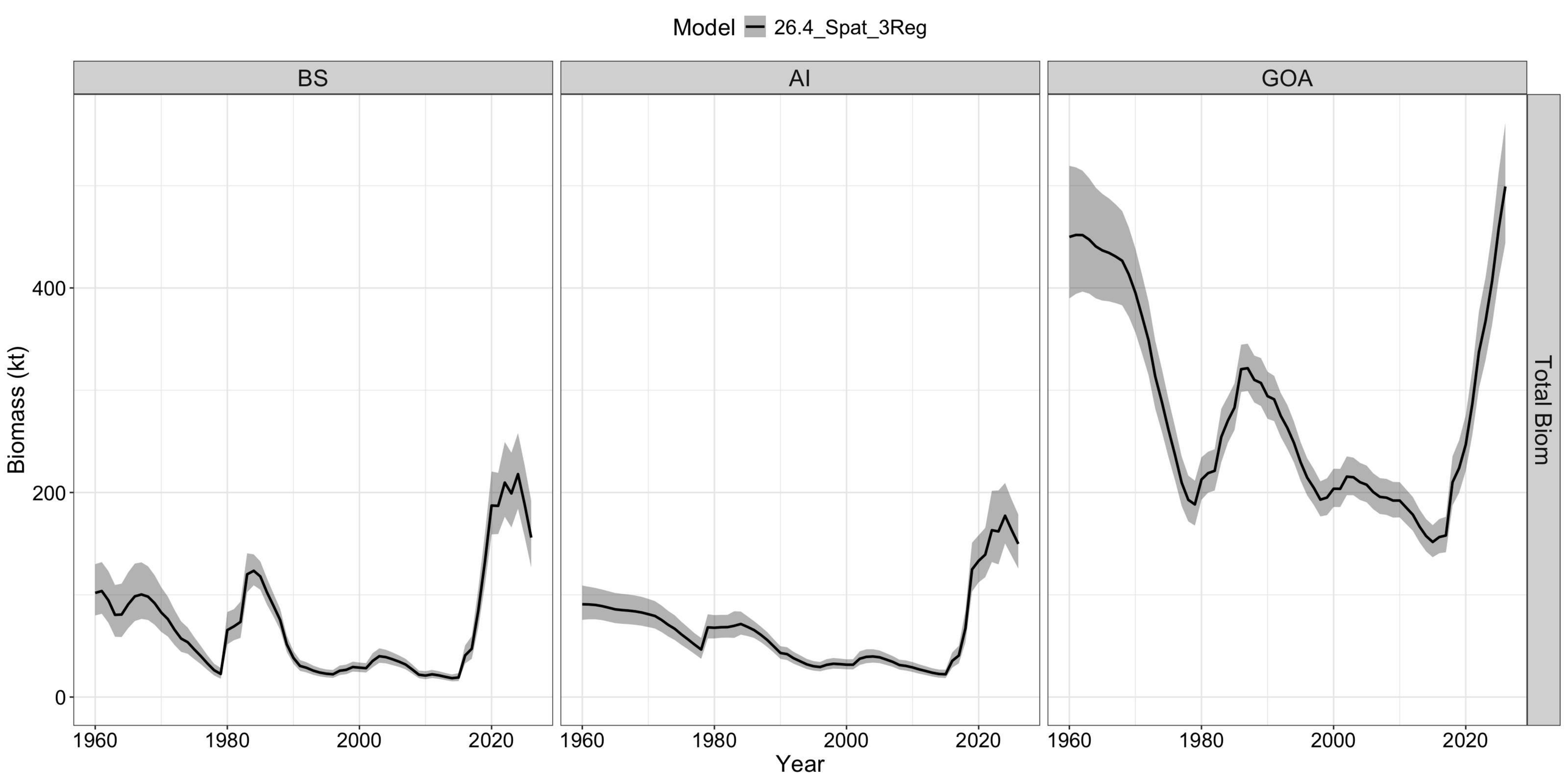

Model — 26.4_Spat_3Reg
BS
AI
GOA
LLF F
TF F
Fishing Mortality
0.0
0.1
0.2
0.00
0.05
0.10
0.15
1960
1980
2000
2020
Year


Model
26.4_Spat_3Reg
BS
AI
GOA
Recruitment
Recruitment (Millions of Fish)
0
25
50
75
1960
1980
2000
2020
Year

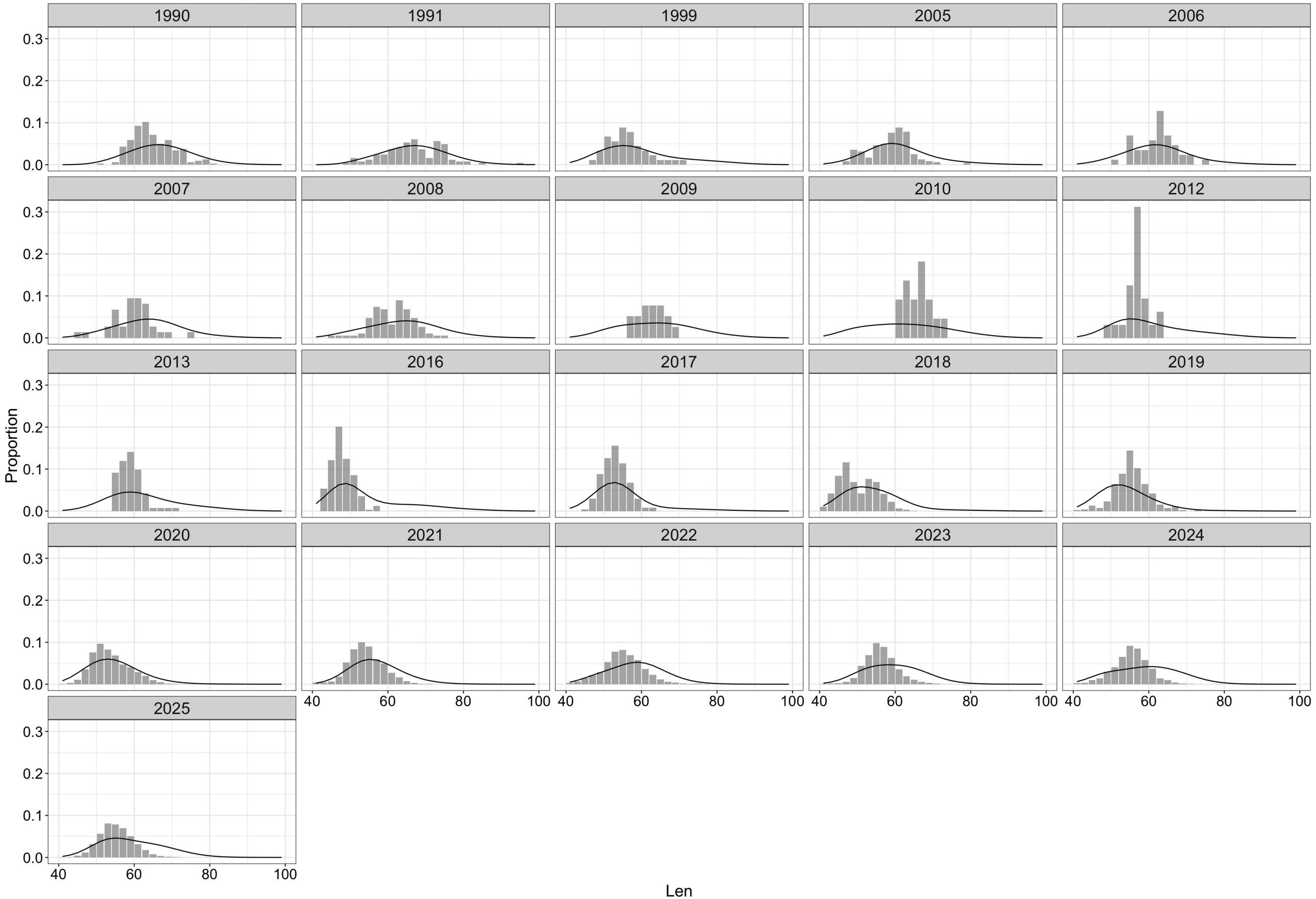

Model
26.4_Spat_3Reg
BS
AI
GOA
Dynamic SSB0
Unfished SSB (kt)
0
100
200
300
1960
1980
2000
2020
Year


Type — Population 1 Dynamic SSB0 -- Population 1 SSB Model — 26.4_Spat_3Reg
BS
AI
GOA
SSB (kt)
Year
0
10
20
30
40
60
100
200
300
1960
1980
2000
2020

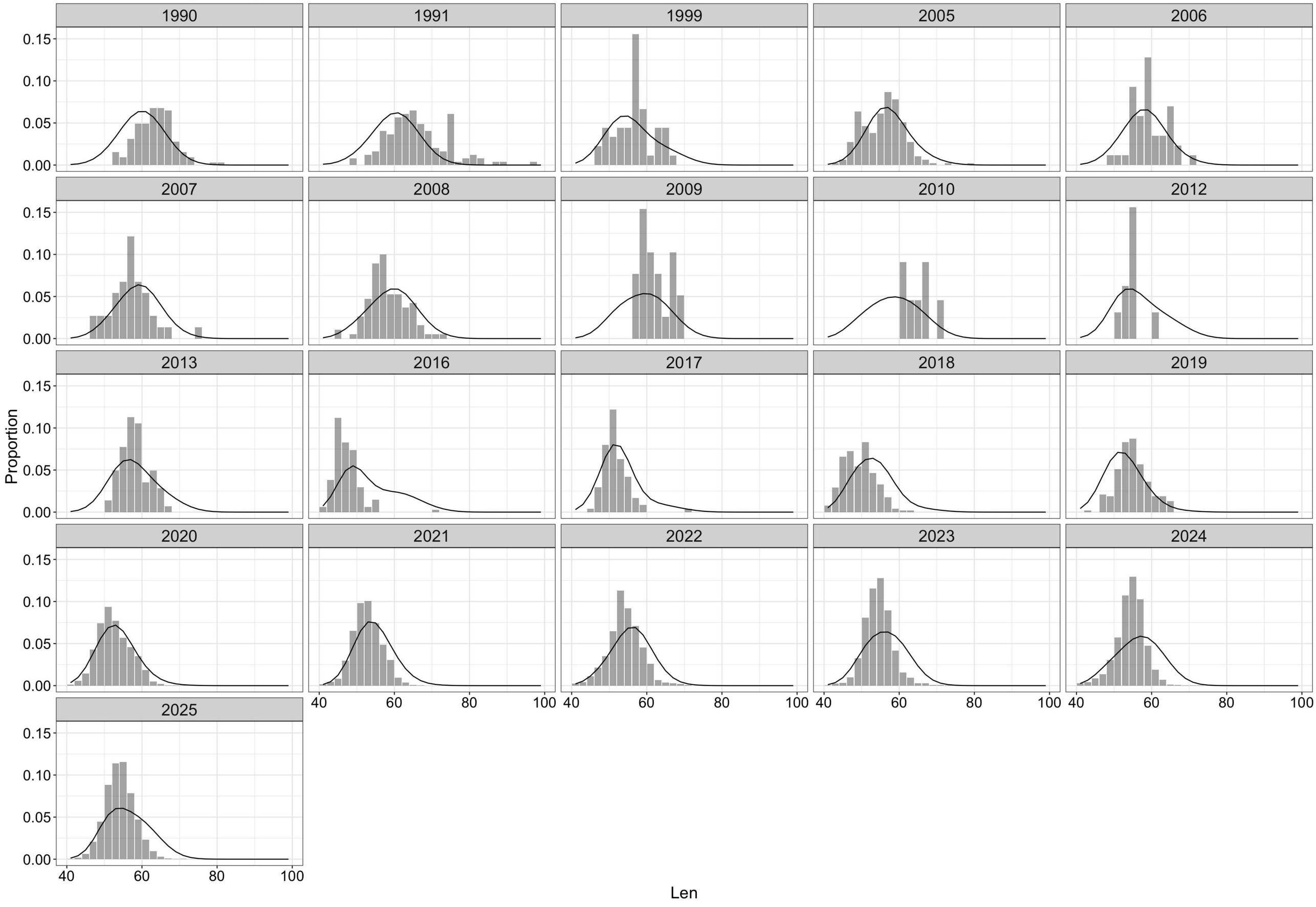

Model
26.4_Spat_3Reg
BS
AI
GOA
SSB
SSB (kt)
200
100
0
1960
1980
2000
2020
Year


Model
26.4_Spat_3Reg
BS
AI
GOA
Total Biom
Biomass (kt)
0
200
400
1960
1980
2000
2020
Year

Trawl Fishery Length Comps Female BS



Trawl Fishery Length Comps Male BS



Trawl Fishery Length Comps OSA BS

abs(Resid) 1 2 3 4 Sign Neg Pos

Female | Male | BS

Len: 100, 80, 60, 40

Year: 1990, 2000, 2010, 2020

BS

SDNR=0.95
(0.96-1.04)

Sample quantiles: 4, 2, 0, -2

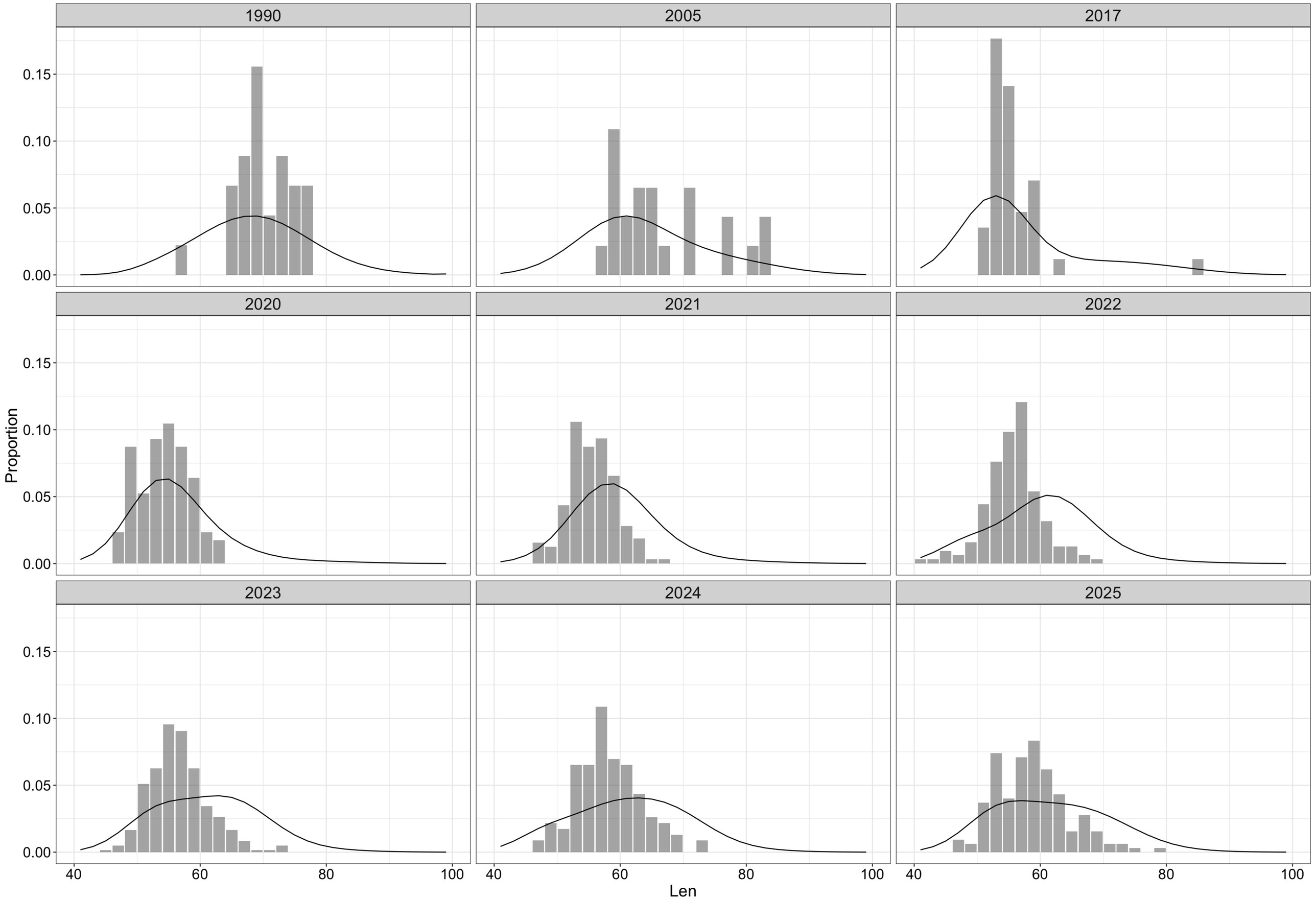
Theoretical quantiles: -2, 0, 2

Female | Male | BS

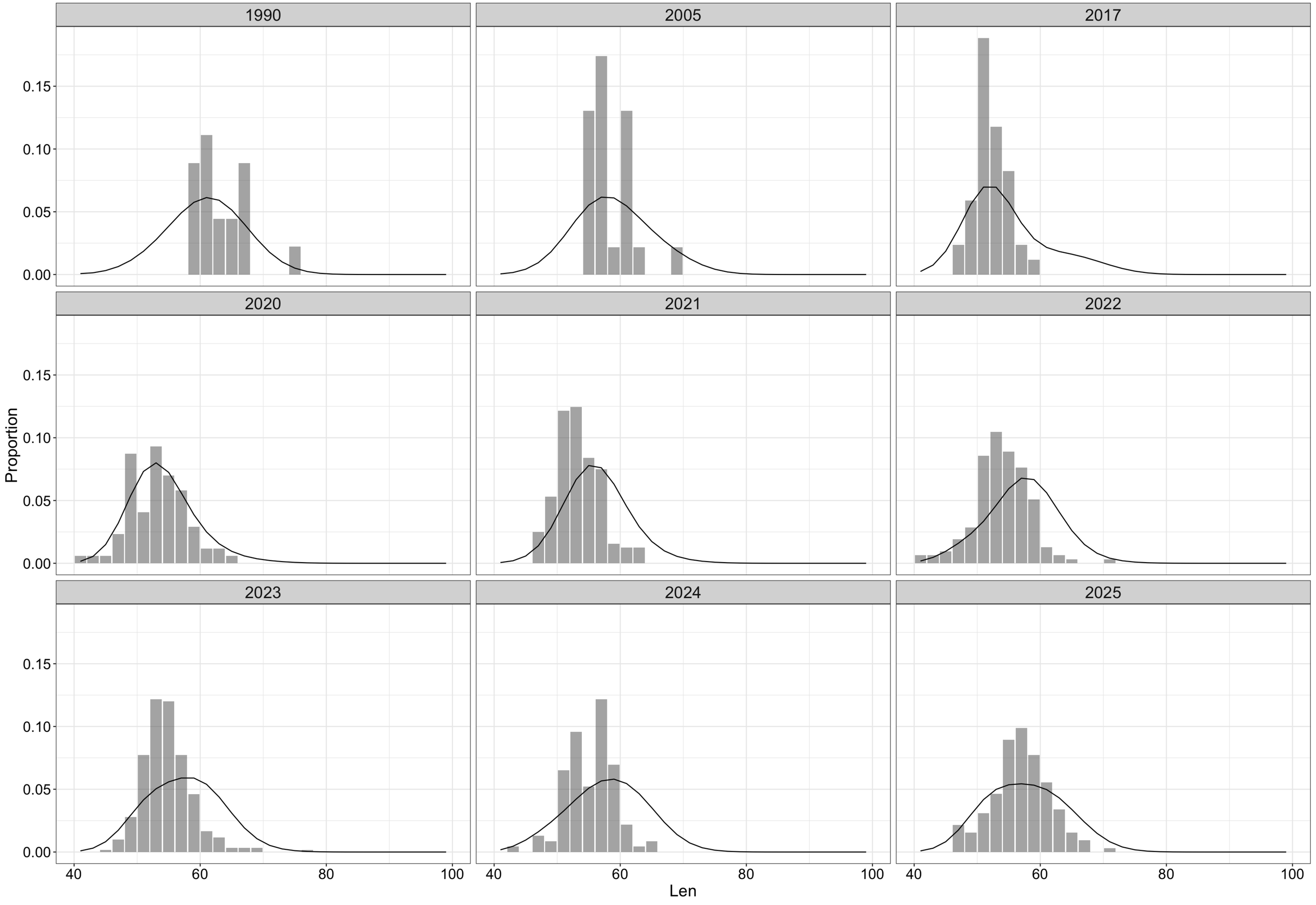
Proportions: 0.06, 0.04, 0.02, 0.00

Len: 40, 60, 80, 100

Trawl Fishery Length Comps Female AI



Trawl Fishery Length Comps Male AI



Trawl Fishery Length Comps OSA AI

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AI

Len

100

80

60

40

1990 2000 2010 2020

Year

AI

SDNR=0.99
(0.94-1.06)

Sample quantiles

2 0 -2

-2 0 2

Theoretical quantiles

Female

Male

AI

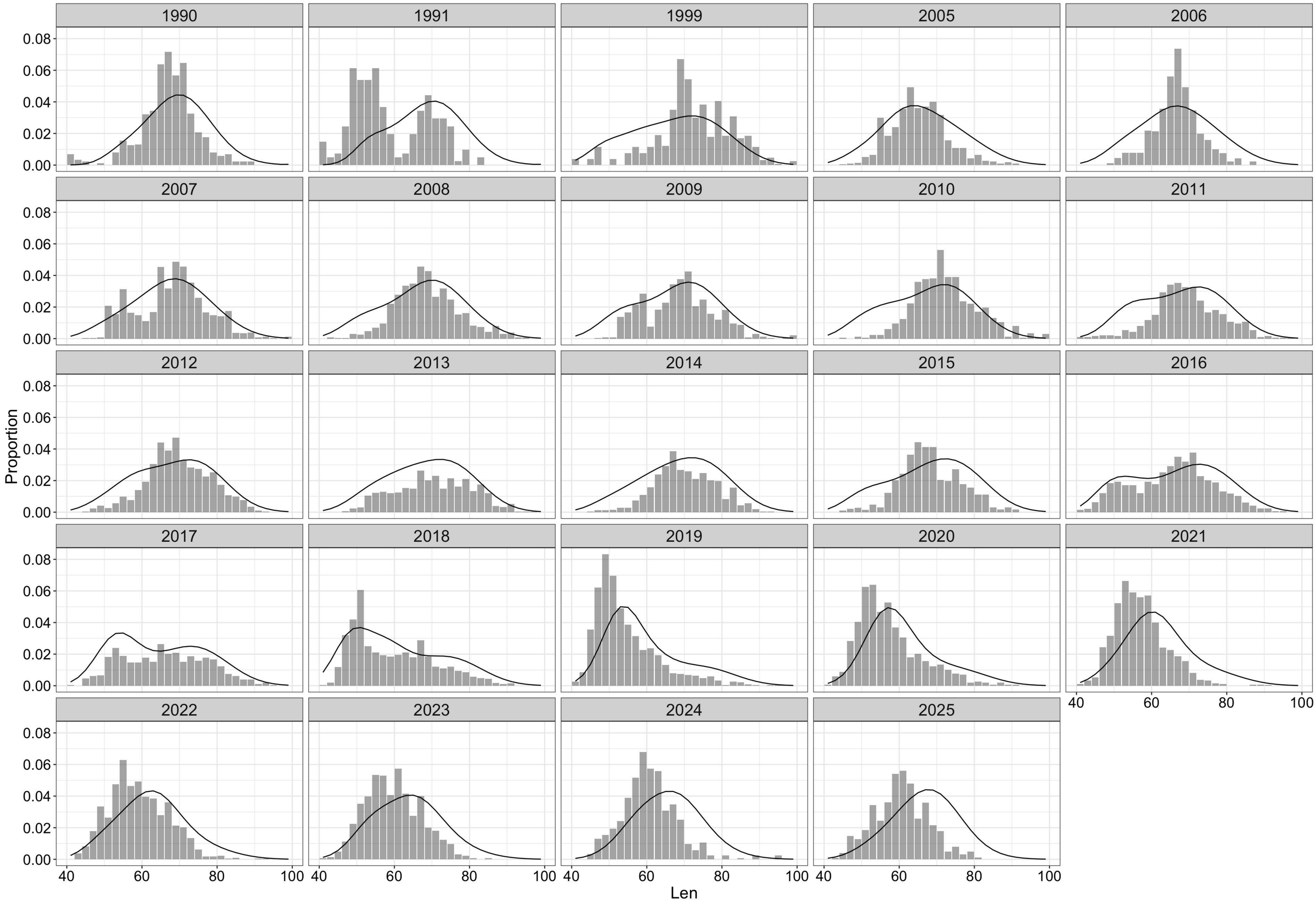
Proportions

0.08 0.06 0.04 0.02 0.00

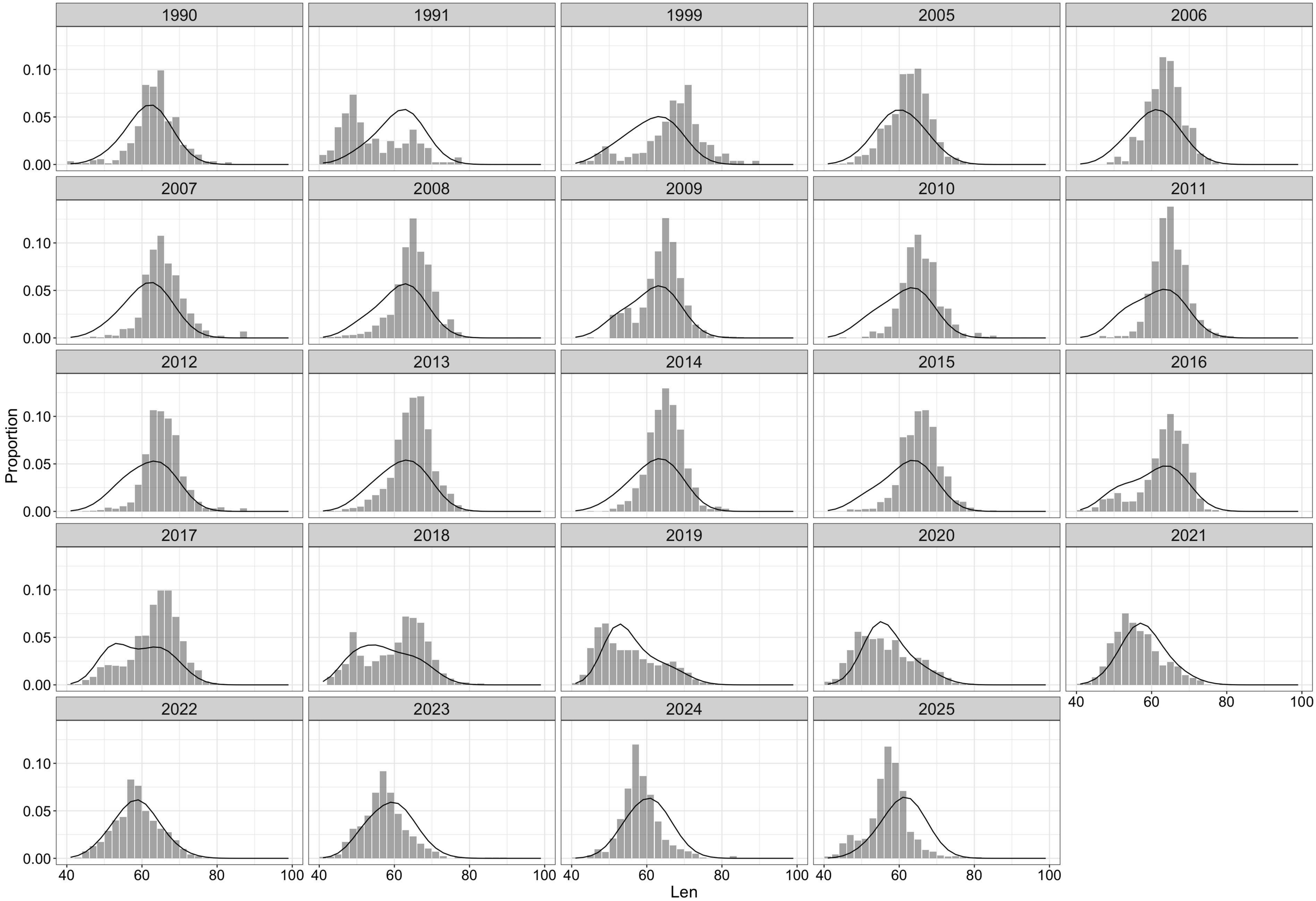
40 60 80 100

Len

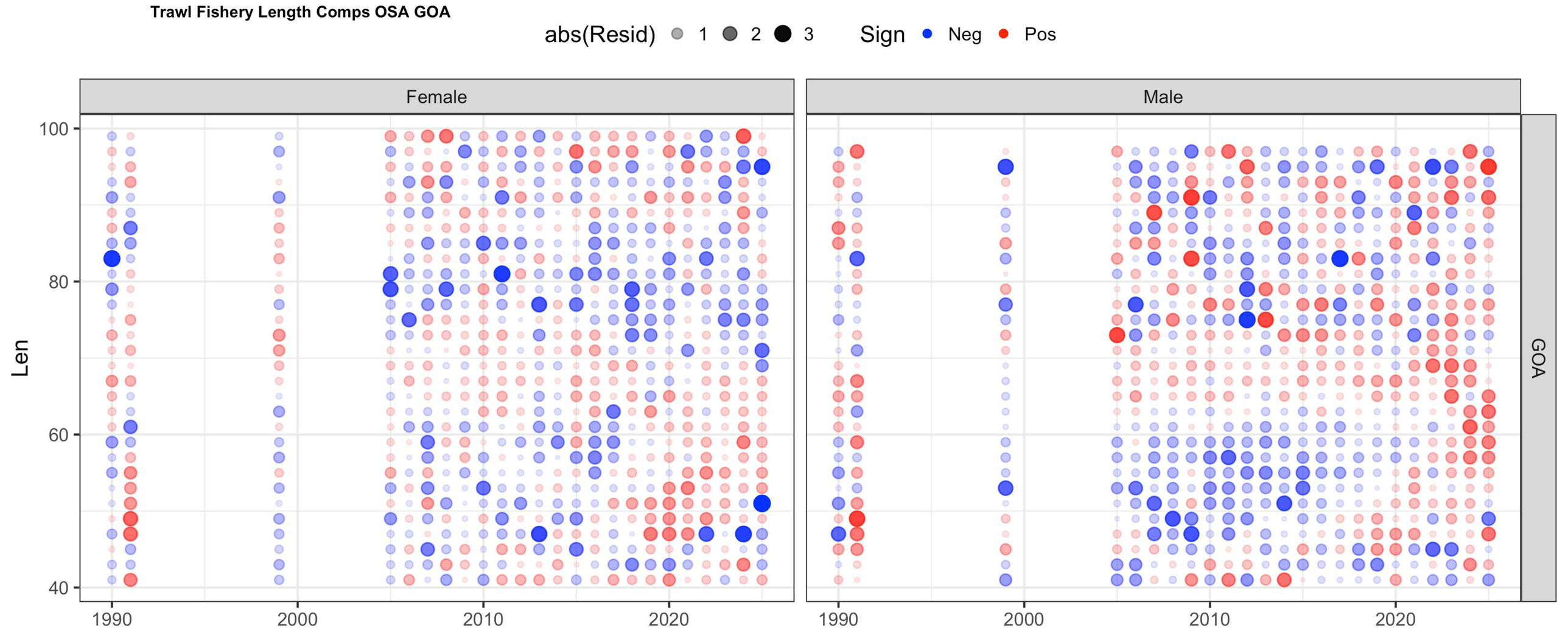
Trawl Fishery Length Comps Female GOA

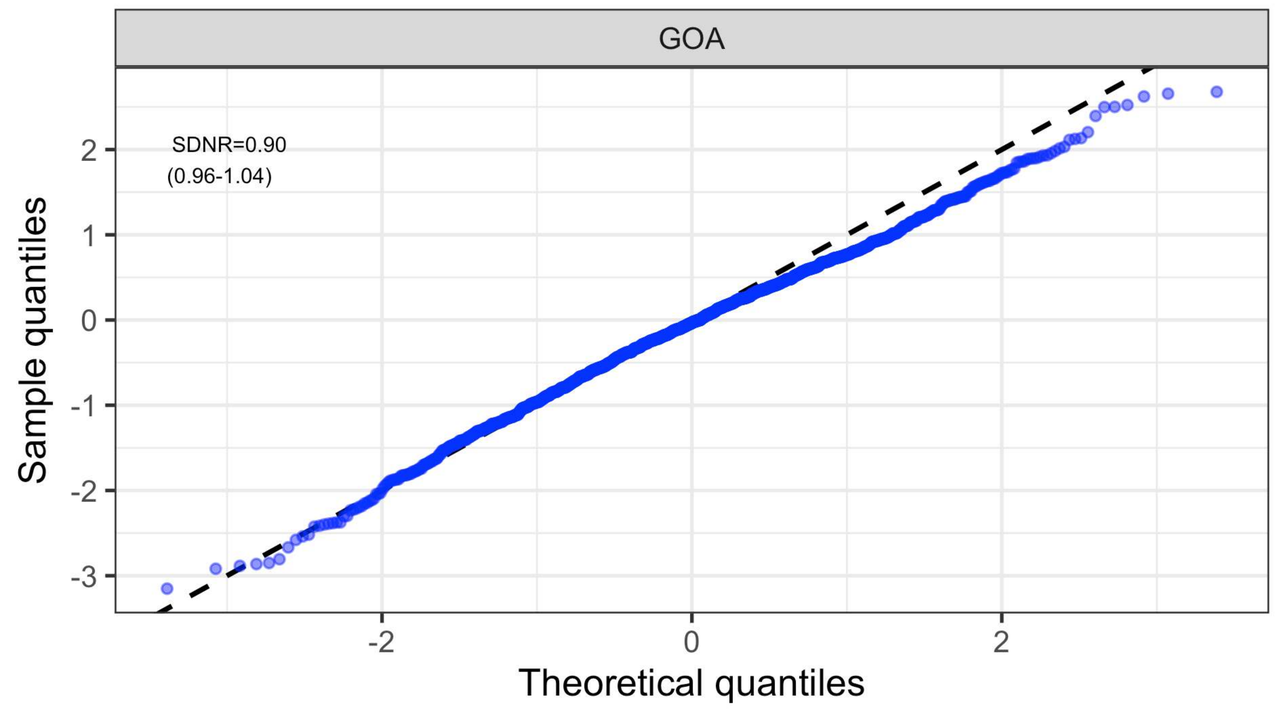


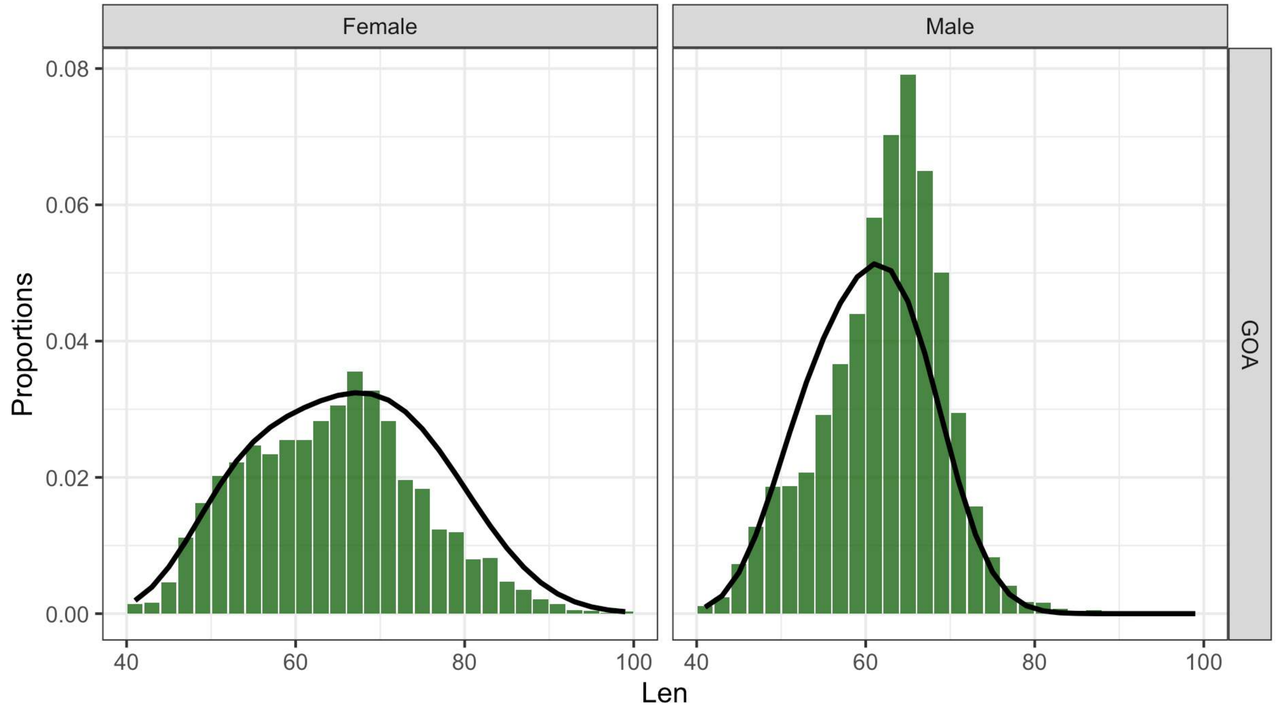
Trawl Fishery Length Comps Male GOA

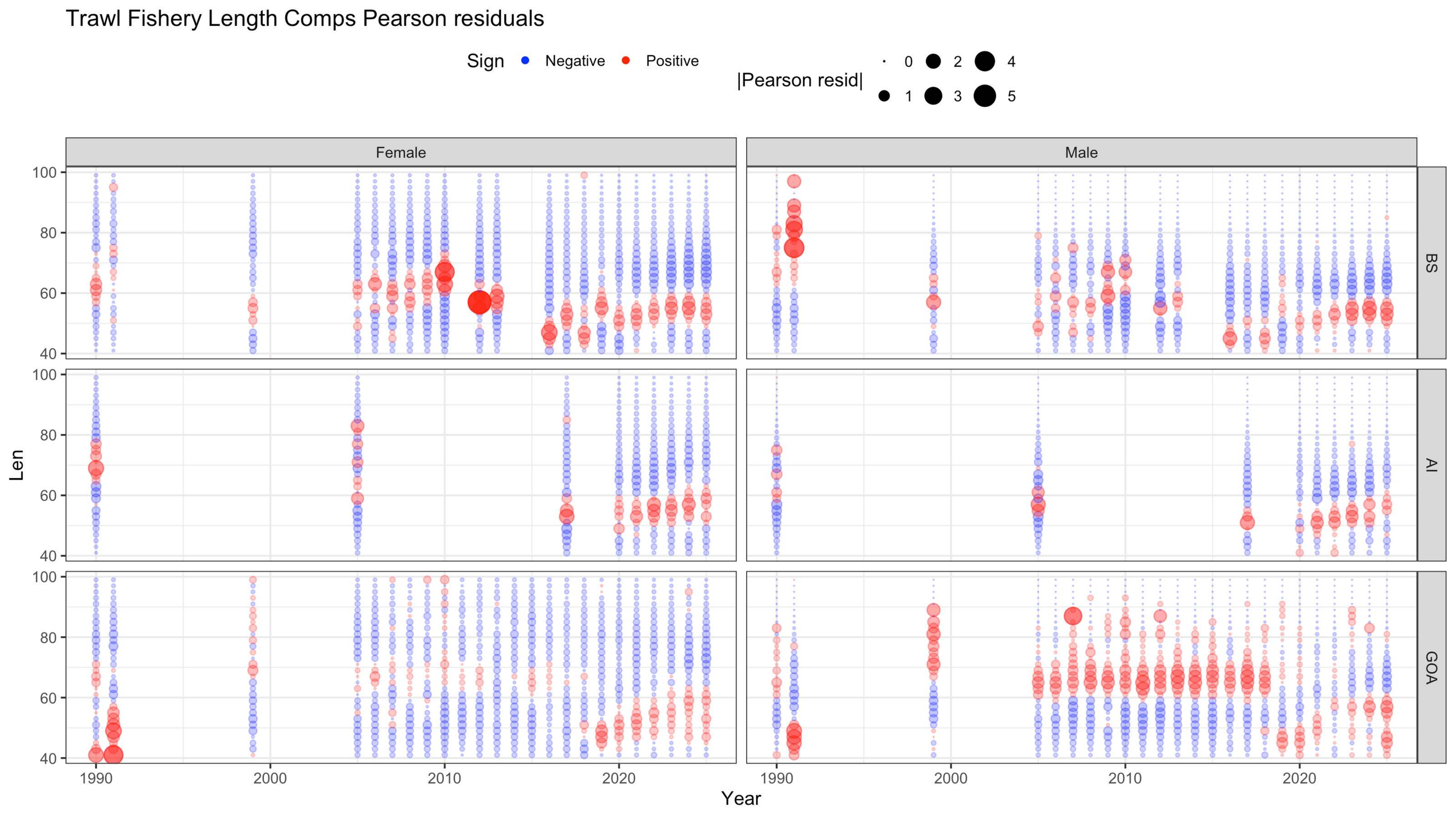


Trawl Fishery Length Comps OSA GOA

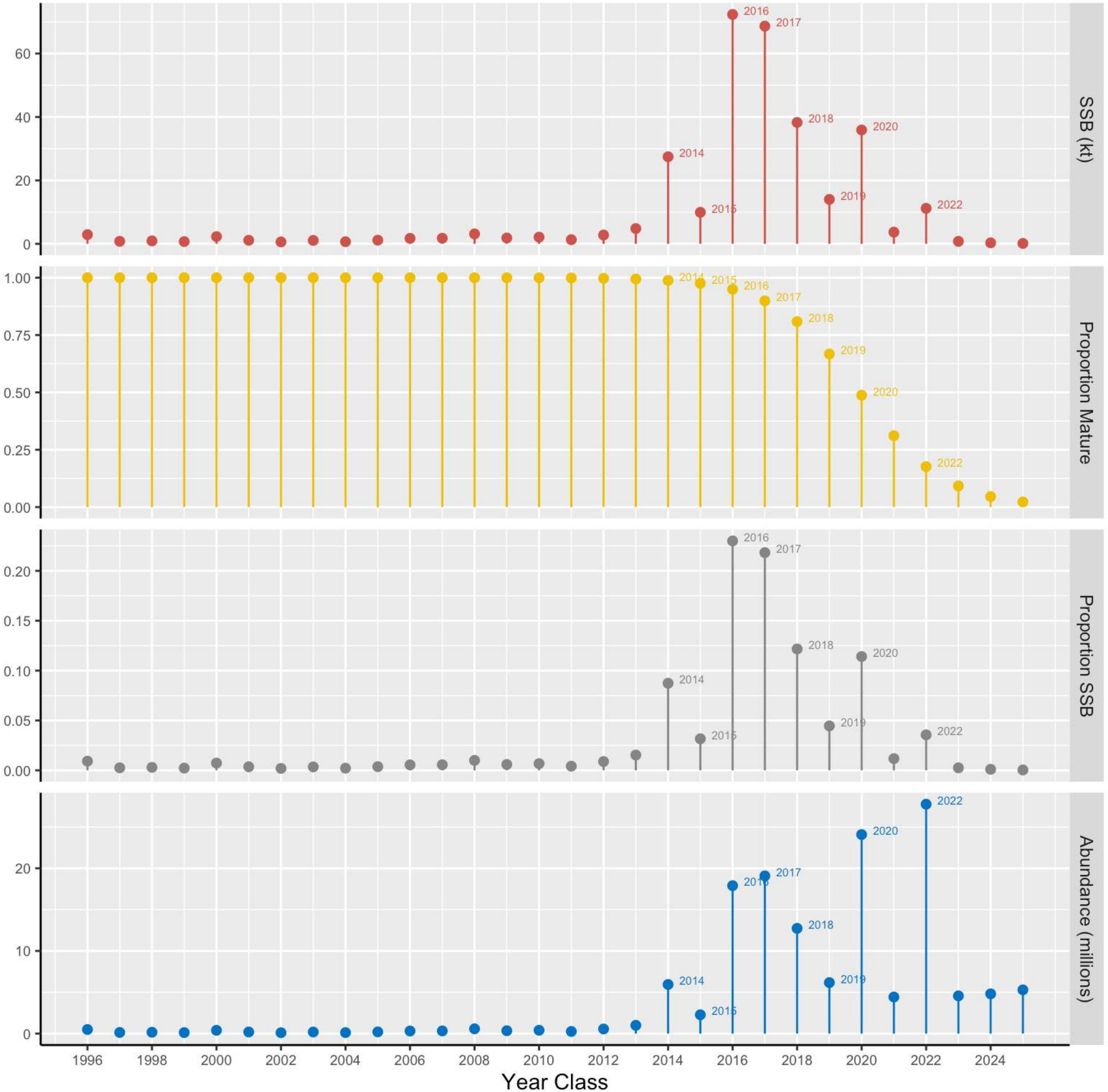







Trawl Fishery Length Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
0
1
2
3
4
5
Female
Male
BS
AI
GOA
Len
Year
100
80
60
40
1990
2000
2010
2020

Contribution to 2027 SSB by Year Class



Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1	NA	NA
BS JPN LL Survey Index	0.25	1	NA	NA
AI LL Survey Index	0.23	1	NA	NA
AI JPN LL Survey Index	0.25	1	NA	NA
GOA LL Survey Index	0.08	1	NA	NA
GOA JPN LL Survey Index	0.07	1	NA	NA
BS LL Survey Ages	NA	1	33.00	33.00
BS JPN LL Survey Ages	NA	1	33.00	33.00
AI LL Survey Ages	NA	1	33.00	33.00
AI JPN LL Survey Ages	NA	1	33.00	33.00
GOA LL Survey Ages	NA	1	33.00	33.00
GOA JPN LL Survey Ages	NA	1	33.00	33.00
BS Fixed Gear Fishery Ages	NA	1	25.00	25.00
BS Fixed Gear Fishery Lengths	NA	NA	NA	NA
BS Trawl Fishery Lengths	NA	1	26.81	26.81
AI Fixed Gear Fishery Ages	NA	1	25.00	25.00
AI Fixed Gear Fishery Lengths	NA	NA	NA	NA
AI Trawl Fishery Lengths	NA	1	20.56	20.56
GOA Fixed Gear Fishery Ages	NA	1	50.00	50.00
GOA Fixed Gear Fishery Lengths	NA	NA	NA	NA
GOA Trawl Fishery Lengths	NA	1	25.88	25.88

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.4_Spat_3Reg	1	32.31797	43.87277	0.030992408	8.33731	13.84588	0.07046	2.33412	0.73663	0.43985
26.4_Spat_3Reg	2	44.52230	63.77559	0.005190415	8.68328	17.88580	0.07046	2.48925	0.69811	0.07366
26.4_Spat_3Reg	3	211.06394	345.22109	0.043110056	33.15566	131.32995	0.07046	1.60713	0.61139	0.61183

Region	ABC_2027	OFL_2027	ABC_2028	OFL_2028	Prop_ABC_2027	Prop_ABC_2028
BS	8343.286	9800.068	6944.740	8158.124	0.166	0.141
AI	8682.118	10197.289	7571.177	8892.900	0.173	0.154
GOA	33154.731	38939.323	34635.509	40679.267	0.661	0.705
Total	50180.134	58936.681	49151.426	57730.292	1.000	1.000

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	NA	NA
M Male (estimated)	0.114	0.114	NA	NA
Average Recruitment (millions)	25.9	25.9	25.1	25.1
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	803134	794807
Projected Female Spawning Biomass (t)	219,714	241,217	314632	333206
B100% (t)	302,672	302,672	407654	407654
B40% (t)	121,069	121,069	163062	163062
B35% (t)	105,935	105,935	142679	142679
FOFL	0.102	0.102	0.083	0.083
max FABC	0.087	0.087	0.07	0.07
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	58937	57730
OFLw (t)	58,532	57,797	58806	57599
max ABC (t)	50,283	49,651	50180	49151
ABC (t)	50,283	49,651	50180	49151
ABCw (t)	50,111	49,482	50049	49020
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S2: Author's F (FABC Ratio)
Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S7: Approaching Overfishing
(FABC -> FOFL)

S1a: Author's F (FABC Ratio)
Year 1, FABC After

S6a: Author's F (FABC Ratio)
Year 1, FOFL After

Fishing Mortality

0.00
0.05
0.10
0.15

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S2: Author's F (FABC Ratio)
Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S7: Approaching Overfishing
(FABC -> FOFL)

S1a: Author's F (FABC Ratio)
Year 1, FABC After

S6a: Author's F (FABC Ratio)
Year 1, FOFL After

Fishing Mortality

0.00
0.02
0.04
0.06
0.08

2010
2020
2030
2040

Year

S1: FABC (F40)

S2: Author's F (FABC Ratio)
Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

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S7: Approaching Overfishing
(FABC -> FOFL)

S1a: Author's F (FABC Ratio)
Year 1, FABC After

S6a: Author's F (FABC Ratio)
Year 1, FOFL After

Total Biomass (kt)

0
250
500
750

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S2: Author's F (FABC Ratio)
Years 1+2

S3: 5-year Average F

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2010
2020
2030
2040

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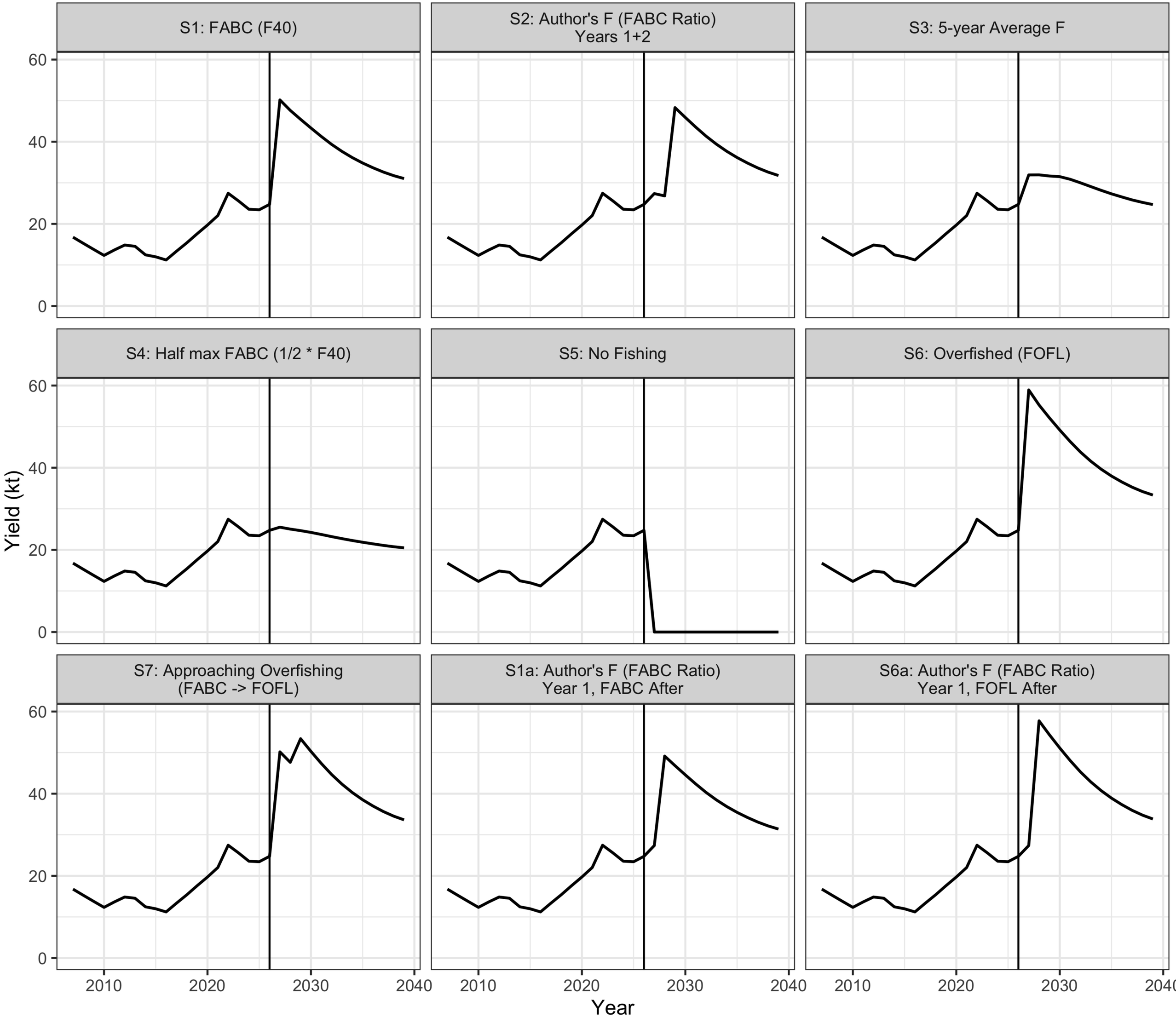
S6a: Author's F (FABC Ratio)
Year 1, FOFL After

Yield (kt)

0
20
40
60

1960
1980
2000
2020
2040

Year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Year 1, FOFL After
Yield (kt)
0
20
40
60
2010
2020
2030
2040
Year

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(FABC -> FOFL)

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Year 1, FABC After

S6a: Author's F (FABC Ratio)
Year 1, FOFL After

Recruitment (millions)

0
30
60
90
120

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S2: Author's F (FABC Ratio)
Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

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Recruitment (millions)

0
30
60
90
120

2010
2020
2030
2040

Year

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Year 1, FABC After

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Year 1, FOFL After

SSB (kt)

0
100
200
300
400

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S2: Author's F (FABC Ratio)
Years 1+2

S3: 5-year Average F

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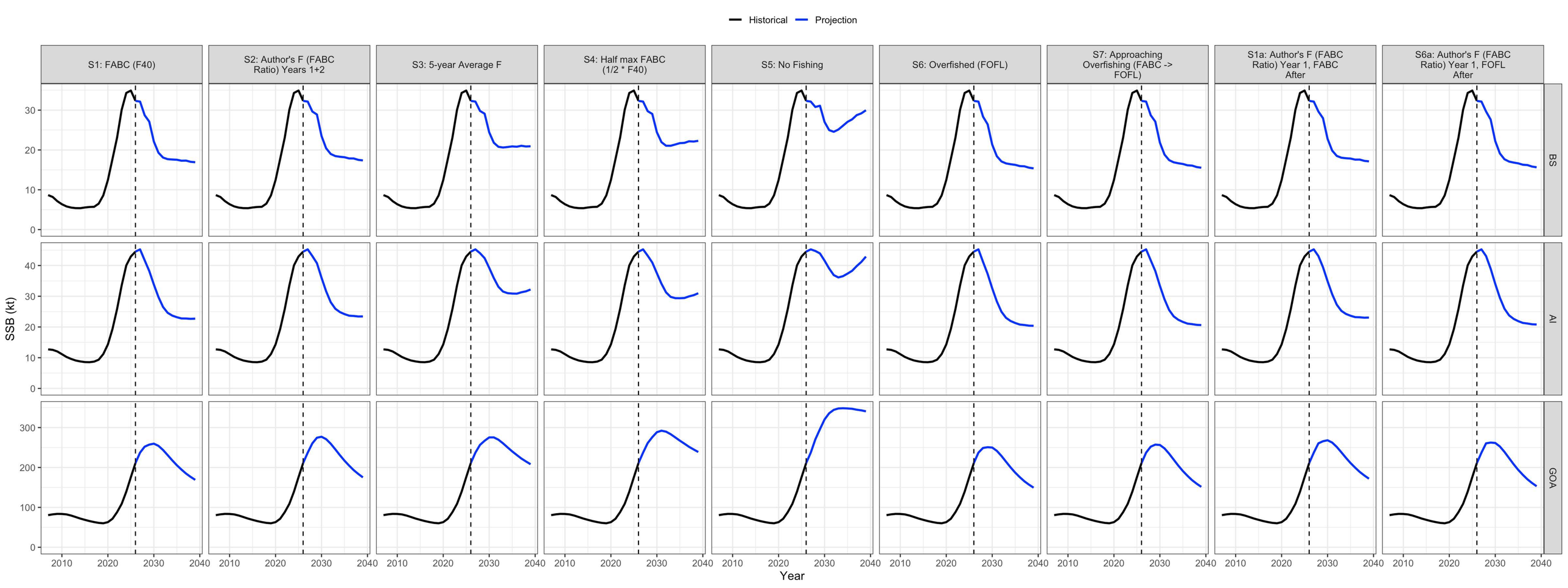
S6a: Author's F (FABC Ratio)
Year 1, FOFL After

SSB (kt)

0
100
200
300
400

2010
2020
2030
2040

Year


Historical
Projection
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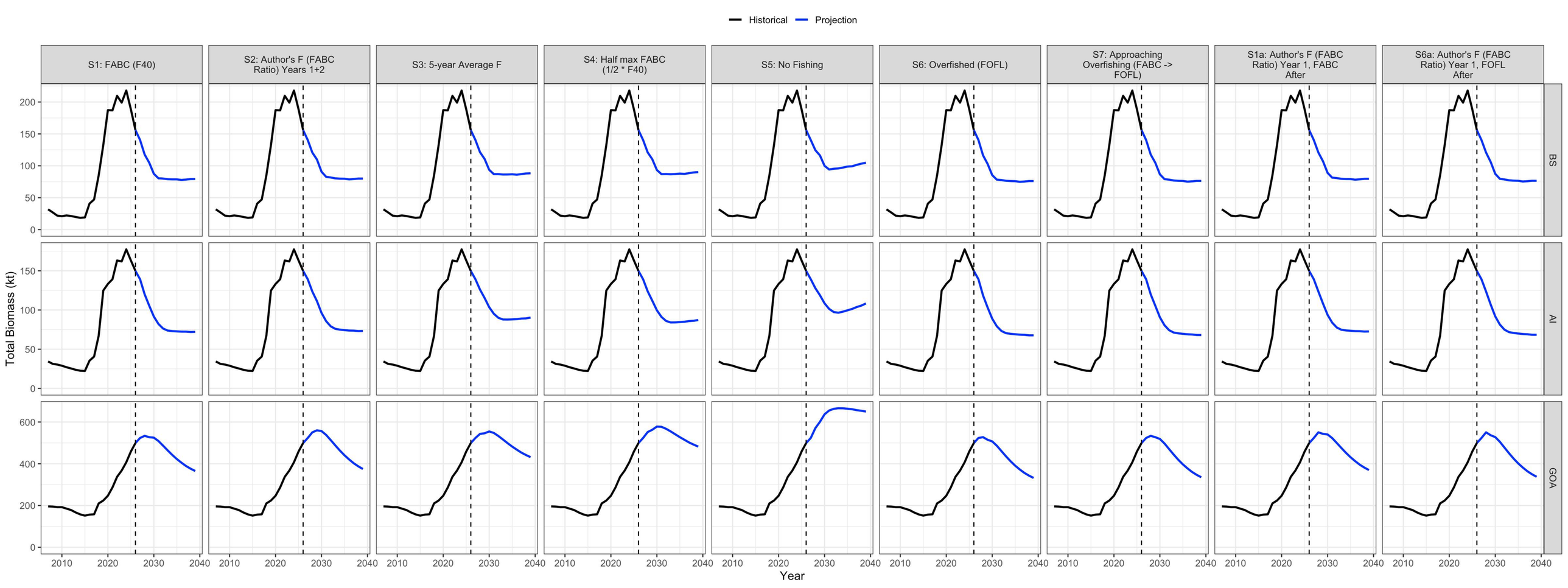
S1a: Author's F (FABC Ratio) Year 1, FABC After
S6a: Author's F (FABC Ratio) Year 1, FOFL After
BS
AI
GOA
SSB (kt)
Year
0
10
20
30
40
100
200
300
2010
2020
2030
2040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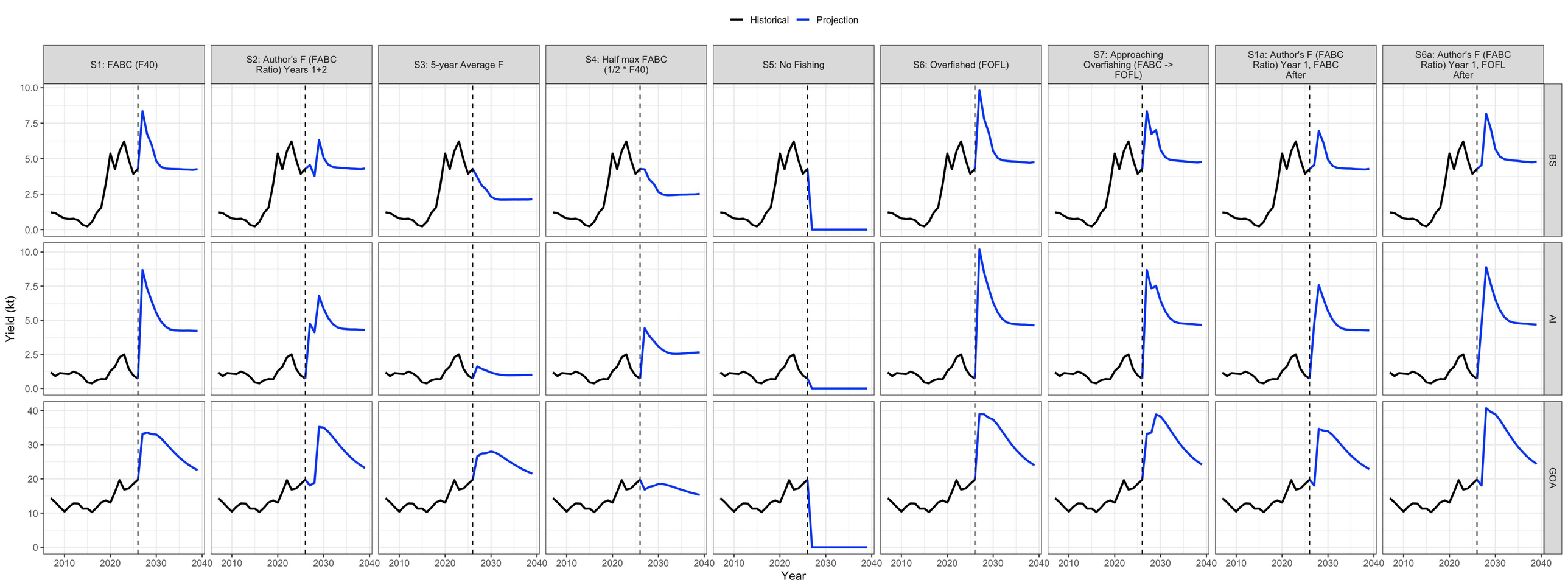
S1a: Author's F (FABC Ratio) Year 1, FABC After
S6a: Author's F (FABC Ratio) Year 1, FOFL After
BS
AI
GOA
Total Biomass (kt)
Year
0
50
100
150
200
400
600
2010
2020
2030
2040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S1a: Author's F (FABC Ratio) Year 1, FABC After
S6a: Author's F (FABC Ratio) Year 1, FOFL After
BS
AI
GOA
Yield (kt)
Year
0.0
2.5
5.0
7.5
10.0
0
10
20
30
40
2010
2020
2030
2040